Concept Design Report



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Executive Summary

CRASH (Controlled Recovery and Safe Handling) is a rocket competition hosted by the University of Central Florida with the goal of introducing students to the engineering principles behind launching, protecting, and safe recovery of fragile payloads. Team two will be designing, building, and testing a L1-class rocket capable of transporting at least two eggs and releasing 5 oz of water, additionally providing information about the flight trajectory. This requires developing an egg capsule that can absorb the loads of launch, parachute deployment, and ground impact while preventing cracking or internal failures. This challenge extends beyond the typical rocketry competition of reaching a set altitude, it gives students a chance to practice constraints faced in the aerospace industry. Protecting delicate payloads is common in industry, like biological payloads or sensitive instruments. Balancing mass, safety needs, performance, manufacturability, prices, and competition scoring. Learning the fundamentals of system architecture, subsystem integration, requirements development, and testing. The team is currently conducting system-level analysis, defining engineering requirements, and down-selecting between payload protection concepts. Recent work includes modeling the expected acceleration loads, evaluating foam damping materials, assessing water-release mechanisms, and performing preliminary structural and recovery simulations. The project is progressing toward a fully integrated rocket and payload system that maximizes survivability while meeting competition scoring objectives.

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1.0 Introduction

The UCF Controlled Recovery And Safe Handling (CRASH) Challenge is designed to introduce students to the core principles that have defined many historic space missions. The challenge emphasizes payload protection, recovery reliability, and the ability to engineer within real-world constraints. The scoring criteria for the challenge is primarily based upon how many eggs are safely launched and recovered by the individual teams, with secondary objectives available to score additional points. Team 2 is attempting to design a modular payload solution to this challenge, such that once the required safety parameters are met, the design can be stacked to accommodate a larger payload quantity.

The purpose of this report is to present the design and justifications for Team Two's rocket system for the CRASH challenge. The rocket is broken down into 5 subsystems: egg payload, water payload, recovery system, aerostructure, and avionics. Each has multiple potential solutions being evaluated to balance mass limitations, safety needs, performance, manufacturability, prices, and competition scoring. Two of the most significant challenges facing the team are budget and weight. While advanced materials and sensors could greatly enhance payload protection, water release mechanisms, flight tracking, and live video capabilities, the team is working with a small and still-uncertain budget. This mirrors real aerospace engineering conditions, where the optimal solution is not necessarily the most sophisticated, but the one that meets mission needs within financial constraints. The weight challenge is directly tied to the competition's shift from I-class motors to H-class motors for this year's event. With a lower impulse motor, mass becomes a critical driver in the rocket's ability to reach the minimum required altitude of 1,000 ft. Failure to meet this threshold results in automatic disqualification, making every ounce of structural mass, payload mass, and subsystem hardware a key design consideration.

This report is organized to guide the reader through the full progression of the project, from initial brainstorming to detailed subsystem development. It begins by outlining the project needs, purpose, and goals, followed by the concepts generated to meet those requirements. The report then reviews relevant industry practices and evaluates how they can be adapted to a smaller-scale rocket system. Each subsystem's design concepts are presented and compared to identify the most suitable solutions. Preliminary engineering analyses of the current designs are then discussed. Plans for testing, potential failure modes, and corresponding mitigation strategies are also documented. The report concludes with a summary of the team's accomplishments to date and an overview of the work planned for the remainder of the project.

2.0 Needs Analysis

The primary stakeholder is the CRASH competition which will evaluate the team's ability to deliver the eggs and water safely under the competition's constraints. As the vehicle itself is being designed specifically for competition, the only direct stakeholder is the competition organizer, the University of Central Florida. The CRASH competition aims to challenge students to find the balance between point maximization, safety, and meeting minimum altitude. Team 2 shall be the sole end user of the final product. However, the principles being applied for this competition vehicle are similar to those which have been used for many prominent missions in

2.0 Needs Analysis (continued)

the past, so unique solutions to the problem of handling delicate payloads can benefit many clients in the aerospace industry. Parallel to industry finding a balance between their own needs and constraints, typically cost and performance. The CRASH competition also puts a strong emphasis on safety, the top priority being the safety of the payload and water, demonstrated by the automatic disqualification of rockets impacting above a speed limit of 25 m/s. The broader relevance of the rocket is the delivery of fragile payloads, common in the real-world space industry. Applications like the transport of biological materials, microgravity science payloads, and precision instrumentation, where the survival of the fragile payloads determines the success of the mission. The CRASH Rocket will contribute to the improvement of payload packaging standards and safe recovery methods.

As there are no direct third party users of the rocket system, all priorities are derived by the competition rules. All "threshold" goals of the competition are considered "needs", as if they are not met, the team is disqualified. All goals that offer bonus points on top of threshold goals will be designated "wants".

1. [Need] Safe Payload delivery to ground (little to no damage: no visible cracks or dents on the eggs, no visible cracks or leaks for the water container). The contest rules require the payload is recovered with at least two eggs fully intact, as well as no leakage having occurred in the water vessel.

2. [Need] Landing within a reasonable distance of the launch zone. The contest rules require the vehicle to be recovered within 800 feet from the launch pad.

3. [Need] Vehicle is recovered in flyable condition. The contest rules require the vehicle to be recovered in a flyable condition.

4. [Need] Undergo a controlled flight (Maximum altitude and landing speed). The contest rules require the descent velocity during the final five seconds of flight to not exceed 25 feet per second, as well as an apogee requirement of between 999 feet and 1500 feet.

5. [Need] Diversity of cargo (eggs and water). The contest rules require a minimum of 2 eggs and 5 oz of water on board.

6. [Need] Ease of access to cargo (time requirements). The contest rules require a payload loading and unloading time of 5 minutes or less.

7. [Need] Timed ejection of recovery equipment. Ejection of recovery equipment in a dual deployment manner gives the team more points and gives the team control over the speed of descent and the distance of landing.

8. [Need] Logging of flight data such as velocity, acceleration and altitude. Logging of flight data is needed to prove apogee requirements are met on launch day. A tracking system also gives the team additional points.

2.0 Needs Analysis (continued)

9. [Need] Backup equipment to ensure data recoverability. Allows the opportunity to be scored even if the primary electronics fail.
10. [Need] Use of sufficiently scaled motor for the task ($\leq$ 320 Ns). The contest rules require the sum of all launch vehicle propulsion systems to contain no more than 320 Newton seconds of total impulse.
11. [Want] Deployment of a parachute during recovery.
12. [Want] Capability of carrying more cargo. To maximize points scored in the competition, the vehicle should be capable of carrying additional eggs and water.
13. [Want] Some form of acoustic or GPS locating system. Additional points will be earned from carrying a location/tracking system.
14. [Want] Employment of a dual-deployment recovery system. Additional points will be earned from the utilization of a dual-deployment recovery system.
15. [Want] Inclusion of a functional camera on the rocket.
16. [Want] Video playback of camera footage.
17. [Want] Live stream video footage from rocket to ground station. Additional points will be earned from the transmission of live video footage from the vehicle to a ground station.

3.0 Technology Assessment

In order to gather a baseline for technologies relevant to the design of our launch vehicle, a series of studies were performed. Each study is focused around a specific subassembly, and details technologies which are applicable to each subsystem component within that subassembly.

3.1 Egg Payload

3.1.1 Understanding Eggs

To design a payload capsule that protects the eggs carried by the vehicle, the physics of the eggs must first be assessed. Eggs are often believed to be most resistant to cracking when positioned vertically; however, according to a study published in Nature Communications Physics, “The Uncrackable Egg: Static and Dynamic Compression Tests” [1], eggs falling horizontally were found to be more resistant to cracks. The study on static compression revealed that the orientation of the egg does not significantly affect shell strength, averaging 46.0 ± 6.61 N for vertical and 45.2 ± 5.52 N for horizontal orientations [1]. In terms of dynamic compression,

3.1.1 Understanding Eggs (continued)

eggs dropped on their equator were less likely to crack than those dropped on their poles when dropped from a great height [1].

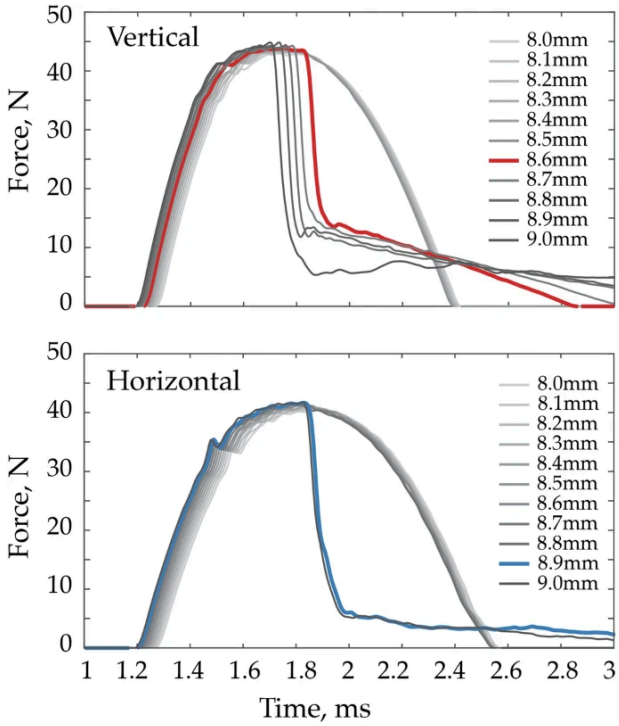


Fig 1. Force–time curves for vertical (top) and horizontal (bottom) drops [1]

A separate study by ADMET [2] tested eggs using a single-column compression machine with foam padding and found that eggs broke under an average load of 53 lb (235.8 N) on their poles and 90 lb (400.3 N) on their sides. The higher strength observed during compression compared to drop tests is due to the egg's shape, which helps distribute force evenly. As ADMET [2] explains, "Like an arch under a bridge, the shape of an egg helps distribute the applied force across the entire surface. Cracking an egg with the edge of a pan is easy because the force is concentrated on one small section of the shell. When force is applied more evenly, eggs appear to be much stronger." Therefore, if we protect eggs by distributing force evenly around them, the chances of cracking decrease significantly.

Although the orientation of the vehicle and capsule cannot be controlled by the team, the orientation of the eggs inside still matters. Because the rocket capsule cannot uniformly protect all sides, positioning the eggs so their most vulnerable areas align with the most dampened regions of the capsule can improve protection and reduce the risk of damage.

3.1.2 Capsule Conditions

During flight, the eggstronauts will be going through multiple stages of acceleration and deceleration. First rapid upward acceleration when the motor ignites, then coasting after burn time. Eventually decelerating until apogee, where it reaches 0 vertical velocity. Then accelerating again downwards until the release of the drogue, then the main parachute. Finally, a rapid deceleration when impacting the ground. This process likely causes continuous vibration and potential collisions with adjacent eggs if not properly separated.

3.1.3 Options for Protection

Dampening methods for fragile cargo include the use of cushioning and energy absorbing materials such as foams, honeycomb structures, and air filled packages. Additional strategies involve structural design features like suspension systems, crumple zones, and other mechanisms that promote energy dissipation and viscous damping to protect delicate payloads from impact and vibration. These methods have varying levels of difficulty in integrating into the vehicle, cost, and effectiveness of dampening.

Foams are the easiest to integrate, lowest cost, and can be very effective in dampening if the correct foam is chosen. When choosing a foam, its density and structural qualities must be considered. For example, an open cell like a memory foam would crush too easily and bottom out under high g. Avoiding a loose styrofoam that would allow eggs to move and still have angled impacts. Additionally, avoiding foams that would be too stiff and transfer the force to the eggs. Polyethylene is closed-celled, has elastic recovery over plastic deformation, and is lightweight while structurally efficient. Within the category of Polyethylene, there are two options. Regular polyethylene (PE) and cross-linked polyethylene (XPLE). Regular PE foam is inexpensive, easy to cut, and offers good shock absorption for one-time or short-term use, though it can lose resilience after repeated impacts. XLPE is denser and more durable, retaining its elasticity over multiple flights and providing smoother surfaces for tight-fitting designs, though it comes at a slightly higher cost and weight. Overall, medium-density PE or XLPE foam strikes the best balance between strength, cost, and energy absorption for the eggs' protection [3].

3.2 Water Payload Assembly

This study shall be focused on the design of the vessel for the water payload. Some important factors considered for this study are weight critical design, leak prevention, and ease of access.

3.2.1 Structural Analysis

The first factor that must be addressed for this memo is the structure of the water vessel. A rigid material, such as plastic or metal, resists deformation under acceleration, and can provide a defined shape to be secured within the rocket body. However, a potential downside to a rigid container is the ability for water to slosh while the rocket is in motion [4].

3.2.1 Structural Analysis (continued)

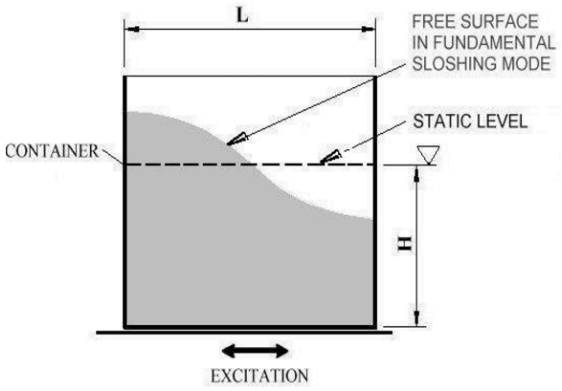


Fig 2. Schematic Presentation of Sloshing [4]

This could lead to shifts in the rocket's center of mass, resulting in instability of the rocket during ascent, which could have devastating effects. In order to minimize this risk, it is crucial that the container is filled maximally, such that there is little freedom for a sloshing effect. In contrast to a rigid container, is a flexible one, such as a plastic bladder, which conforms to the water volume. This approach can essentially negate the slosh effect completely, but would be paired with some challenges of its own. A flexible container would be more difficult to securely mount within the rocket body, likely requiring additional supportive structure. A soft-body vessel would also be much more susceptible to damage, which may result in leakage, compromising the entirety of the project. Due to the added risk involved in implementing a soft vessel, a hard plastic would likely be the best option.

For the design of this vessel, the lowest risk approach would be purchasing a consumer-off-the-shelf product (COTS). There are many COTS lab bottles which are designed to be leak-proof, utilizing threaded caps and O-ring seals [5]. However, care should be taken to ensure that the chosen bottle shall be fully compatible with the payload bay. The selection should also be of appropriate volume, such that it will not be underfilled, to reduce sloshing, but is also capable of containing a pre-agreed upon amount of water to maximize points scored.

3.2.2 Leak Prevention

As a leaking vessel shall result in no credit, it is of the utmost importance that the vessel is designed to mitigate all risk of leakage. Therefore, the water container shall use some form of proven leak-proof cap system. Many lab-grade bottles utilize compressive O-ring seals to remain leak-proof in harsher environments than what will be experienced during the challenge required for this project [6]. Care must be taken to ensure the seal is sufficiently tight, such that it will not fail due to intense vibration, which is likely to be experienced during launch.

3.2.3 Ease of Access

Another major requirement for the water payload is that it must be quickly accessible for judging. This necessitates a design that can be readily opened or removed. The simplest way to achieve this would be to treat the vessel as a removable module which would be seated in a

3.2.3 Ease of Access (continued)

dedicated payload bay, designed for easy access. The exact specifications of such a module will be highly dependent on many other factors correlated to the entire design of the rocket. In the event a removable module is not practical for the rest of the design, the vessel can be fitted with ports for loading and unloading.



Fig 3. DURAN® GL 45 Bottle Connection Cap [7]

3.2.4 Integration

The water payload must be integrated into the rocket's structure such that it remains secure throughout the entirety of the flight and recovery phases. In the case of this project, the mass of the water will be very significant. If the vessel is improperly mounted it could affect the stability of the rocket, or even directly cause damage to the frame or electronics. Therefore, the vessel shall need to fit tightly into a well supported payload assembly. This assembly will require bulkheads to protect the payload from the ejection charge [8]. The heat generated from chute ejection could cause damage to the water vessel, compromising the entirety of the project if the proper precautions are not in place.

3.3 Recovery & Propulsions

This study will focus on the background info for the propulsion and recovery systems of a model rocket that will assist us in meeting the project requirements for our rocket. The rocket motor must have ample propulsion in order to reach its target altitude, and also possess correctly sized parachutes in order to slow its descent to a reasonable velocity.

3.3.1 Motor Analysis

Most model rocket propulsion is typically achieved through the use of solid rocket motors which our team will be using as well. The motors most commonly used in model rocketry are produced by Estes, Aerotech, and Cesaroni. Aerotech and Cesaroni both offer reloadable motors, but those motors will not be taken into consideration as they will require us to buy both a motor tube and fuel reloads, and tend to be about 3 times more expensive than a single use motor.

Rocket motors are classified according to their total impulse, with low-power motors typically being A-D, mid-power motors being E-G, and high-power motors being H and above.

3.3.1 Motor Analysis (continued)

As determined by the project requirements, the impulse limit of the rocket engine we select is a class H. Due to the altitude requirements of the challenge requiring our rocket to reach a minimum apogee of 1000 feet while carrying a payload of eggs and water, we will most likely be using an H motor. Due to our rocket's need to carry a payload, we also need a stronger motor as increased weight from the payload will necessitate the need for higher lift provided by the motor, to keep our rocket's thrust to weight ratio as close to 5:1 as possible, as recommended by the Tripoli Rocketry Association. Additionally, using the most powerful motor possible will allow us to carry a larger payload of eggs and score more points. It is also important to note that an H motor requires an L1 certification for purchase.



Fig 4. Motor size comparison

Motors come in multiple sizes, with the motors in the E-H range coming in the following diameters: 24mm, 29mm, 38mm. H class motors are only produced in 29mm and 38mm variants. A larger diameter motor will require a larger diameter body tube, which will also allow us to fit a larger payload inside the rocket. Figure 4 shows five different motor sizes next to each other. It is important to note that as motor power increases, the body of the motor also increases in length. This means that a more powerful motor will result in less cargo space within the body of the rocket. A way to solve this issue would be to use a larger diameter motor, which will allow for a shorter rocket motor of the same power.

3.3.1 Motor Analysis (continued)

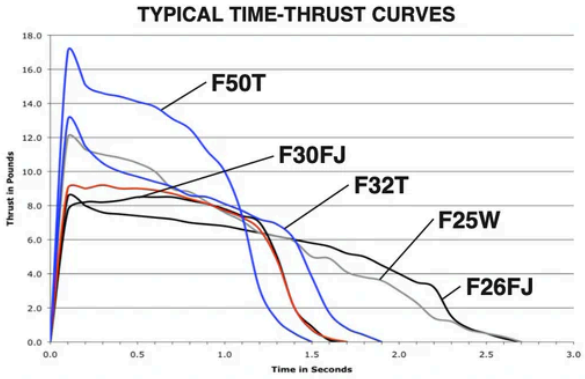


Fig 5. Motor Performance Data and Thrust curves

Another aspect of the motor that our team wants to look into is the burn time. A rocket motor with a short burn time will have more instantaneous thrust, which will lead to faster acceleration, but will also cause increased stresses on the payload and airframe which increases the chance for failure on launch. A rocket with a long burn time will accelerate slower and smoother, leading to reduced structural stress and a lower likelihood of airframe failure or payload damage. Figure 5 graphs the thrust versus time of each motor. In the graph we can see that the F50T uses the majority of its power in the first 0.2 seconds of flight. This thrust curve would be suboptimal for our rocket as it could possibly lead to damage to our payload. A thrust curve similar to the one of the F26FJ would be significantly more desirable as the acceleration over time would reduce the chances of damage to the payload. Another important thing to note about thrust time is that a rocket with a motor that burns quickly is significantly less susceptible to being affected by external forces such as wind. A rocket with a motor that burns slowly is also more likely to deviate from its estimated trajectory.

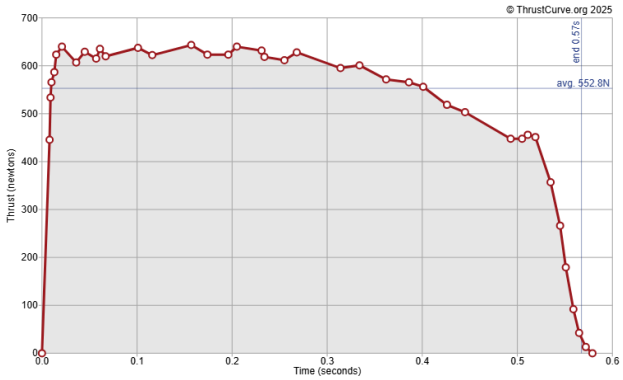


Fig 6. AeroTech H550ST-14A

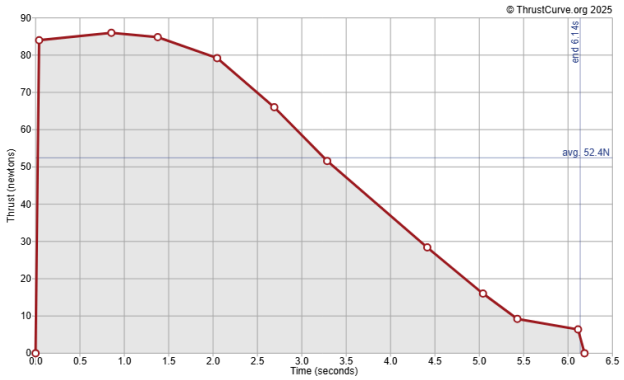


Fig 7. AeroTech H45W-PS

From all the data collected, we want to use a motor at the top of the H impulse class. This means that we want a motor with an impulse as close to 320 Newton-seconds as possible. Looking at only single use motors, that leaves the options being the AeroTech H550ST-14A, and AeroTech H45W-PS. Both are 38mm motors with impulses of 320 Ns, but the H550ST-14A has

3.3.1 Motor Analysis (continued)

a burn time of 0.6 seconds and the H45W-PS has a burn time of 6 seconds. The H45W-PS is also a plugged motor which would require a second ejection charge.

3.3.2 Recovery Analysis

Safe recovery of model rockets is very important, as a projectile shaped object falling out of the sky at a high speed is dangerous for the people below. To ensure safe recovery and a slow descent, most model rockets employ the use of a parachute to slow them down enough to mitigate damage to the rocket or nearby objects.

In our case, we need a parachute that can slow the rocket down enough so that when it impacts the ground, both the rocket and payload are undamaged. As defined in the project requirements, a decent rate exceeding 25 feet per second during the final 5 seconds of flight will not be scored, so our descent must be slower than that. However we also have a constraint that the rocket cannot travel more than 800 feet from the launch site. A large parachute, while ensuring the safest landing, may also result in the rocket drifting a greater distance from the launch site. Our goal is to find a balance between the slowest descent and the least possible amount of drift.

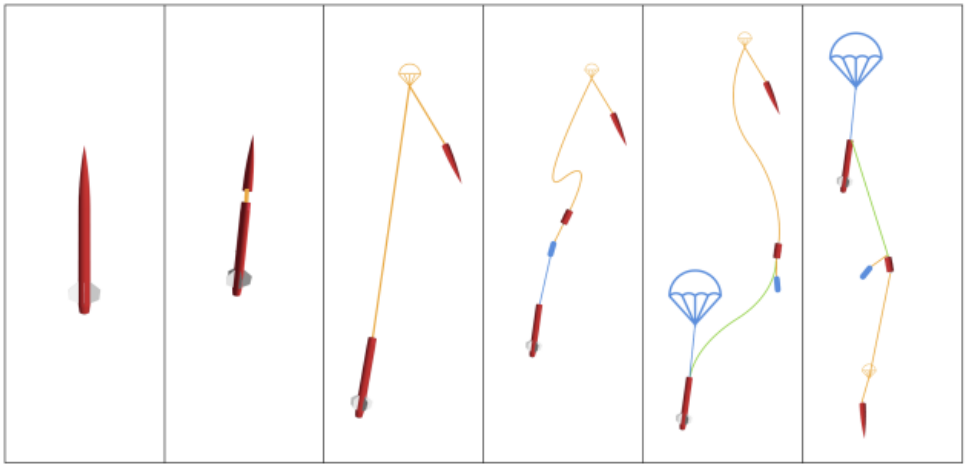


Fig 8. How the drogue and main parachutes are deployed in sequence

One of the best ways to create this balance is by having the parachute deploy at a lower altitude. The most common way this is done in model rocketry is with dual deployment. This method utilizes two separate parachutes, a drogue and a main, to slow the rocket's descent. A drogue parachute is a small parachute meant for deployment at high speeds, that can reduce the falling speed of the rocket, without letting it drift too far, but the drogue itself is not large enough to slow the rocket down to an appropriate and safe landing speed. Once the rocket reaches a low enough altitude, we can have it deploy the main parachute. This parachute will be significantly larger in order to slow the rocket down to a safe landing speed which will ideally lead to no damage to the rocket and payload. Late deployment of the main parachute will allow for the desired balance of slowing the rocket while not allowing it to drift beyond 800 feet. Figure 8 models the deployment process of both the drogue and the main parachutes.

3.3.2 Recovery Analysis (continued)

Another important consideration about the parachute is the need for a shock cord. The shock cord is what connects the parachute to the rocket, and it must be strong enough to absorb the force of slowing the rockets down without snapping. When the parachute is ejected from the rocket, the shock cord will rapidly unravel as the parachute expands and increases drag. Once the shock cord is fully unraveled and the parachute is fully deployed, the falling velocity of the rocket will instantly decrease, and this force will go directly into the shock cord.

Shock cords used in model rocketry are most commonly made out of nylon or kevlar. The pros of nylon are that it is cheap, flexible, and has a high elasticity. The flexibility allows for the shock cord to be easily packed into the rocket without taking up much space, and the elasticity allows the shock cord to stretch in order to absorb the force from parachute deployment. However, nylon may not be strong enough to absorb the impact of slowing down a rocket with a heavy payload such as ours and may snap upon parachute deployment. Kevlar on the other hand is also flexible, has a very high heat resistance, and it is extremely strong. The heat resistance is beneficial because it will be near the motor which will get very hot, and the strength of kevlar will prevent it from snapping when the parachutes get deployed. However, this strength comes at an increased cost, which will have to be factored into the budget. Additionally, due to the high strength of kevlar, it is not uncommon that the cord can cut into the rocket body when the cord absorbs the shock of parachute deployment. To mitigate any damage to the body of the rocket by the shock cord, we will utilize a small piece of TPU that encases a section of the shock cord and prevents it from cutting into the body tube.

Due to all these factors, our team may find the best success with a 48-60 inch main parachute and a 12-20 inch drogue parachute, connected to the rocket via a kevlar shock cord.

3.4 Aerostructures

This study focuses on the comparative analysis of materials used in the manufacturing of a high-powered rocket. Primarily focusing on the main body structure, fins, and the nose cone; the three general structure components of initial rocket design. The evaluation performed in the study considers flight performance factors in terms of each component's material selection, as well as factors of material strength, reliability, safety, and manufacturing. Each material will have different properties in how it reacts to each factor including, machining processes during manufacturing and the exposure of initial launch conditions during launch testing. Individual factors, once changed, will then have a lasting effect on flight performance which is critical to a successful and safe launch. A proper balance of all these parameters must be taken into account during rocket design to maximize performance and reliability.

3.4.1 Body Tube Material Comparison

There are a number of materials typically used when constructing and machining the main body tube. In the past, cardboard has mainly been used for certification launches for its cheaper cost during fabrication. Other material considerations would include fiberglass, a blue phenolic tube (high-strength paper tube), and carbon fiber.

Cardboard is relatively reliable when used for model rocketry as it is a cost effective and easily attainable option. This material does come at a disadvantage for its lack of reliability and

3.4.1 Body Tube Material Comparison (continued)

reusability in some cases. Compared to other materials, cardboard tends to be weaker and more susceptible to deformation under high-impact loads. This is crucial to correlate with what type of propulsion is required for the rocket design. Determining the desired thrust of the designed rocket, then reduces your options for material selection. Experiments conducted to find maximum thrust that cardboard can withhold show compressive strength tests of cardboard tubes until failure which provides maximum compressive strength. **Figure 9** shows three cardboard samples tested to measure maximum compressive strength.

Sample	**Length** [mm]	**Thickness** [mm]	**Area** [mm²]	**Compressive Strength** [MPa]	**Elastic Modulus** [MPa]
1	151.84	2.06	323.38	10.5441	1540.5075
2	151.12	1.91	300.26	10.6399	1683.6040
3	152.76	2.06	323.05	11.0509	1354.4987
Mean	151.9067	2.0066	315.5633	10.7449	1526.2034
Std. Dev	0.82203	0.08798	13.25410	0.269241	165.01827

Fig 9. Cardboard Samples & Respective Compressive Strength **[**9**]**

These compression force measurements allow for a stress/strain comparison to better understand the limit of thrust for cardboard. **Figure 10** details those comparisons and identifies maximum thrust permitted by various materials.

Material	σ_{max} **[MPa]**	A_c **[mm²]**	**Max Thrust [N]**
Cardboard	11.0509	300.26	3319.28
Blue Phenolic	29.5440	516.66	15269.43
Fiberglass	175.1379	513.81	90018.35

Fig 10. Material Samples & Calculated Maximum Thrust **[**10**]**

Although cardboard is the most likely candidate when it comes to rocket body tube design, more advanced materials can provide greater reliability and strength. A material referred to as “blue tube” is one example of this. Blue tube balances cost and performance of materials when performing model rocketry because it tends to be cheaper than fiberglass, have similar traits of cardboard, but also have a greater strength than cardboard. Rocket component purchasing companies such as Apogee Components often refer to blue tubes as a close alternative to phenolic or fiberglass tubes because the strength of the materials is nearly similar, but when it comes down to manufacturing, handling and fitting blue tube to the overall design is cheaper and more convenient **[**11**]**. Using similar experimental procedures, blue tube’s maximum withstanding thrust is detailed in **Figure 10**.

3.4.1 Body Tube Material Comparison (continued)

A more costly material which is known for its resistance, strength, and light weight would be carbon fiber. Although this is a great option for rocket design due to its high durability, it can also be quite difficult to handle during the manufacturing process. With high material cost, comes high machining cost because carbon fiber is so abrasive. Typical machining involved in rocket manufacturing could include milling and sawing, whose tools would be deteriorated by the carbon fiber, rising machining cost and time **[**12**]**. Something else to consider in this extensive process is the risk of reducing integrity and possibly deforming the carbon fiber by utilizing extensive machining. Although this material is an all around ideal option for performance, cost and manufacturing time are critical aspects to developing a new rocket.

Finding alternatives that can supply rocket design teams with a durable and effective material without costing rocketry teams valuable time and money is an essential part in the engineering process.

3.4.2 Fin Material Comparison

Rocket fins are crucial components to a rocket design's aerodynamics. The overall shape and number of fins can drastically change flight performance by altering altitude, speed, and stability. However, the question of material must be answered prior to defining shape.

Smaller model rockets often use simple Styrofoam fins and even nose cones, but high-powered rockets require material that is more durable and reliable to handle increased thrust. It is also important that the strength of the material is high because the return impact of a rocket could be detrimental to the fin's integrity. Typical fin material for high-powered rockets are plastic or wood. Plywood specifically has a strong history of being a low-cost with a good strength-to-weight ratio for their use. Good quality plywood is important so there is a reduced risk of cracking or splitting during manufacturing or launch. Plastic also being a good option can give more leniency towards design, as some materials can be 3D printed from an engineered design. Both materials have their setbacks however. With an increased thrust with a high-powered rocket motor, there is an increased chance of flutter in both plastic and wooden fins. The flutter is when the fins oscillate due to vibrations that may cause the material to fail. Because they have a lower elastic modulus and lower strength than say that of an aluminum fin **[**13**]**. This flutter then creates a higher overall drag experienced by the rocket, hindering flight performance. Refer to **Figure 11** for strength measurements of materials including plywood and 3D filament (plastic).

3.4.2 Fin Material Comparison (continued)

Material	Type	TYS (psi)	UTS (psi)	UFS (psi)	E (ksi)	rho lb/cu.in.
Plywood	birch		6300	10750	1400	0.025
Garolite	G10		41000	75000	2700	0.065
	G11		43000	80000	3000	0.065
3D filament	PLA	6800	7300	12000	580	0.047
Aluminum alloy	1100	5000	13000		10300	0.098
	3003-H14		16000		10100	0.099
	5052-H22	22000	31000		10100	0.097
	5052-H34	25000	34000	48000	10100	0.097
	6061-T6	38000	43000	61000	9900	0.098
	2024-T3	48000	65000	90000	10500	0.101
	7075-T6	72000	80000	122000	10300	0.101

TYS = Tensile Yield Strength
UTS = Ultimate Tensile Strength
UFS = Ultimate Flexural Strength
E = Elastic Modulus
rho = mass density

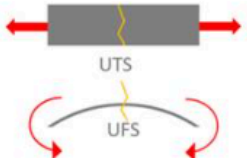


Fig 11. Material Samples & Strength Measurements [13]

Another material option for fins could be fiberglass. Fiberglass allows for slightly more flexibility when it comes to designing fins due to its chemical and physical properties, permitting strong material properties [14]. Although the shear modulus for fiberglass is similar to that of wood and plastic, it has the potential to have a higher strength due to its chemical properties creating a strong and tight bond. The downside of using fiberglass in manufacturing comes with the application process. The process of constructing materials using fiberglass is quite extensive and daunting as it takes long periods of time for it to cure and ensure a solid bond. It is also important to re-shape the fiberglass fins to be smooth for proper airflow because cured fiberglass can be rough and may be slightly deformed from the designed shape.

The most ideal and cost effective material choice for high-powered rocket fins would likely lean towards plywood or plastic as they are easier to manufacture and control the overall design that is critical to the aerodynamics of the rocket.

3.4.3 Nose Cone Material Comparison

The nose cone in rocket design is something that greatly affects flight performance during a launch. It is known that the nose cone or fairing is the leading component of every rocket design, implying that it significantly influences the airflow around the rocket and how the rocket will perform the remainder of the flight.

For high-powered rocket models, a wide variety of material types would suffice for safe flight. Carbon fiber being one of those options has been denoted as something that would increase tooling costs and labor times. Fiberglass is another strong material selection; however, the demanding manufacturing and perfection process can be longer than a completely usable material that has slightly less strength. This consequently leads to two of the best choices in nose cone material selection including, plastic and wood.

3.4.3 Nose Cone Material Comparison (continued)

The shape of the nose cone itself is the primary concern for this design process based on target speed, location of center of gravity, drag, and target altitude. **Figure 12** represents multiple nose cone designs applicable to high-powered rockets.

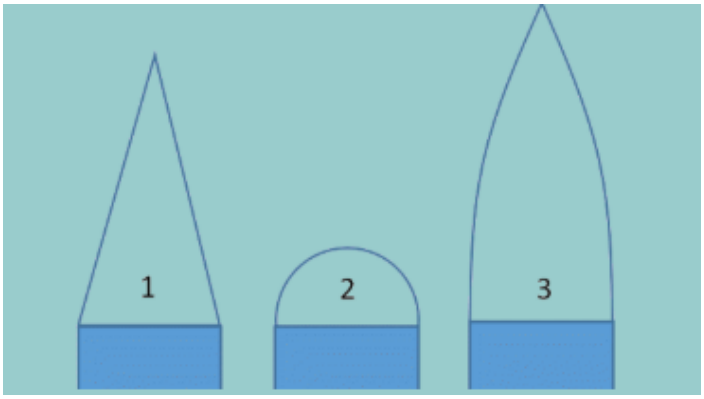


Fig 12. Nose Cone Designs **[**15**]**

With the material selection being primarily between plastic and wood in this case, the main concern is manufacturability. In **Figure 11** it is made clear that both of these materials have similar tensile strengths with plastic being almost 1000 psi stronger. When manufacturing a nose cone, the most precise and easily manipulatable option would be to 3D print using 3D filament. Since plastic tends to be lighter and more durable, there is less of a chance of cracking or deformation during flight or rocket assembly. Manipulating plywood in a way that bends to the shape of the designed airfoil causes residual stress and strain which could lead to damages in the future.

Nose cone design comes down to the design of the shape itself and using a 3D filament and a printer to precisely match dimensions called out in a CAD drawing allows for a more structurally sound and reliable component. Wood creates unnecessary risks to nose cone manufacturing that could be avoided.

3.5 Avionics

This section focuses on providing relevant background information on the avionics portion of the rocket vehicle. This includes responsibilities such as data acquisition, flight event control, recovery devices, and camera equipment.

3.5.1 Data Acquisition

The requirements for the project stipulate that the rocket “shall carry data logging equipment capable of recording altitude over time and descent velocity over time” for the duration of the flight. In addition, the data logging equipment must be accompanied by a backup “redundant data logging system capable of recording apogee” that can serve to validate the target apogee of 999-1500 feet.

3.5.1 Data Acquisition (continued)

The most suitable device for this task is called an altimeter. These are devices designed to record the dynamic profile of the flight including acceleration, velocity and altitude. This is achieved by incorporating various sensors that work in conjunction to collect data and calculate various flight characteristics. Typical components of an altimeter are a microcontroller unit, inertial measurement unit, accelerometer, gyroscope, and barometer.

The microcontroller unit (MCU) is the brain of the altimeter. It is responsible for data computation and interpretation. It is central to the altimeter's operations, interfacing with all sensors and input/output capabilities onboard. For example, the MCU can take the signals provided by the accelerometer and gyroscope and convert them into the vehicle's current orientation, acceleration and speed while in flight. As this data is collected by the MCU, it can format and store that crucial flight data onto onboard flash storage for later offloading and usage after successful recovery.

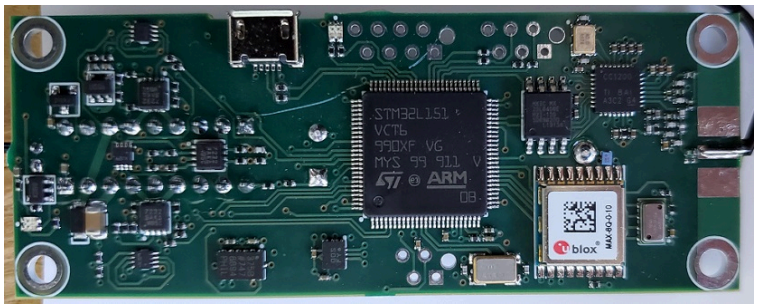


Fig 13. The Altus TeleMega Altimeter, utilizing the STM32L151 MCU. [16]

The accelerometer is a device that detects acceleration on multiple different axes. This is done by using a capacitive microelectromechanical system to measure differences in capacitance in the sensor. The capacitor contains a diaphragm that flexes when undergoing acceleration, changing the electrical potential of the capacitor, as depicted conceptually below. [17] This is then interpreted as a signal and sent to the MCU to formally calculate the acceleration of the vehicle along a particular axis.

3.5.1 Data Acquisition (continued)

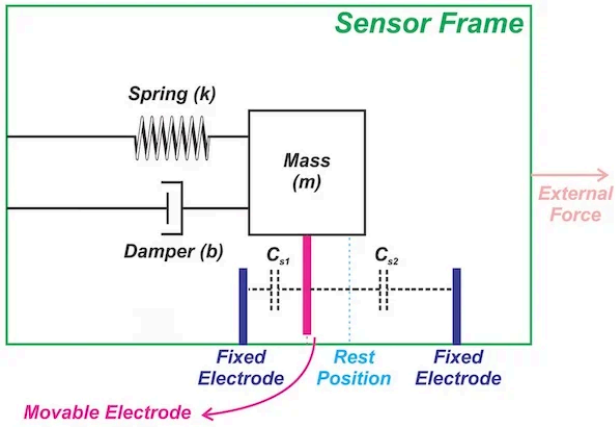


Fig 14: The mechanical process depicting the function of a capacitor sensor frame.

The gyroscope is a sensor used for detecting changes in orientation of the vehicle. This is performed by detecting vibration via the coriolis effect on sensing arms attached to a stator (as depicted in Fig. 3) which can then be converted into an electrical signal. This signal can then be converted into rotational data by the MCU, giving the altimeter system accurate data about how the rocket is oriented in flight. The combination of both this system and the accelerometer creates what is known as a 6-axis inertial measurement unit, or IMU. [16]

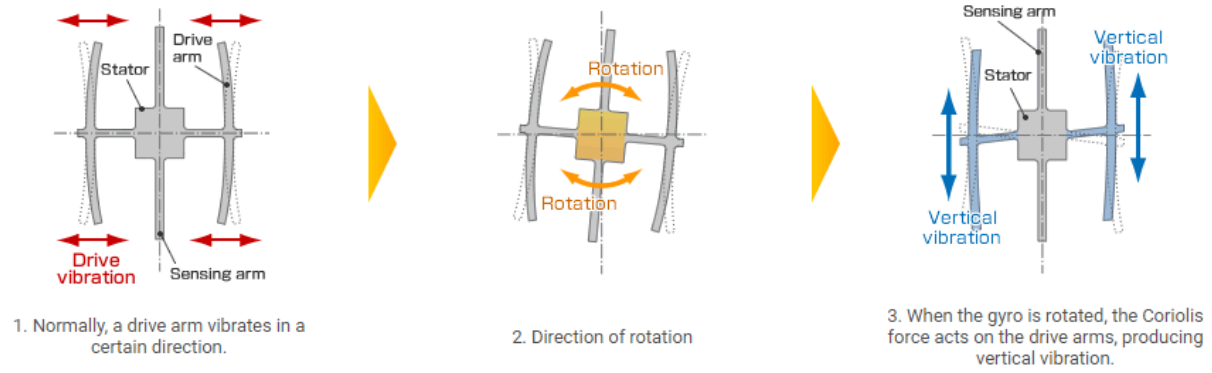


Fig 15: The mechanical process depicting the Coriolis effect at work in a gyroscopic setup.

Finally, the barometer is the last major standard sensor on an altimeter. The barometer is an electrical component that converts differences in air pressure into electrical signals. As a rocket increases in altitude during flight, the static air pressure decreases. [17] This decrease in pressure is then scaled and interpreted by the MCU to determine what altitude the rocket has achieved. This is based on the reference point the altimeter was calibrated to on the ground.

With all of these flight characteristics collected, the flight can be characterized throughout the duration of the entire flight. More significantly, this data can be used to initiate flight events like those associated with dual-deployment.

3.5.2 Dual-Deployment

When a rocket is launched to high altitudes, the predictability of its descent becomes very skewed. Factors such as wind and launch angle can make the time in the air and reasonable landing zone for the vehicle wildly varied. One of the most essential requirements of the challenge is the 800 foot radius landing zone for the rocket. This creates a unique design challenge: how do you guarantee that even in the event of wind, the rocket will not drift far out of the landing zone upon descent? The most reasonable answer is the employment of a dual-deployment recovery system.

To characterize the rocket's flight, it can be broken down into 3 main phases: ascent, apogee, descent. The ascent starts at the ignition of the rocket, and has the initial acceleration caused by the burn followed by the deceleration of the vehicle in the coasting phase. As the rocket slows down due to the force of gravity, it will reach the peak of its flight, known as the apogee. This is the most opportune time for the deployment of recovery equipment like parachutes, as forces in the z-axis of movement have equalized for a brief moment. A timed ejection charge separates the rocket at this point, allowing the vehicle to separate and its chutes to be set free from its internal compartment. The parachute allows the terminal velocity of the vehicle to be reduced down to safe speed, which in this case is required to be below 25 meters per second. At this speed, it can slowly land on the ground without risking damage. [18]

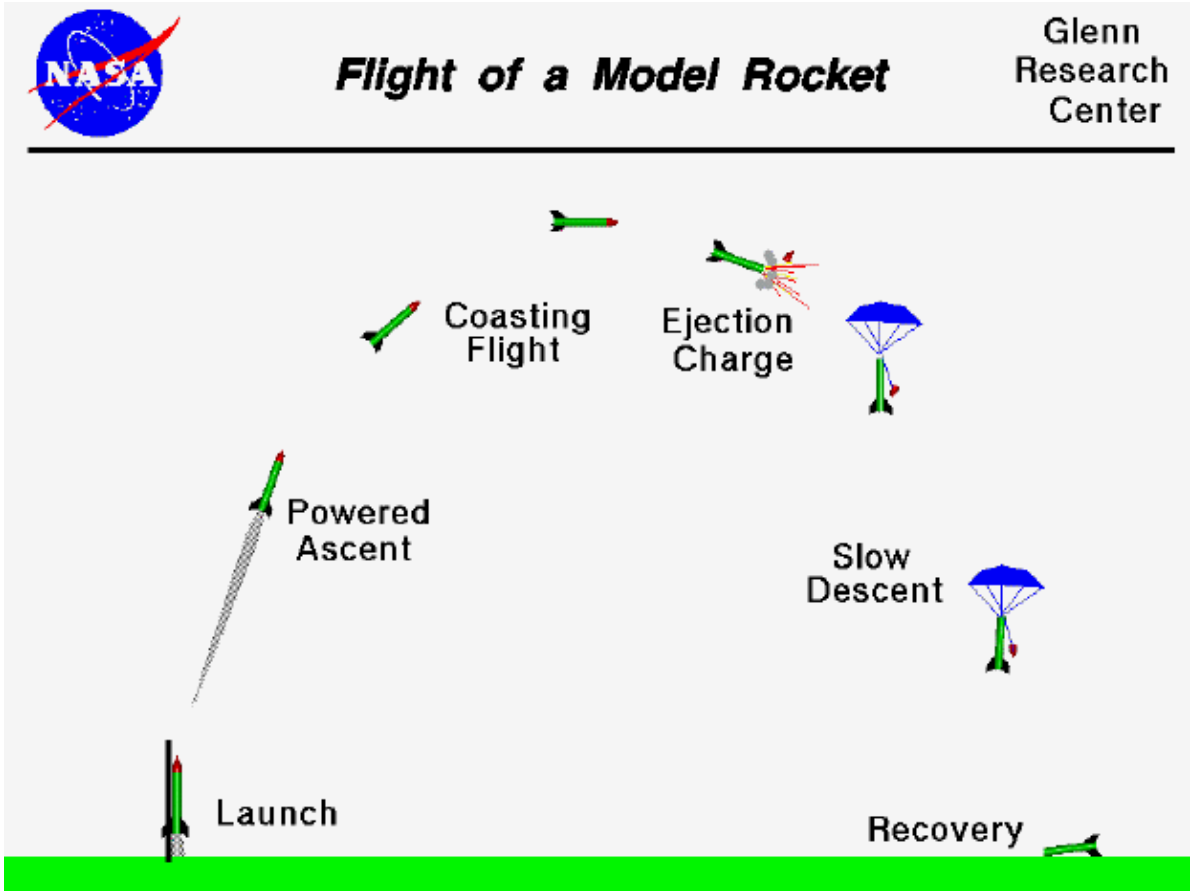


Fig 16. The basics of a rocket flight

3.5.2 Dual-Deployment (continued)

The main problem lies in the parachute deployed at apogee. A chute large enough to slow the vehicle to below the safe ground-hit speed threshold will also be large enough to dramatically increase the drag of aerodynamic forces on the vehicle. This means that the rocket stays in the air longer during descent, and is also affected significantly by the push of wind. This means that even at average wind speeds, the rocket can be carried vast distances from the intended landing zone until finally hitting the ground.

The dual-deployment approach remedies the increased drag by deploying a much smaller "drogue" parachute at apogee. This chute must be big enough to break the initial acceleration of descent, but small enough for the rocket to fall more directly to the ground without much wind drag. The terminal velocity with only the drogue deployed would be too fast to guarantee a safe ground-hit speed, so a secondary "main" parachute must be deployed separately at a specified altitude to lower the velocity of the vehicle into the acceptable range. Past the deployment of the main parachute, the rocket may descend to the ground as normal. This results in a much more direct descent, as depicted below. [18]

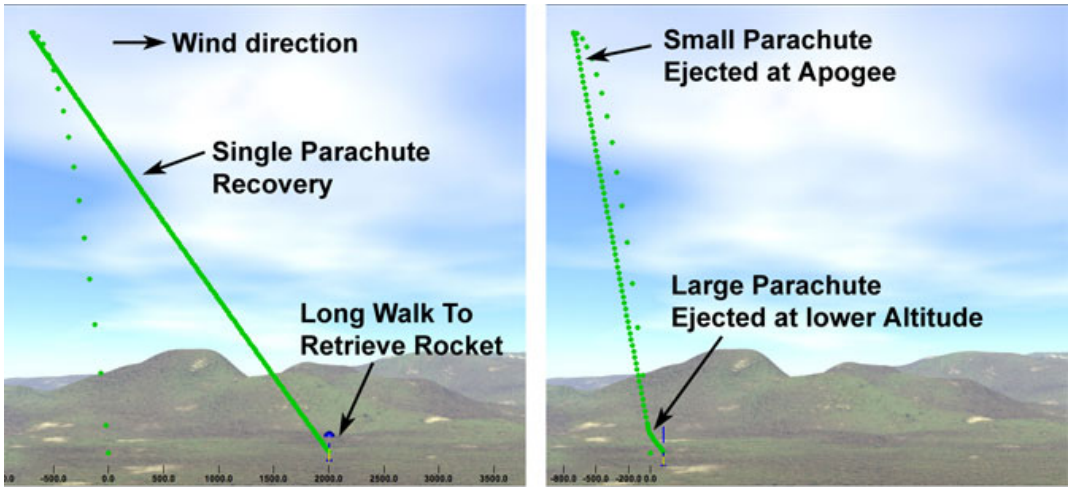


Fig 17. A comparison of a singular deployment flight (left) and a dual-deployment flight (right)

3.5.3 Recovery Devices

While recovery should be much easier to achieve with a dual-deployment system, there is still not a guarantee for an easy recovery. Terrain, wind drag, or cloud cover may obstruct an easy recovery. However, there are devices that can be added to improve the search for the vehicle post-flight, including auditory or positioning system aides.

Auditory devices include electronics such as "beeper" beacons. These are simple modules designed to make a loud "beeping" sound during descent and after the vehicle hits the ground. [16] Depending on how loud the device is, it can increase the range that the rocket can be located from by adding an auditory cue for the recovery team. These are especially effective when used in launch areas with tall grass that can obscure the vehicle visually.

The limitation of such a system is the battery life and decibel rating of the device used. As the vehicle can sit on the pad for an extended time.

3.5.4 Live Video System

A live video system is the largest potential avionics hurdle of the project. While optional, there is a high point value assigned to the creation of an onboard camera system that is capable of playback of flight footage or the livestreaming of that footage to the ground station.

The integration of a camera would require the creation of a slot to give view of the outside of the vehicle, and the usage of a camera small enough to not disturb the integrity of the airframe. Similarly to the data logging system, there would need to be the inclusion of an onboard SD card digital storage to store that footage for recovery after flight.

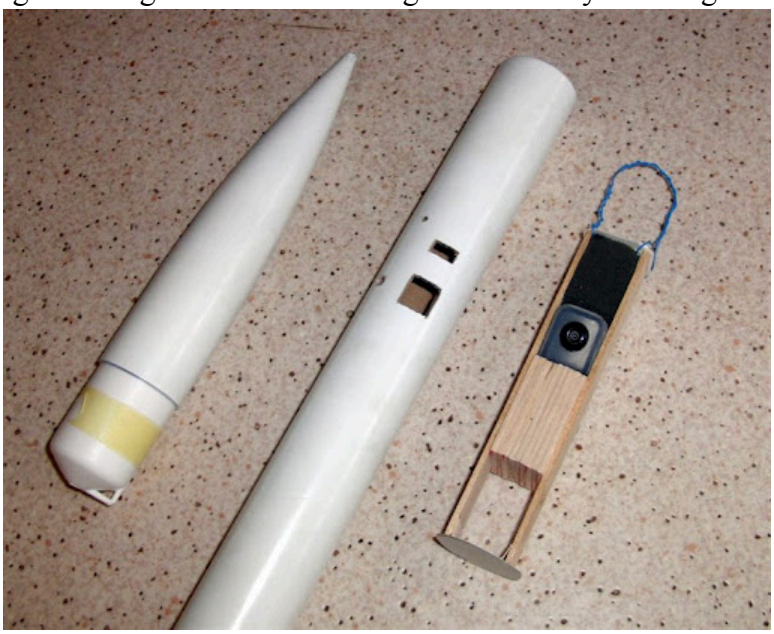


Fig 18. Rocket with side cutout for integrated camera system in electronics bay

The major challenge for this section of the project is the livestreaming component. To stream video, a similar RF system of communication between the vehicle and the ground station would need to be established. As video is a much more dense data format to transmit, more unorthodox RF frequencies must be used to retain data quality. This would require a HAM Radio License to keep within legal regulation. On top of that, further ground equipment and antennas are required. On top of complexity, the cost for a livestreaming system would increase dramatically.

4.0 System Requirements

Identifying and addressing system requirements early in the design process is important for understanding the parameters that the final design must satisfy. The key requirements for each subsystem of the project vehicle are detailed in the following sections.

4.1 Egg Payload System Requirements

The main purpose of the CRASH competition is to safely transport fragile payloads with an L1 class rocket. Eggs have a history of needing protection during transportation, although through more conventional means like trucks and planes. There are many methods of egg protection in the egg production industry, straw in a crate, filler trays made of wood pulp or plastic, or smaller cases made of polystyrene. All with the goal of dampening vibrations and impact forces. Even with these protections practiced in industry, the standard is still quantity over quality. It is understood that not all eggs will survive transportation, but the cost of transporting every egg perfectly is greater than simply packing more eggs and eating the cost of those that don't survive the journey. A lesson that may or may not be applied to our payload.

Benchmarks from the egg transport industry give us these requirements:

- The protection system shall reduce vibration
- The protection system shall reduce impact forces

A key function of the Egg Capsule is to protect the eggs inside by dampening vibration and, more importantly, to dampen impact forces. The capsule will need to protect the eggs from the forces generated by the launch and landing of the rocket. Another key function of the capsule is to be opened, loaded, then closed quickly. The competition requires the team to be able to load the eggs in under 5 minutes, and unload under 5 minutes. So the capsule must have a simple and fast loading and unloading mechanism. Loading the eggs quickly into the capsul, and also loading the capsule quickly into the rocket. The capsule's final key function is to carry a minimum of 2 eggs; more eggs give bonus points, but 2 must be carried at a minimum.

Knowing the key functions and the information gathered from the technology assessment about the strengths and limitations of eggs gives us the following key system requirements:

1. The egg protection system shall reduce vibration forces to a maximum of 7.5 m/s^2 .
2. The egg protection system shall reduce impact forces to a maximum of 200N.
3. The egg protection system shall allow for eggs to be loaded and unloaded from the rocket within 5 minutes.
4. The egg protection system shall accommodate a minimum of two 2 eggs.

In conclusion, the egg capsule will allow eggs to be loaded and the capsule inserted into the rocket within five minutes. During flight, the system protects the eggs from launch forces, in-flight vibrations, and ground impact. After landing, the capsule can be quickly extracted and fully unloaded within five minutes.

4.2 Water Payload System Requirements

The CRASH competition has set standards regarding the water payload which must be followed to be deemed as a valid entry in the competition. The combination of these standards as well as requirements set by Team 2 have been joined to create a list of requirements for the water payload assembly for the project vehicle.

1. The water mass carried by the assembly shall be a minimum of 5 fl oz.
2. The total weight of the water payload assembly shall be no more than 15 oz.
3. Water carrying vessels shall be able to be loaded or unloaded from the vehicle in 5 minutes or less.
4. Water carrying vessels shall be free of cracks, dents, and warping.
5. Water carrying vessels shall be carefully sealed to ensure no leaks are present during flight.
6. The entire assembly shall be capable of withstanding the maximum forces predicted to be experienced during flight.
7. The design shall ensure water carrying vessels cannot become projectiles in the event of structural failure.
8. The water payload assembly shall not interfere with the operation of the other vehicle systems.

This set of requirements ensures the water payload system is capable of successfully fulfilling its role for the competition and is also meeting the standards set by the team.

4.3 Propulsion Requirements

Since our project is to build a rocket, the most important part of the project is that our rocket actually flies and demonstrates the functionalities of a rocket. In order for our rocket to lift off from the launch site and then propel itself to an altitude of at least 1000 feet given by the project requirements, it will require a solid fuel motor. The project requirements given for the propulsion system are as follows:

1. The motor shall be fully intact and able to be removed and replaced.
2. Airframe and motor mount shall be free of extensive scorching or burning.
3. Motor shall be securely fastened with the motor mount pre & post flight.
4. Motor shall not be in an impulse class higher than H.

4.4 Recovery Requirements

An integral part of this project as well as one of the main foci, is our rocket's ability to safely return to the ground with minimal damage to the airframe and payload, such that the rocket remains in flyable condition after landing. This will be done using a parachute connected to the rocket body via a shock cord. The project requirements given for the recovery system are as follows:

1. Target descent velocity shall be less than 25 ft/s.
2. Target landing position of the rocket shall be 800 ft or less from the initial launch site.
3. Parachutes shall be fully intact and untangled, free of any holes, shredding, and any damage to the suspension lines.
4. Shock cord shall be fully intact; free of any fraying, extensive burns, and any loose knots/ties.
5. Deployment of recovery systems shall result in no damage to the airframe such as zippering on the body tube.

4.5 Aerostructure System Requirements

While transporting fragile payloads, it is crucial to provide a safe and reliable vehicle that carries out customer requirements. Reliability and safety comes down to material selection and rocket stability during flight. Throughout research, strong and cost effective material was identified and applied to requirements that meet customer needs. Primary goals include proper stability and altitude which are heavily influenced by weight and rocket design. Rocket design comes down to a couple of primary components including the body tube, nose cone, and fins which are all outward facing components during flight. The material chosen for these components ties in reliability because of the strength required to withstand launch and impact. Compromising material could lead to the inability to meet the requirement of building a rocket that is able to be recovered in flightworthy condition. Understanding the key functions, the following requirements have been set for this launch.

1. The target nose cone shall maintain an elliptical shape.
2. Nose cone shall be fully intact pre-flight, free of any material loss, denting, and cracking.
3. Connection to the body tube shall be fixed and rooted under pressure.
4. Target body tube diameter shall accommodate the size of the payload protection system.
5. Body tube thickness shall be ≥ 0.050 inch.
6. The body tube shall have at least 2 rail buttons securely fastened with a ¼ inch diameter.
7. The body tube shall be fully intact, free of any cracks, bubbling, or crushing.
8. The fin quantity shall be greater than 2.
9. Fins shall be fully intact pre & post flight with a strong connection to the body tube and experience no bending, warping, extensive crack propagation, or major chips that could affect the launch vehicle's stability. Small chips, scratches, or dents along fin's edge are permissible as long as the overall integrity is not at risk.

4.5 Aerostructure System Requirements (continued)

10. Bulkheads & motor mounts shall be fixed and intact; free of any cracking, denting, or any other types of deformation.
11. Center of Gravity (CG) shall be at least one body diameter in front of its Center of Pressure (CP).
12. Launch vehicle stability shall fall between the range of 1.5 - 2.0. Anything outside of this range may lead to failure.
13. Entirety of model shall be compliant with the FAA Certificate of Waiver or Authorization (COA).
14. Aerostructure components shall be made up of materials that withstand the force caused by the propulsion system.
15. Aerostructure components shall withstand ignition conditions.

Overall, aerostructure is the sole proponent of a lightweight, reliable, and fully integrated force bearing system. The design shall maintain proper structural integrity under aerodynamic loads, vibration, ignition temperatures, and variational pressure. The design must stay in flightworthy condition post flight upon inspection by comparing flight checklists and requirement parameters.

4.6 Avionics Requirements

Avionics performs tasks ranging from flight event control to data logging for analysis. In order to perform these tasks it requires many small, sensitive and intricate devices working in conjunction in a high-stress environment. In order to perform such complex responsibilities, the avionics system must employ batteries to power all electrical components, flight controllers to initiate flight events and basic data acquisition, ejection charges to initiate separation, and system redundancy for flight-critical devices. In addition, further data logging devices like cameras and recovery devices like beepers can be added to increased functionality.

With a wide range of devices and recovery techniques that can be employed in rocket avionics, the selection of components for the operational configuration of the rocket was balanced based on the necessity, affordability, feasibility, and point potential. All those considered, the most optimal configuration would be a flight controller and camera setup, along with redundant ejection charges and auditory recovery devices. The flight data would also be backed up by a self-contained altimeter in the event of a data logging failure from the primary flight computer.

1. The avionics bay shall weigh no more than 20 oz when fully configured.
2. The avionics system shall record flight data at a minimum frequency of 10 Hz for post-flight analysis.
3. The avionics system shall be capable of collecting data for a maximum of 5 minutes of flight.
4. The avionics system shall operate on a nominal voltage of 9 Volts.
5. The avionics system shall not exceed an amperage load of 500 mA at any point during operation.
6. The avionics system flight data shall be able to be extracted directly from the flight controller.
7. The avionics system shall include a backup recording device for flight characteristics.

4.6 Avionics Requirements (continued)

8. The avionics bay shall integrate vent ports to accommodate onboard barometers.
9. The avionics system shall employ a dual-deployment recovery setup.
10. The avionics system shall initiate the deployment of a recovery device at a minimum of 500 feet of altitude of flight descent.
11. The avionics system shall allow for the use of e-match pyrotechnic devices for recovery equipment initiation.
12. The avionics system should incorporate an onboard camera facing outward from the vehicle.
13. The avionics system should capture and store camera footage onboard the vehicle.
14. The avionics system onboard footage should be able to be extracted via microSD.
15. The avionics system shall be armed externally while the vehicle is on the launch pad.
16. The avionics system shall employ the use of an arming switch accessible through vent ports located on the side of the avionics bay.
17. Onboard electronics shall be properly connected, free of any burns or damages leading to the electronics being unreadable/unusable before or after flight.

In summary, the avionics system is set up to provide reliable control over flight events and data logging through redundant design. This design is constructed to perform across the scope of the projected operation time of the vehicle, as reflected in the avionics requirements.

5.0 Concept Generation and Selection

5.1 Egg Payload Capsule

When brainstorming for ways to design the capsule for the eggs, the team approached it by focusing on one need at a time that would restrict the design. Starting with capsule placement in the rocket for the balance of the rocket, the need to load and unload the eggs within 5 minutes, how the eggs will be cushioned, and egg quantity and orientation.

First, the placement of the capsule was decided to be in the upper part of the rocket. This gives space for the parachutes to be near the coupler and to better balance the weight of the rocket. The idea is to place it close to the nosecone.

5.1 Egg Payload Capsule (continued)

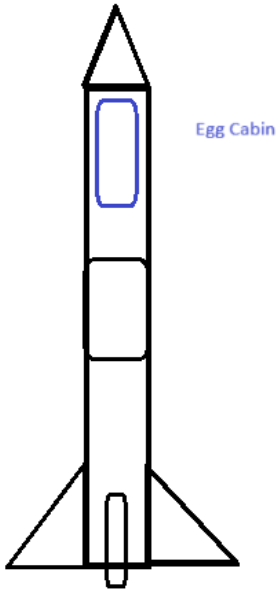


Fig 19. Depicting the placement of the Egg capsule in the vehicle

Then, brainstorming began on how the capsule can be loaded and unloaded. One idea was a door for the payload, and another was to make it slide in and out of the capsule body. The ability to slide the payload into the capsule makes loading and unloading much easier. Especially after further brainstorming on the protection system. The design of opening the capsule with the end cap is simpler to load, simpler to manufacture, and less likely to get stuck in the rocket body tube.

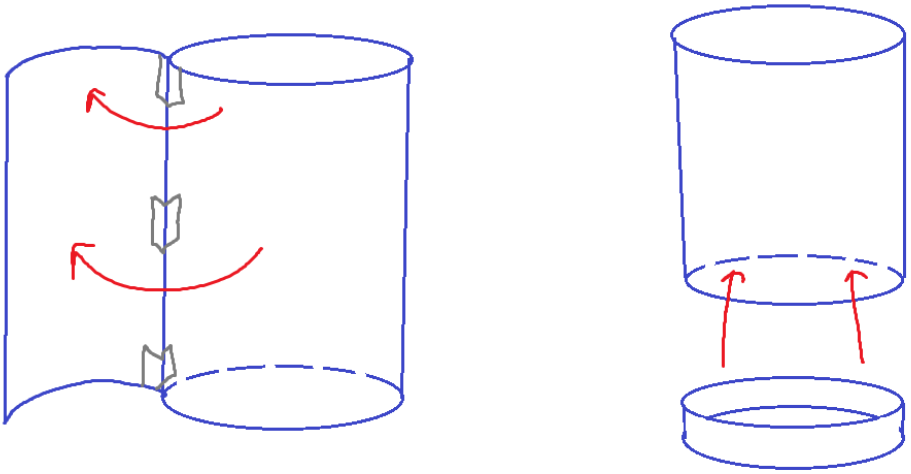


Fig 20. Depicts on the left an opening door concept for the capsule and on the right a lid concept for the capsule

5.1 Egg Payload Capsule (continued)

Table 1. Decision matrix for egg capsule opening mechanic

Decision matrix	Fast Loading Time (1-5)	Manufacturing Ease (1-5)	Lower Chances of Failures (1-5)	Total
End Cap Opening	5	5	4	14
Door Opening	3	2	2	7

After further development of the end-cap concept, the end cap will be secured to the capsule sleeve using a set of evenly spaced notches attached to the sleeve wall with epoxy. These notches provide engagement points where the end cap can be fastened without penetrating the internal foam or compromising the egg compartment. Because the capsule interior is primarily foam and eggs, it cannot be secured like a standard bulkhead. Instead, screws will pass through the sleeve at each notch and be tightened with nuts on the exterior, allowing the end cap to be removed and reinstalled quickly as needed.

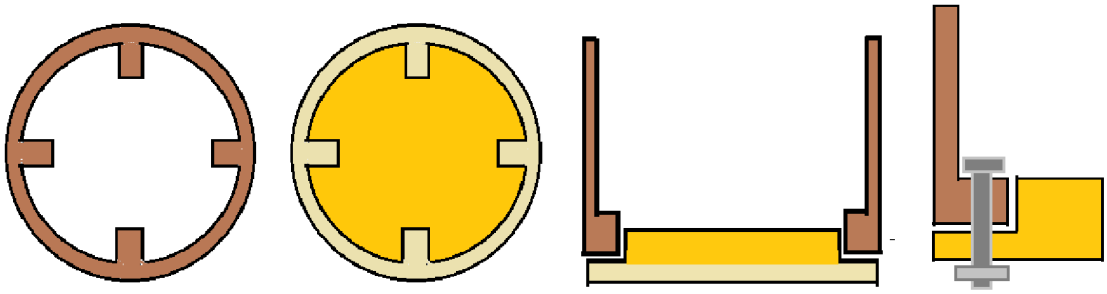


Fig 21. Left depicts the internals of the sleeve, center left depicts the wood endcap, center right depicts how the sleeve and endcap would sit, and the right image depicts how the screws and nuts are attached.

The most varied brainstorm was the method that the eggs will be cushioned, ranging from simply having only foam to an elaborate connection of foams and springs. Foam-only variation has only foam to cushion the eggs, with the eggs stacked in pucks to easily load and unload. This is the easiest to manufacture and to work with. A variation of this idea includes adding a foam puck to the top and bottom of the cylinder that would act as a sacrificial crush zone to absorb even more impact. Reducing force transfer to the eggs, but also lowering the amount of egg space in the capsule. For both versions, the easiest form of loading the eggs would be to cut the cylinder foam into pucks with egg-shaped divots that would contain half the egg on the top side and half the egg on the bottom side. This also allows the top and bottom puck to be easily replaced by a crumple zone puck if more cushioning becomes prioritized. The pucks would be stacked onto the endcap with a rod in the center. Using pucks to make the capsule also allows the team to change the number of eggs in the payload, adding more pucks to lengthen the capsule or decreasing the pucks to save weight. Making this a modular design.

5.1 Egg Payload Capsule (continued)

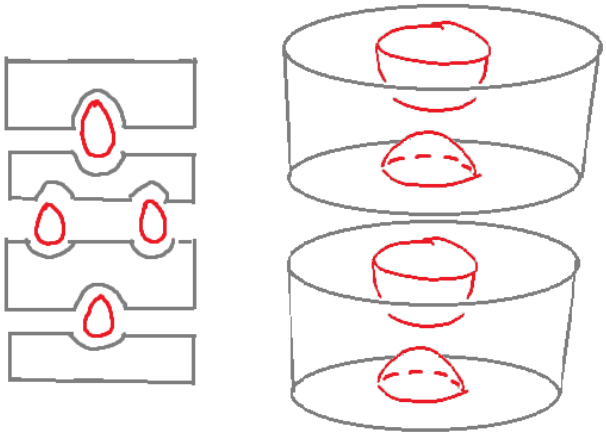


Fig 22. Depicts the concept of the puck stacking, giving space on the top and bottom of each puck for an egg. Alternating egg stacking for maximizing foam spacing between eggs.

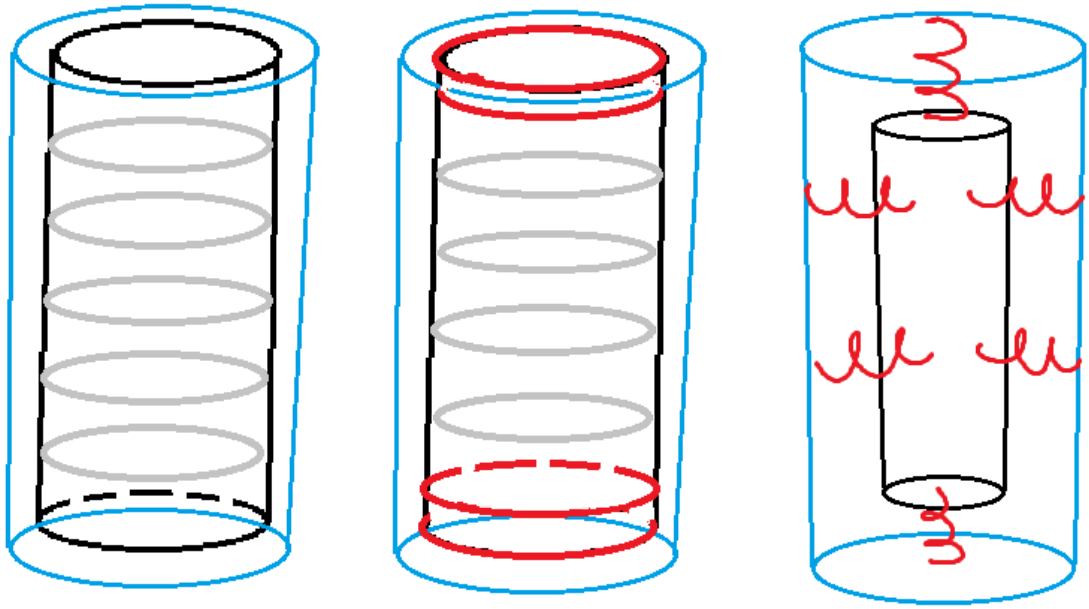


Fig 23. Depicts 3 options for capsule configuration. Left depicts foam only, middle has foam and a crumple zone, and the right is a suspension concept

The suspension variation includes adding springs to all parts of the wall, top, and bottom of the capsule. This will take up significant space in the rocket, would be much more difficult to manufacture, and cost more. Suspension is also not guaranteed to prevent the main foam body protecting the eggs from striking the walls, especially during ground impact. How the eggs will be placed in a suspended foam enclosure was also not yet thought through. Between the

5.1 Egg Payload Capsule (continued)

variation, foam capsule, foam with crumple zone capsule, and suspension capsule. The suspension capsule has been ruled out due to its cost and complication.

Table 2. Decision matrix for egg capsule impact reduction

Decision matrix	Fast Loading Time (1-5)	Manufacturing Ease (1-5)	Lower Chances of Failures (1-5)	Low Cost (1-5)	total
Foam Only	5	5	3	5	18
Suspension	2	2	1	2	7

The orientation and quantity of eggs are constrained by the available space within the rocket's body tube and the amount of foam required for adequate protection. Maximizing the number of eggs would necessitate stacking them vertically, which unfortunately corresponds to their weakest orientation under launch loads. A horizontal arrangement, while accommodating fewer eggs, provides additional cushioning space and offers greater structural resilience during launch. The final orientation will be determined through preliminary engineering analysis, which will assess the necessary spacing between eggs based on foam properties, capsule mass limitations, and whether the team chooses to prioritize egg quantity or overall protection.

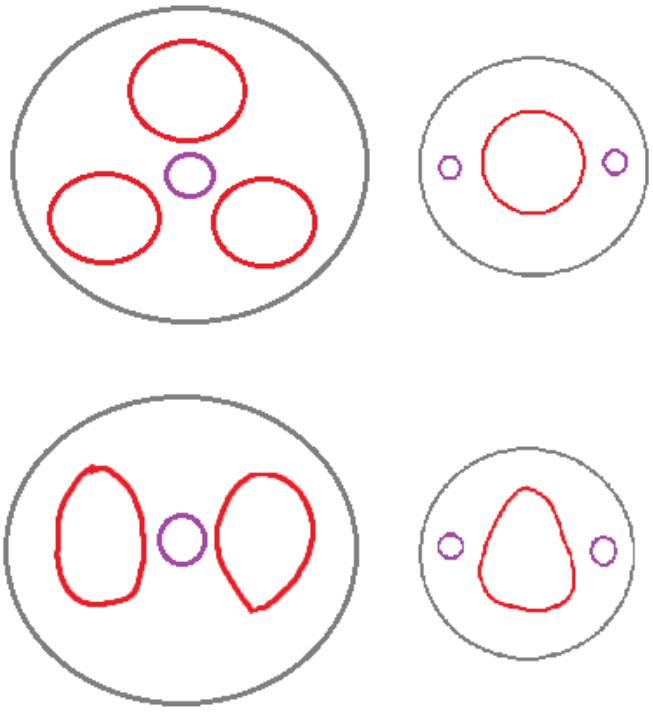


Fig 24. Depicting possible orientations for eggs (red) with a connecting rod (purple). Top left depicts 3 vertical eggs around a central rod. Top right depicts a single vertical egg with two rods. Bottom left depicts two horizontal eggs with a single rod. Bottom right depicts a horizontal egg with two rods.

5.2 Water Payload Assembly

To begin generating concepts for the water payload assembly, the team discussed the general concept during meetings, and made sketches on a whiteboard to easily convey ideas. This informal brainstorming allowed everyone to pitch ideas regardless of any initial constraints. The main criteria which was used to determine the best system concept was mass, manufacturing complexity, and ease of access, also considering that leak mitigation is a top priority. There were about four main unique variations that were discussed before deciding on the best concept for a final design.

The first iteration was the simplest, adding a water tank directly adjacent to the egg payload bay near the top of the vehicle. This concept came to mind first as it followed a very conventional layout, with payloads stored in the upper body section. However, this section of the rocket was already denoted as a prime location for other internal components such as the egg payload, avionics, and recovery systems, so a dedicated water tank here would likely force an increase in the overall rocket length, adding mass. A leak occurring with the water in such close proximity to the avionics bay could also damage electronics, which would be catastrophic for the vehicle and competition as a whole. Because of these drawbacks, the team decided to consider other options, and discovered that increasing packing efficiency within the vehicle could potentially save on mass, so alternative locations for the water vessels were explored.

It was then decided that utilizing the empty space around the motor tube towards the aft of the vehicle could be optimal for optimizing the space within the vehicle. To maximize the space, a vessel shaped as a cylindrical shell, with space for the motor tube to fit in the center, was theorized. The primary advantage of this design was its extremely efficient use of the otherwise unused volume. However, as early estimations for the vehicle specifications carried along, it was discovered that the amount of water to be carried would not justify such a complex and risky design.

The concept which was decided to be the best for this application was four individual vessels, spaced between the fins, within the aft bay. This design allows for the most flexibility to any changes of specifications through making only slight adjustments to the design dimensions. Through the implementation of a simple support structure which will have slots to house each vessel, the design shall be modular enough to accommodate changes to water mass. The structure will consist of multiple rings which will be able to easily slot into and be removed from the rocket body, only requiring a few screws to be fully secured inside its respective compartment. The main downside to this design is the increased points of failure, with each vessel requiring its own sealing mechanism the amount of potential leak paths are greater than with using one large tank, however mitigation to these risks is discussed later in the report.

One concept which was briefly discussed was the use of external water tanks, which would have been contained within separate structures outside the main rocket body, such as pods blended into the fins. At first this approach had some appealing features, like being easily accessible, and any leaks would be occurring outside the body, away from any critical electronics. However a deeper analysis showed some large disadvantages. External tanks would add a considerable amount of drag, lowering altitude, and complicating a stability analysis. Assembling these tanks with proper supports would also increase mass. Due to competition constraints, the amount of added mass and additional manufacturing complexity made this option much too risky to be further considered.

5.2 Water Payload Assembly (continued)

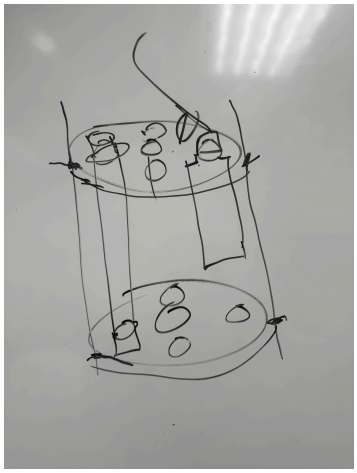


Figure 25. Rough Sketch of Water Payload Assembly

After comparing all these variations, the best overall option for the project was determined to be the third design, with four vessels surrounding the motor assembly. It gives purpose to otherwise wasted space, and is a decent location for pre- and post-flight access for filling and inspection. Should the team proceed with formulating a water deployment system, having the unload points located on the very bottom of the vehicle would be optimal for allowing the water to dump using gravity.

5.3 Motor Selection

As stated in 3.3.1, in order to maximize our score, we want to be able to carry as many eggs as possible. To do this, we want to find the most powerful motor that meets the project requirements. Using an AeroTech motor matrix that lists the specifications for all the motors that Aerotech produces, we created the following matrix.

Name	Diameter (mm)	Length (in)	Total Impulse (Ns)	Average Thrust (N)	Peak Thrust (N)	Thrust Duration (s)	Propellant Weight (g)	Total Weight (g)	Time Delay (s)
H13ST-P	29	8.00 (203mm)	211.2	13	50.4	15.2	116.4	203.4	Plugged
H135W-14A	29	8.00 (203mm)	226	135	152	1.7	121	211	14
H169WS-13A	29	8.00 (203mm)	236	169	249	1.3	112	202	13
H182R-14A	29	8.00 (203mm)	218	182	192	1.2	115	207	14
H195NT-14A	29	8.00 (203mm)	236	195	254	1.16	107	196	14
H14ST-P	29	9.06 (230mm)	265	14	47.6	18.3	134	230	Plugged
H45W-PS	38	7.98 (202mm)	320	53	87	6	180	364	Plugged
H100W-14A	38	6.12 (155mm)	234	100	131	2.3	121	261	14
H219T-14A	38	6.12 (155mm)	234	219	267	1.1	109	256	14
H283ST-15A	38	6.12 (155mm)	201	283	327	0.7	97	247	15
H550ST-14A	38	7.98 (202mm)	320	550	667	0.57	145	315	14

Fig 26. AeroTech single use motor data [19]

This new matrix isolates the only viable motors for our project. All motors are single use and do not use any “sparky” propellant. “Sparky” propellant has magnesium shavings in it, which ignite and glow, causing an eye-catching effect, but they are a fire hazard and as such are banned for use in this project. As stated earlier, the main focus in picking a motor is having the most power possible; we highlighted both motors with 320 Newton-seconds of impulse (H45W-PS and H550ST-14A) in bright green as they are the most desirable. Possible choices for weaker motors were highlighted in a duller green. It was also mentioned in 5.3.1 that the main differences between these two possible motors is the burn time and that the H45W-PS is a plugged motor. OpenRocket testing showed that the H550ST-14A is a much better option, as the H45W-PS did not propel the rocket at a high enough speed for a safe stability margin off the rail, and because of this, we have decided on using the AeroTech H550ST-14A.

5.3 Motor Selection (continued)

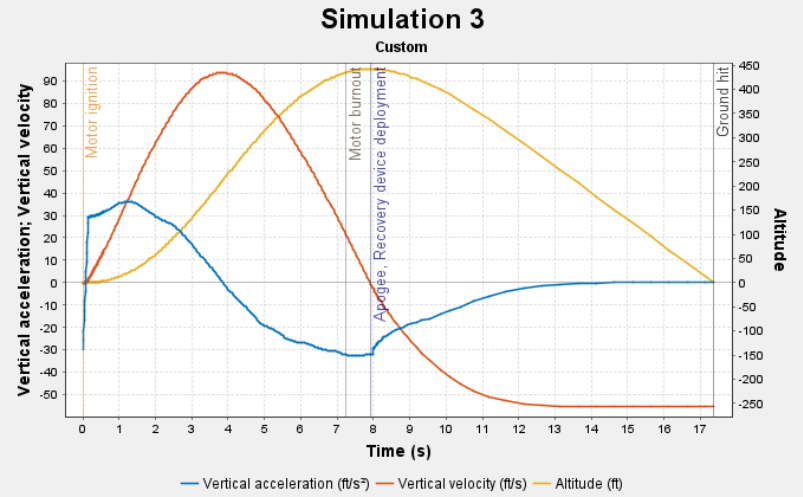


Fig 27. H45 Simulation plot showing low instantaneous velocity

5.4 Recovery Component Selection

As per the project description, our team will score 100 points by using a dual deployment recovery system, so we knew from the start that we were going to target that recovery method. Through research done for the tech memo, parachute estimates were obtained by viewing prior testing to see what parachute sizes were used. We looked at rockets of a similar size and weight to what we were planning to build, and upon analyzing the recovery system used by those rockets, we came to the conclusion that the size of our parachutes would be around 18 inches for the drogue and 48 inches for the main. These estimated sizes are also backed by the standard that for a dual deployment system, the drogue parachute is typically ½ to ⅓ the diameter of the main parachute.

The maximum descent velocity in the final 5 seconds of flight as stated in the project requirements is 25 feet per second, and the team decided that we wanted to target a descent velocity of 18 feet per second. We chose this velocity as it would give us some leeway if a failure were to happen, such as the parachute not fully deploying or a cord snapping, as well as minimizing the impact on the egg payload upon collision with the ground. We are also given the project requirement that the landing position of the rocket cannot exceed 800 feet from the launch site, so the parachute we use cannot be too big as it would lead to our rocket drifting farther, especially if there is wind on our launch day.

In order to come to a more accurate conclusion of what parachute sizes we should use for our rocket, we utilized OpenRocket simulations. Tests in OpenRocket were done via trial and error, selecting different sizes for both the drogue and main parachute until a final descent

5.4 Recovery Component Selection (continued)

velocity of 18 feet per second was achieved. Simulated testing has found that our rocket would impact the ground at 18.6 feet per second, while only drifting 340 feet from the launch site when using a 21 inch parachute for the drogue and a 78 inch parachute for the main.

Apogee	Velocity at deployment	Optimum delay	Max. velocity	Max. acceleration	Time to apogee	Flight time	Ground hit velocity
1095 ft	57.8 ft/s	7.74 s	279 ft/s	577 ft/s²	8.27 s	52.7 s	18.6 ft/s

Fig 28. Rocket flight data

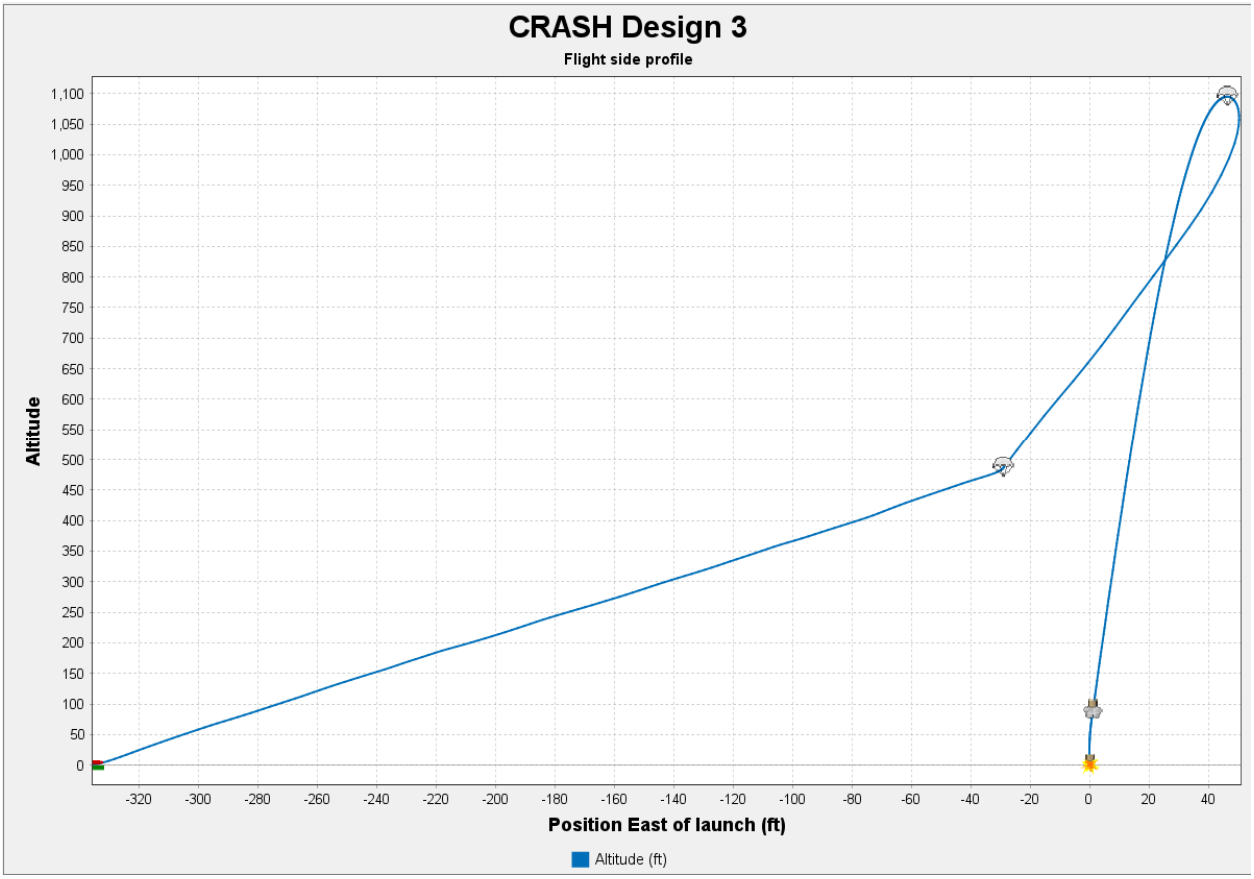


Fig 29. Rocket Flight Profile

Another decision the team made pertaining to recovery is the choice of the shock cord. Shock cord materials and their pros and cons were addressed in section 3.3.2. We decided that we would want to use a kevlar shock cord for our rocket due to the significantly higher tensile strength. This is mainly due to the increased weight of our rocket as a result of carrying a payload of eggs, which means that significantly more forces will be put into the shock cord upon parachute deployment. We believe that the increased cost of a kevlar shock cord over a nylon shock cord is more than justified for the increased safety margin it will give our rocket as well as a higher likelihood of egg survival and the rocket being recovered in a flightworthy condition.

5.5 Avionics System

Avionics setups on high power rockets have been fleshed out by a large field of hobbyists that have established the conventions of computer-directed rocket flight, particularly for dual-deployment setups. The approach to generating a baseline configuration was by cross referencing the "pass" criteria of the competition with the established methods of avionics. Then those aspects can be used to create operational concepts for the vehicle that satisfy the requirements that this project sets out to fulfill.

5.5.1 Baseline Operational Concept

The primary driver for the design was the needs versus wants analysis, as seen in section two of this document. A crucial need of the vehicle is a landing within 800 feet of the launch zone. This poses a particular challenge, as a big issue with rocket recovery is the pushing of the vehicle during descent by strong winds, carrying it further from where it initially launched. There are a few tactics that can be used to mitigate this (such as launch angle) but when coupled with the additional need for the rocket to descend at a rate of 25 feet per second or less, it becomes much more of a challenge. To get a descent rate that slow, you must employ a much larger parachute to slow the rocket's descent. As a tradeoff, it will take considerably longer to descend, and thus is under the forces of the wind which push it further from the launch site. The proposed solution to this problem is to introduce a dual-deployment setup that will be directed by a flight controller stored in an avionics bay.

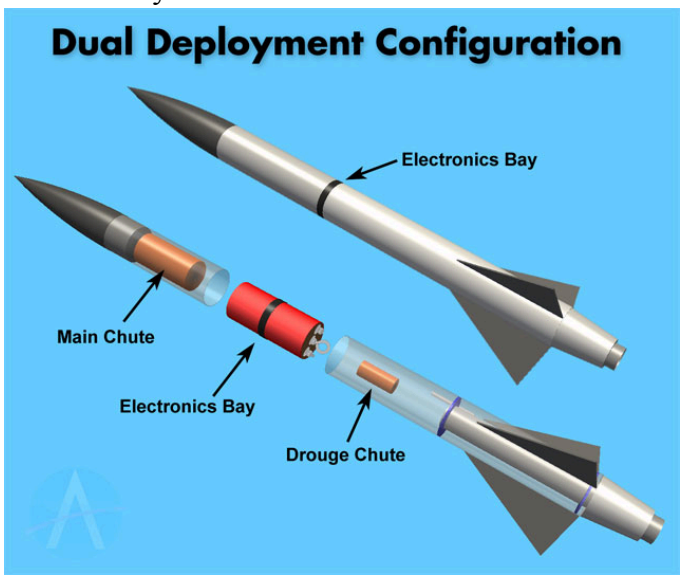


Fig 30. Arrangement of typical dual-deployment rocket setup. [20]

The setup shown above allows for a separation of recovery responsibilities across two different recovery bays, one above and one below the avionics bay. The lower bay holds the drogue chute, and is separated from the rest of the vehicle at apogee from a timed ejection charge that is typically already part of the motor. The main chute is stored in the top recovery bay, which is typically set to separate and deploy at a set altitude (around 500-700 feet) upon descent [20]. Control over this deployment is assigned to a flight controller housed within the avionics bay. This deployment event occurs when the flight controller detects a barometric pressure

5.5.1 Baseline Operational Concept (continued)

corresponding to the preset altitude of deployment, at which it sends a high current signal to an electrical match that ignites a black powder charge in the upper recovery bay. The rapid gas expansion from the charge separates the top section of the rocket to deploy the main parachute, slowing the vehicle down to a proper ground-hit velocity.

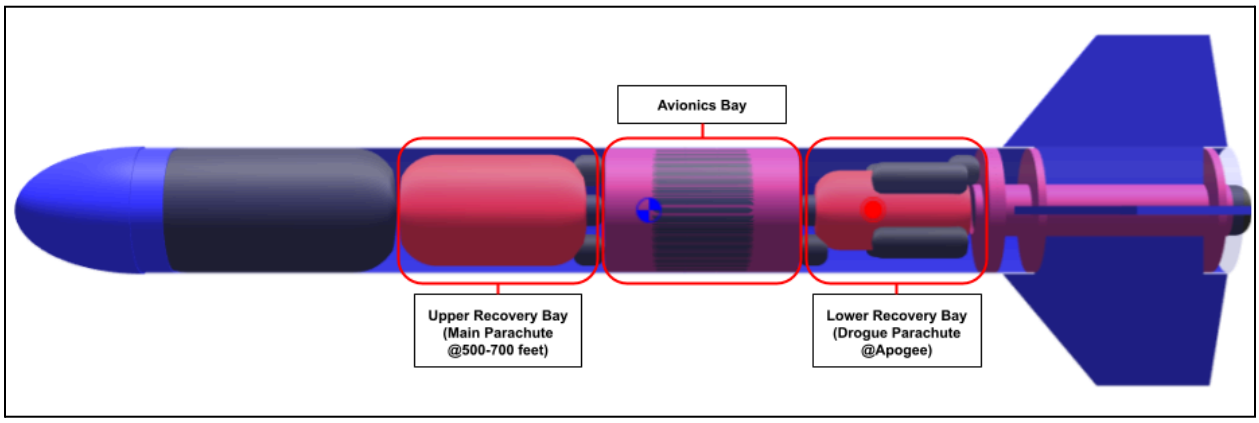


Fig 31. Diagram of recovery and avionics bays.

With the incorporation of a flight controller, it will be working double duty as a data logging device. Using its built-in barometer, accelerometer and gyroscope, it can track the velocity, acceleration and altitude across the duration of the flight. This data also needs to be readily available for access in the post-flight analysis. In addition, this device must be backed up by a redundant, self-contained data logging system. This would serve as a reliable, less volatile commercial-off-the-shelf (COTS) backup in case the data from the main flight controller is corrupted or lost.

5.5.2 Additional Components

In order to bolster the reliability and point earnings from the avionics, additional components have been considered to aid the baseline operational concept. The setup described above is sufficient to carry the rocket throughout flight. However, in terms of risks, it introduces multiple failures with no backup. With the threshold requirements introduced by the rules of the CRASH Competition, it is grounds for disqualification if any of those components does not function. This adds the need for redundancy in critical functions of the rocket, such as ejection charges that initiate flight events.

Other additional components would be ones that are classified as “Wants” in section 2.0 of this document, which are goals designed to earn additional points in the competition. Some examples of this would be the inclusion of a camera system, location devices, and livestreaming capability. All of these components come at additional cost and complexity when added to the baseline design, thus increasing potential risk. In order to determine the viability of each of these components, a study was run to compare each additional feature with its projected necessity, affordability, feasibility and point potential.

5.5.2 Additional Components (continued)

Component Additions						
Component	Necessity (1-5)	Affordability (1-5)	Feasibility (1-5)	Point Potential (1-5)	Total	Integrate? (10+)
Backup Flight Controller	3	1	4	1	9	**No**
Backup Drogue Ejection Charge	3	4	4	1	12	**Yes**
Backup Main Ejection Charge	2	4	4	1	11	**Yes**
Camera System	2	3	3	4	12	**Yes**
Live Video System	1	1	1	5	8	**No**
Auditory Recovery Device	4	4	5	3	16	**Yes**
GPS Location Device	3	1	2	3	9	**No**

Table 3. Additional Component Viability Matrix

Based on the study, the most optimal features to incorporate to increase reliability and point potential for the avionics section are redundant ejection charges on both recovery bays, a camera system, and an auditory recovery device. These should compliment the baseline operations of the vehicle to ensure the most wants and needs are met.

5.6 Rocket Design

Overall rocket design is the most sensitive and important attribute to this design process. This is the vehicle that carries the necessary payloads and carries out the goals of the mission. When it comes to design of the rocket itself, identifying weights and sizes is crucial as we balance altitude targets and stability. Besides weight and size, something that is easily adjustable and heavily impacts both stability and altitude are the nose cone and fin type. Working all of the attributes together must be strategic in a way that meets all customer needs.

Early on in the design process it was identified that the rocket design must integrate some sort of dual deployment system to counteract the weight and thrust caused by the rocket and its propulsion. Under the competition specifications, it is required of the participants that for a safe and successful flight, all rockets must obtain a decent velocity of 25 ft/s during the final 5 seconds of flight. With this implementation, it further made it a challenge as to where the payloads shall sit securely during flight. A dual deployment system meant that an added ejection charge would need to be added in a separate stage of the rocket, limiting safe space for the payload.

Initial concept designs for this project outlined an elongated body tube with enough room to house all payloads necessary to complete a successful launch. This was the main concern as other aspects of the aerostructure such as fins and the nose cone could be adjusted to meet altitude and stability requirements.

5.6 Rocket Design (continued)

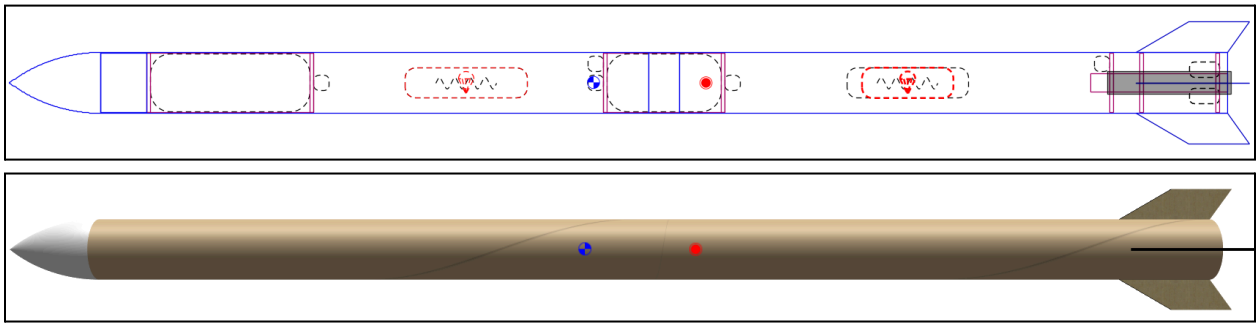


Fig 32. shows the initial elongated rocket design

The elongated rocket design combined designs of the egg payload bay, water payload bay, and overall positioning of recovery, avionics, and propulsion systems. To give ample room for the egg payload bay and move it further from an additional charge packed above the avionics bay to promote dual deployment, the egg payload seemed best situated towards the tip of the rocket. The second obstacle of tackling dual deployment was the location of the water payload bay. To safely house and not compromise the integrity of the water payload, it seemed more beneficial for the water to be placed at the tail end of the rocket, near the motor and fins. Although this seems to be an obscure placement, this location allows for easy access to the payload bay and is sufficiently protected by the heat caused within the motor because of the surrounding components which maintain integrity during ignition and flight.

A problem with the elongated rocket design was balancing altitude and stability with an Aerotech H195NT-14A motor which was originally set as the propulsion system for this project. While tackling the issue of reaching the minimum altitude specified by the competition guidelines, it was determined that a more powerful rocket must be needed to reach the desired apogee. Upon placing an Aerotech H550ST-14A motor in the design, altitude and stability seemed to balance out and meet requirements.

To allow for more flexibility and fix the altitude and stability issues noted earlier, the team suggested switching to a shorter body design. By making the fuselage more compact, the mass is pulled closer together and helps the rocket stay more stable during flight. A shorter airframe also uses less material, which makes it easier and more cost effective to build. Even with this big change in shape, the simulations showed that the shorter design performed almost the same as the longer tube when paired with the higher-powered motor. Key factors like peak altitude, flight stability, and flight performance all stayed within acceptable ranges. Overall, the shorter design offers a more efficient and cost-effective option without giving up performance.

5.6 Rocket Design (continued)

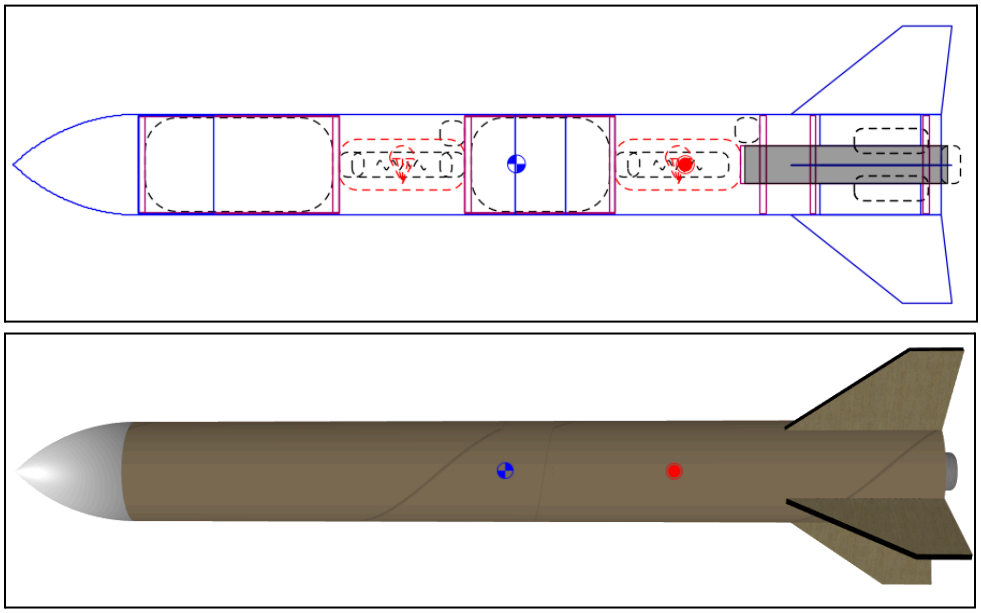


Fig 33. shows the compact rocket design.

The above rocket design involves an ogive nose cone shape which is typically best suited for rockets under supersonic flight conditions. The ellipsoid nose cone shape is another design choice which is more suited for subsonic flight but underperformed in this specific situation. The reason for this change was to adjust for the lack of altitude and stability. Changing the design to ogive, provided more viable flight conditions.

Overall, the design referenced in figure 33 is not the ideal design case for the customer/competition requirements. Simulations projected altitude that narrowly cleared the required minimum height, stability also narrowly cleared the minimum requirement, leading to false hope in the design. On top of these nearly failing parameters, fin design was a concern for the team as the fin length spanned past the end of the body tube which creates greater risk of fins chipping and/or breaking upon landing. To combat this problem, a new design was set forward.

Major changes to the following design included implementing a fin shape that effectively mitigated the risk of breaking the fin components before, during, and post flight. Adhering to the characteristics of subsonic flight and designs with best performance under those parameters, an ellipsoid nosecone was designed instead of an ogive nose cone. Previously this was changed to adhere to competition flight requirements, but the new rocket design has more nominal flight conditions.

5.6 Rocket Design (continued)

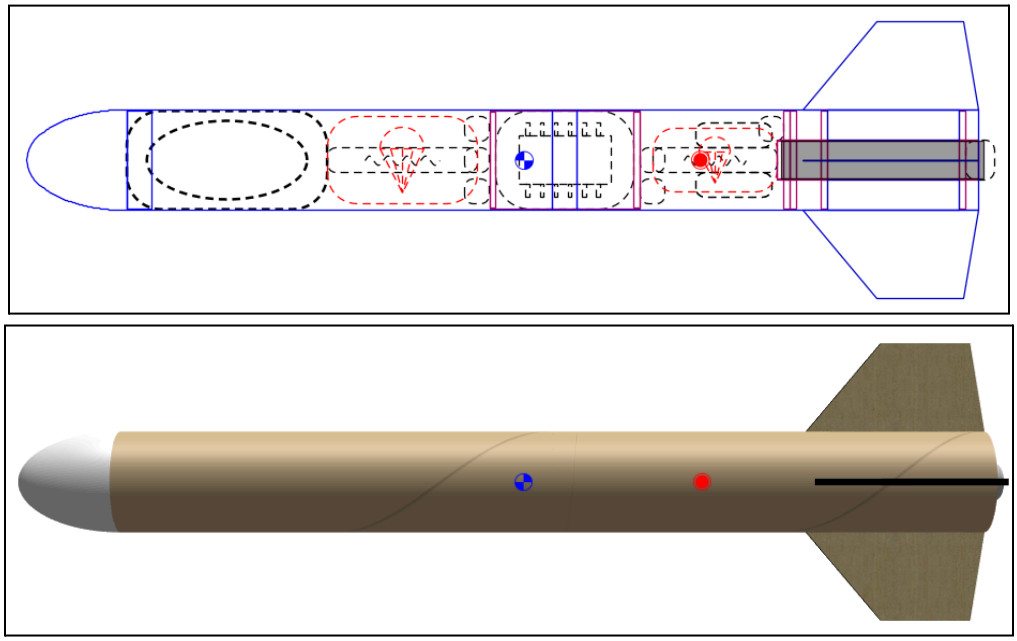


Fig 34.shows the final compact rocket design.

Referring back to the materials research done for the aerostructure components, it is clear that the most cost effective and overall generally effective material for the body tube is cardboard which is durable as proven in previous experiences. Alternative materials for the body can come at a higher cost and require more difficult machining, while potentially adding unnecessary weight to the vehicle itself. Material for the nose cone will be a PLA material that is easily 3D printed to obtain the ideal shape required for launch. This is a light and practical option as adjustments in design and adjustments in weight balancing can be done easily. Between the fins, bulkheads, and centering rings the shared material will be birch plywood at a quarter inch thickness. This provides sound structural support under intense and rapid force caused by launch and landing.

6.0 Preliminary Engineering Analysis

6.1 Optimizing foam cushioning

In order to find the weight of the egg payload, the distance needed between eggs, and the orientation of the eggs, the selection of potential foams needs to be analysed first. Method for calculating optimum foam density and distance between eggs is derived from the MIT Structures lesson Cellular Solids: Structure, Properties and Applications [21].

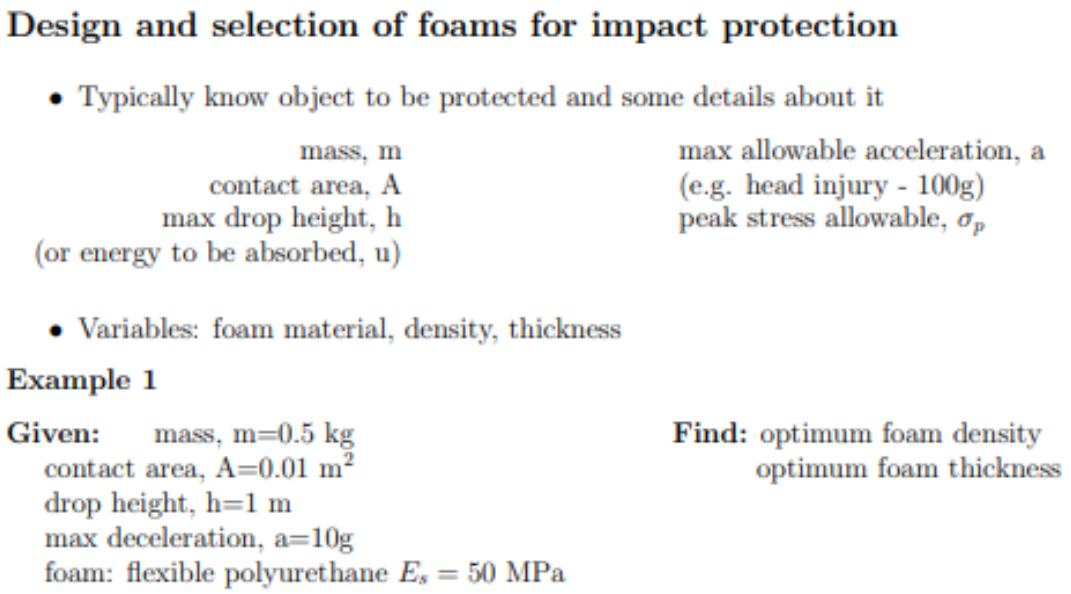


Fig 35. Design an selection of foam for impact protection [21]

6.1 Optimizing foam cushioning (continued)

Example 1:
Find Foam Density and Thickness

Table 8.2 Example 1: selection of foams

Specification of the problem		
Mass of the package object, $m = 0.5\,\text{kg}$		
Area of contact between foam and object, $A = 0.01\,\text{m}^2$		
Velocity of package on impact (drop height $h = 1\,\text{m}$), $v = 4.5\,\text{m/s}$		
Energy to be absorbed, $U = mv^2/2 = 5\,\text{J}$		
Maximum allowable package force (based on deceleration of 10g), $F = ma = 50\,\text{N}$		
Maximum allowable peak stress, $\sigma_p = F/A = 5\,\text{kN/m}^2$		
Solid modulus in foam (flexible polyurethane), $E_s = 50\,\text{MN/m}^2$		
Maximum allowable normalized peak stress, $\sigma_p/E_s = 10^{-4}$		
Iterative procedure		
1st Iteration	$t_1 \gg t$	$t_1 \ll t$
Initial choice of t_1	1 m	0.001 m
Resulting strain-rate, $\dot{\epsilon} = v/t_1$	$4.5\,\text{s}^{-1}$	$4.5 \times 10^3\,\text{s}^{-1}$
Resulting (W/E_s) at $\sigma_p/E_s = 10^{-4}$	5.25×10^{-5}	7.4×10^{-5}
Energy absorbed per unit volume, W	$2620\,\text{J/m}^3$	$3700\,\text{J/m}^3$
2nd Iteration		
Revised t_2 (from $U = WAt$)	0.19 m	0.14 m
Revised $\dot{\epsilon} = v/t_2$	$24\,\text{s}^{-1}$	$32\,\text{s}^{-1}$
Revised (W/E_s)	6.6×10^{-5}	6.7×10^{-5}
Revised W	$3300\,\text{J/m}^3$	$3350\,\text{J/m}^3$
3rd Iteration		
Revised t_3 (from $U = WAt$)	0.15 m	0.15 m
Optimum density, ρ^*/ρ_s (Fig. 8.8)	A little below 0.01	

Fig 36. Example problem 1: Find Foam Density and Thickness [21]

6.1 Optimizing foam cushioning (continued)

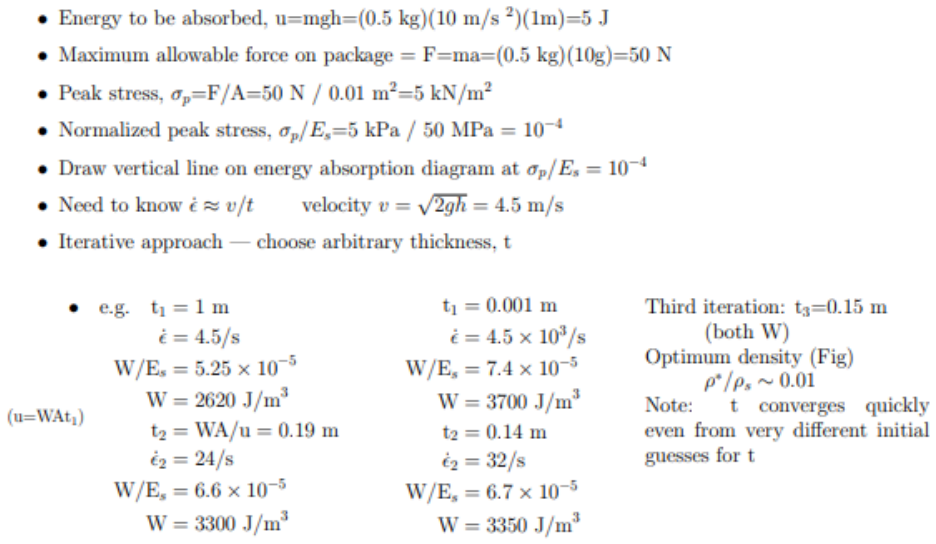


Fig 37. Solution to Example 1 [21]

The iterative process described in the example was translated into a computational code developed for this project to automate and streamline the evaluation of foam materials. This code also allows the ability to choose between charts for Polyethylene Foam Sheet (Closed-Cell PE) or Cross-linked Polyethylene Foam (XLPE), data was collected from Foam Factory's foams physical data sheets [22] [23]. Young's modulus values for the foam were approximated from the manufacturer's compression-deflection data (ASTM D3575) by computing an effective secant modulus in the 25–50% strain range. A Poisson's ratio of 0.25–0.30 was assumed, consistent with reported values for conventional closed-cell polyethylene foams.

Table 4. Comparison of PE and XPLE Foams

Name of foam	Density (lb/ft³)	Compressive Strength (lb/in²) at 25%	Compressive Strength (lb/in²) at 50%	Young's Modulus, E (psi)	Poisson's Ratio, v
Polyethylene Foam Sheet (Closed-Cell PE)	1.2	7	12	~26	0.25
Polyethylene Foam Sheet (Closed-Cell PE)	1.7	7	14	~28	0.25
Polyethylene Foam Sheet (Closed-Cell PE)	2.2	8	14.5	~31	0.25
Polyethylene Foam Sheet (Closed-Cell PE)	4	16.5	20	~53	0.25
Cross-linked Polyethylene Foam (XLPE)	2	8.2	17.1	~34	0.25–0.30
Cross-linked Polyethylene Foam (XLPE)	3	10.5	19.8	~41	0.25–0.30
Cross-linked Polyethylene Foam (XLPE)	4	21.2	31.8	~74	0.25–0.30

6.1 Optimizing foam cushioning (continued)

Link to webpage: https://camilaarmartinez.github.io/foam-impact-tool/



Fig 38. Foam Impact Design Calculator

Increasing the allowed force on the eggs reduces the distance of the foam needed. If restricted to the original limit of 35N acting on the egg, a distance of above 10cm for PE and 4cm for XPLE. A limit raised to 100N requires around 5cm for PE and 1.5 cm for XPLE. A limit raised to 150N requires around 3.5cm for PE and 1 cm for XPLE. A limit raised to 200N requires

6.1 Optimizing foam cushioning (continued)

around 2.7 cm for PE and 0.7 cm for XPLE. The study by ADMET [2] tested eggs using a single-column compression machine with foam padding and found that eggs broke under an average load of 53 lb (235.8 N) on their poles and 90 lb (400.3 N) on their sides. This is a much higher force than the study cited for M2 and the tech memos. This study is more relevant to the project, focusing on the distribution of force on the egg instead of the egg's resistance to dropping. A padding of 10cm PE for a maximum force of 35N is impractical and not space-efficient. With a body tube diameter of 4 inches, the maximum space between eggs and the wall is 3.13cm.

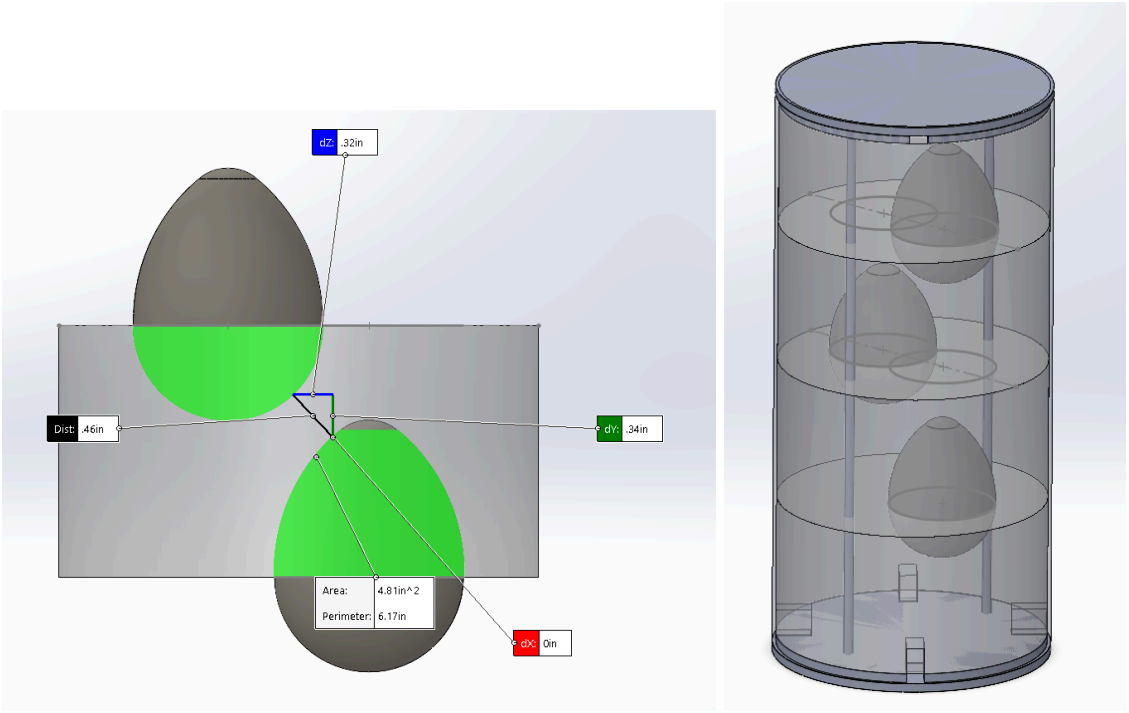


Fig 39. 2in XPLE 2

This version spaces the egg from the wall 1.5cm, in a 2-inch puck, leaving 1.16cm between the eggs internally. 2 inches XPLE would cost $57.99 for a sheet. The spacing between eggs is allowed under the 150N range. With a max height of 8 inches for the sleeve of the capsule, there would only be 3 eggs carried.

6.1 Optimizing foam cushioning (continued)

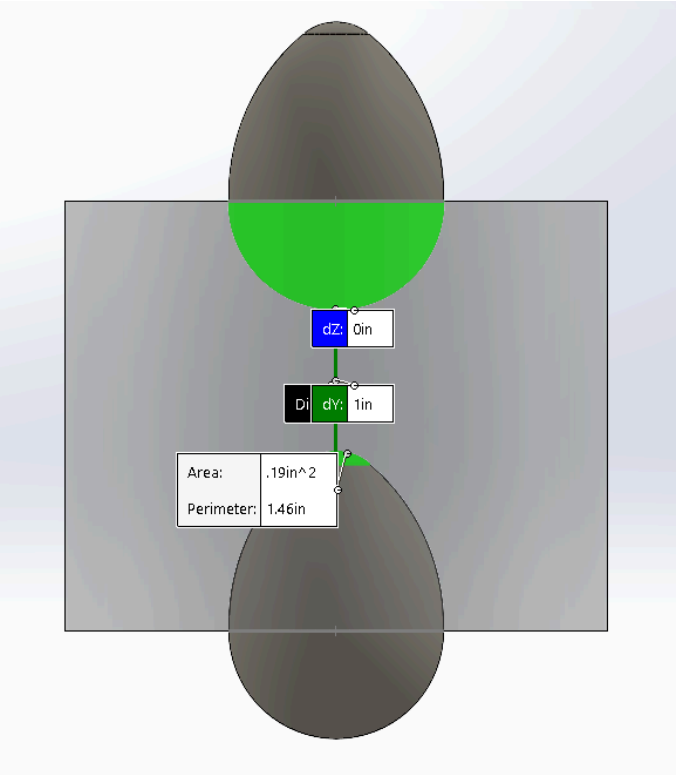


Fig 40. 3in XPLE 2

This version places the eggs in the center of a 3inch puck, with a space of 3.13cm of foam from the walls and 1 inch between eggs. This XPLE foam will cost $77.99. This variation comfortably complies with the 100N range. With a max height of 8 inches for the sleeve of the capsule, there would only be 2 eggs carried.

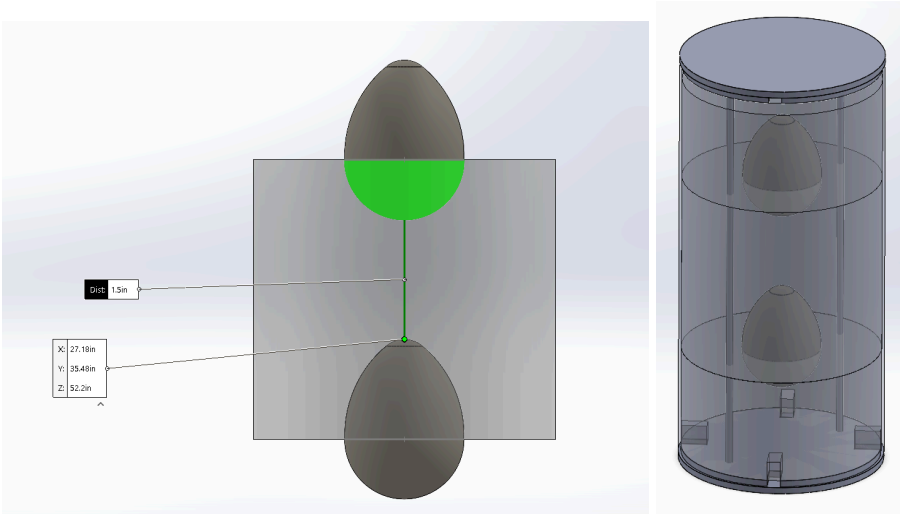


Fig 41 3.5 in PE 2.2

6.1 Optimizing foam cushioning (continued)

The cheapest version using PE 2.2 would be slightly below the optimal 3.4cm for 150N, having 3.13cm of foam between the eggs and the wall. With the option of cutting a premade cylinder instead of sheets, the height of the pucks can be adjusted to the optimal 3.4cm of spacing with a height of 3.5 inches. This PE cylinder foam will cost $14.99. With a max height of 8 inches for the sleeve of the capsule, there would only be 2 eggs carried.

To lower costs and manufacturing, another option if we are only launching two eggs, a vertical dual egg protector from aerotech can be purchased for $11.50 [24].



Fig 42. Depicting the apogeerocket Vertical Dual Egg Protector (BT-80) [24].

6.2. Egg Capacity Optimization

When disregarding cushioning optimization, the design can instead be optimized for maximum egg capacity within the capsule. For each additional egg over two that is safely recovered the team earns 10 additional points. So for point maximization potentially having more eggs, assuming some but not all break, will earn more points than only launching two.

6.2. Egg Capacity Optimization (continued)

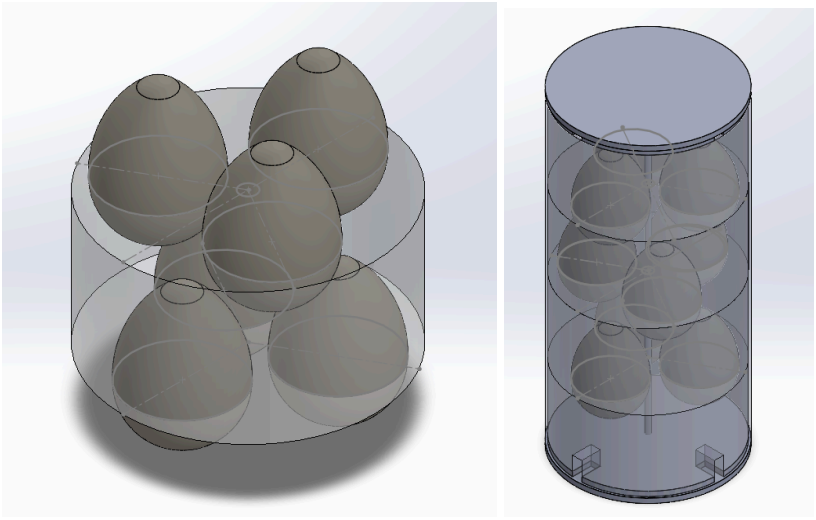


Fig 43. 2in PE 2.2 EGG 3

Maximized to have 3 eggs per layer in 2-inch pucks, 9 eggs total fitting in the coupler.

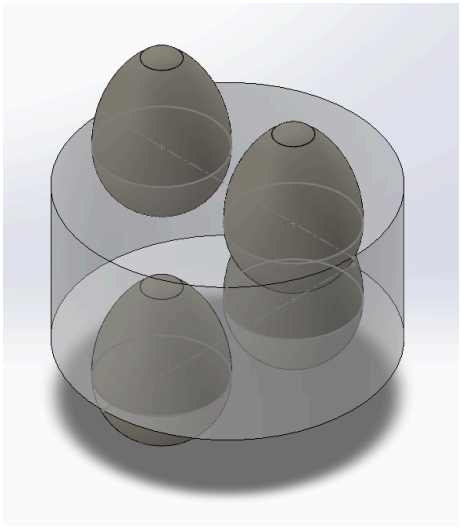


Fig 44. 2in PE 2.2 EGG 2

Maximized to have 2 eggs per layer in 2-inch pucks, 6 eggs total fitting in the coupler.

6.2.1 Egg Capsule Decision Matrix

Table 5. Egg Capsule variation measurements

Variation	Egg capacity	Cost	Weight (foam + eggs)
XPLE 2 2 in	3	$57.99	7.76 oz
XPLE 2 3in	2	$77.99	5.79 oz
PE 2.2 3.5in	2	$14.99	5.97 oz
PE 2.2 2in EGG 3	9	$14.99	19.69 oz
PE 2.2 2in EGG 2	6	$14.99	13.83 oz
Dual egg protector from Aerotech	2	$11.50	5.94 oz

Additional weight for the Egg payload includes the coupler, aluminum centering rod, U-bolts, and 2 plywood bulkheads.
Coupler: 6.38 oz
Aluminum rod (¼" × 8"): 0.61 oz
Two U-bolts: 2.04 oz
Two plywood bulkheads: 2.32 oz
Totaling to 11.35 oz.

Table 6. Egg Capsule Decision Matrix

Variation	Egg capacity (1-5)	Low Chance of breaking (1-5)	Cost (1-5)	Weight (1-5)	Manufacturing (1-5)	Total
XPLE 2 2 in	1	4	1	4	2	17
XPLE 2 3in	1	5	1	5	2	20
PE 2.2 3.5in	1	5	5	5	5	25.5
PE 2.2 2in EGG 3	5	1	5	1	5	25.5
PE 2.2 2in EGG 2	4	1	5	2	5	23.5
Dual egg protector from Aerotech	1	5	3	5	5	23.5

The egg capacity is weighed times 3, the chances of breaking are weighed times 2, and manufacturing is divided in half. PE with egg capacity of 2 eggs, the competition minimum, and the PE with egg capacity of 9, have the same score in the decision matrix. The version of maximum eggs would weigh 31.04 oz, and the maximum cushioning would weigh 17.32 oz. Because both options use the same material and require purchasing a minimum 17-inch cylinder, both versions can be manufactured and tested before launch at no extra cost. This allows the team to choose the final launch version based on test results.

6.3 Water Payload Analysis

An analysis of the water payload was performed to gather a baseline for the forces which must be sustained by the subassembly during flight. This information is important as under-designing could result in disqualification from the competition due to water spillage. The primary objective of this analysis is to ensure that the water vessels, the seals, and the supporting structure are able to withstand the forces sustained during peak acceleration. To proceed with the analysis, several assumptions about the design shall be defined.

Newton's second law [25] was used to calculate the total force to be experienced by the supports due to the water payload. Based on prior simulations of the proposed launch vehicle, the max acceleration to be experienced is 173.4 m/s^2, or about 18g. The combined mass of each complete vessel and its payload is estimated as 0.0552kg, or 1.95oz. Each individual module will produce a force equal to the peak acceleration multiplied by its mass.

$$173.4\left(\frac{m}{s^2}\right) \times 0.0552\,kg \approx 9.6\,N$$

Therefore with four vessels a total force of 38.4 N, or about 8.6 lbf, is the total axial load that the support system has to carry during launch.

Another force which could test the structural integrity of the assembly is the pressure experienced by the caps of the vessels. The density of water is 1000 kg/m^3 [26], and the height of the vessel is 3 inches, or 0.0762 m. With these, the simplified Stevin's Law [27] was used to formulate the hydrostatic pressure on the caps during peak acceleration.

$$1000\left(\frac{kg}{m^3}\right) \times 173.4\left(\frac{m}{s^2}\right) \times 0.0762\,m \approx 13.2\,kPa$$

This combined with the pressure force due to the mass of the water over the area of the cap will give the max pressure on the cap. The vessels will have a standard 20-410 neck fitting, giving an inner cap diameter of 20mm. The mass of the water, multiplied by max acceleration, divided over the area of the cap shall give the pressure force.

$$0.0354369\,kg \times 173.4\left(\frac{m}{s^2}\right) \div \frac{\pi}{4}(0.02\,m)^2 \approx 19.6kPa$$

The results from both of these equations is a max pressure on the cap of about 32.8 kPa, or about 4.8 psi.

The results of these basic preliminary calculations confirm that the subassembly can be designed to withstand the peak launch loads with significant margin. Findings showed that the overall pressure loads are minor, and the total force induced by the payload is not much more significant. This means that a failure due to mechanical weaknesses in the design is highly unlikely, and the focus should be shifted towards ensuring a quality assembly of the payload to reduce further risk.

6.4 Avionics Design Analysis

In order to optimize the baseline configuration for the avionics bay, the components and integration have been weighed against each other to see which setup best suits the needs of the vehicle. The choices under consideration would be for those of the flight controller, backup altimeter, flight controller battery, camera, and auditory locator device.

6.4.1 Flight Controller

When selecting a flight controller for a rocket, the goals of that rocket must be taken into account. Based on the needs set about by the requirements, the flight controller must have a particular sampling rate, number of pyro outputs for flight events, and should be reasonably priced. To gauge this, a study was conducted comparing typical consumer COTS flight controller models based on key characteristics.

Table 7. COTS Flight Controller Comparison Matrix

Flight Controller Comparison				
Flight Controller	Pyro Outputs	Sampling Rate (Hz)	Current Draw (mA)	Price (Including add-ons)
Altus TeleMega	6	100	150	$400.00
Altus EasyMini	2	100	7	$80.00
RRC3/UDC	3	20	35	$91.95
Blue Raven	4	500	18	$175.00

Based on the data, the flight computer that best satisfies the conditions of the vehicle is the RRC3 with UDC connector for configuration. It has the minimum number of required pyro channels to accommodate the primary and backup ejection charges planned. Aiding its appeal is also the reasonable price point and extensive use history in the hobby.

6.4.2 Backup Altimeter

The backup altimeter is an additional self-contained device that is able to record apogee of the vehicle. This is to provide backup data that ensures you reached the target apogee in the case that the data collected by the flight controller is corrupted or inaccessible. Self-contained is defined as a unit that does not require an external battery. Below is a matrix comparing various COTS altimeters that could aptly fulfill the need of the competition rules.

6.4.2 Backup Altimeter (continued)

Table 8: COTS Backup Altimeter Comparison Matrix

Backup Altimeter Comparison				
Altimeter	Apogee Logging?	Velocity Logging?	Acceleration Logging?	Price
JollyLogic AltimeterOne	Yes	No	No	$60.00
JollyLogic AltimeterTwo	Yes	Yes	Yes	$80.00
Missile Meter	Yes	No	No	$49.50

Based on the table above, the optimal choice would be the JollyLogic AltimeterOne. The main reasoning behind this is ease of use and minimum requirements for the unit. The AltimeterOne is self contained unlike the Missile Meter, which has its lithium ion battery and board openly exposed. The AltimeterOne on the other hand clears the minimum need for the apogee to be logged per the rules of the competition.

6.4.3 Battery

Based on the flight controller selection in the previous section, a battery that can best fit the mission time can be selected. One of the conditions that cannot be predicted is the idle time of the rocket on the pad once active. The operation time until the battery runs out of power can be calculated using the equation seen below.

$$\text{Battery Life} = \frac{\text{Battery Capacity (mAh)}}{\text{Load Current (mA)}}$$

Battery power-on time equation

Based on the equation, with an RRC3 using a 9V 550mAh Duracell battery (which it is most accustomed to) the flight computer has an operational time of ~15.25 hours. This is more than enough battery required for the duration of the flight, as long as the battery does not incur any accidental excessive discharge before the time of operation for flight.

6.4.4 Camera

There are two primary approaches to mounting a camera on a rocket, either being external or internal. The internal camera would be one using a pinhole through the side of the tube while the other is directly mounted to the side. Both pose aerodynamic challenges, as a pinhole camera introduces an irregular hole in the side of your tube that can allow abnormal pressure fluctuations. Likewise, side mounted cameras introduce a bulge to the outside of the rocket body that introduces additional drag to the body of the rocket.

6.4.4 Camera (continued)

At first, the idea was to incorporate the camera into the avionics bay and create a pinhole design. However, being able to turn on the camera itself would quickly become a problem as access to the bay is limited once the rocket is entirely assembled. In contrast, a side-mounted external camera (as long as it is secured properly) allows for easy access and configuration. The priority from that point is keeping the profile of the camera low and ensuring it is as self-contained as possible. To compare potential options, a small study on readily available camera devices as shown below.

Table 9. COTS Camera Comparison Matrix

Camera Comparison		
Camera	Internal Battery?	Cost
HD Mini Security Camera	Yes	$19.48
Estes Universal Astrocam	Yes	$48.50
RunCam Split V4	No	$82.99
RunCam 5	Yes	$119.99

Based on the study run, the Estes Universal Astrocam is the preferable choice. It balances price with ease of use, both being intended for rocketry and coming with its own mounting equipment. While the Mini Security Camera is very cheap, it would require the development of a customized mounting system to smooth out its side profile. It also only has motion activation, which may be unreliable unless extensively tested to verify applicability to rocketry.

6.4.5 Auditory Locator Device

Auditory devices allow for easy location of the vehicle in low visibility scenarios such as tall grass or trees. They can vary in complexity, with units designated for rocketry or more rudimentary devices that are meant for commercial use. A study was run to compare some of these options, which is displayed below.

Table 10. COTS Auditory Location Device Comparison Matrix

Auditory Location Device Comparison			
Locator	Decibel Rating	Internal Battery	Price
Kosin Personal Alarm	140	Yes	$2.33
RB02 Beeper	108	No	$27.50
BeepX Sonic Beacon	105	No	$38.50

Based on previous experience and pricing, the personal alarm device serves as a simple and reliable device that can easily be clipped onto existing recovery equipment. It is also very cheap and very loud. With the intended landing distance, it would be more than sufficient for aiding in recovery.

6.5 Open Rocket Analysis

Using the desired rocket body design with a shorter body tube to prioritize altitude and stability, simulation and analysis were performed. Crucial data for launch conditions and in flight performance was evaluated using OpenRocket. Using average spherical approximations, and inputting typical weather conditions of Palm Bay, Florida, a simulation was performed. Data was collected through Weather Spark to determine average wind speed conditions on March 21st during early morning the past three years. [28]

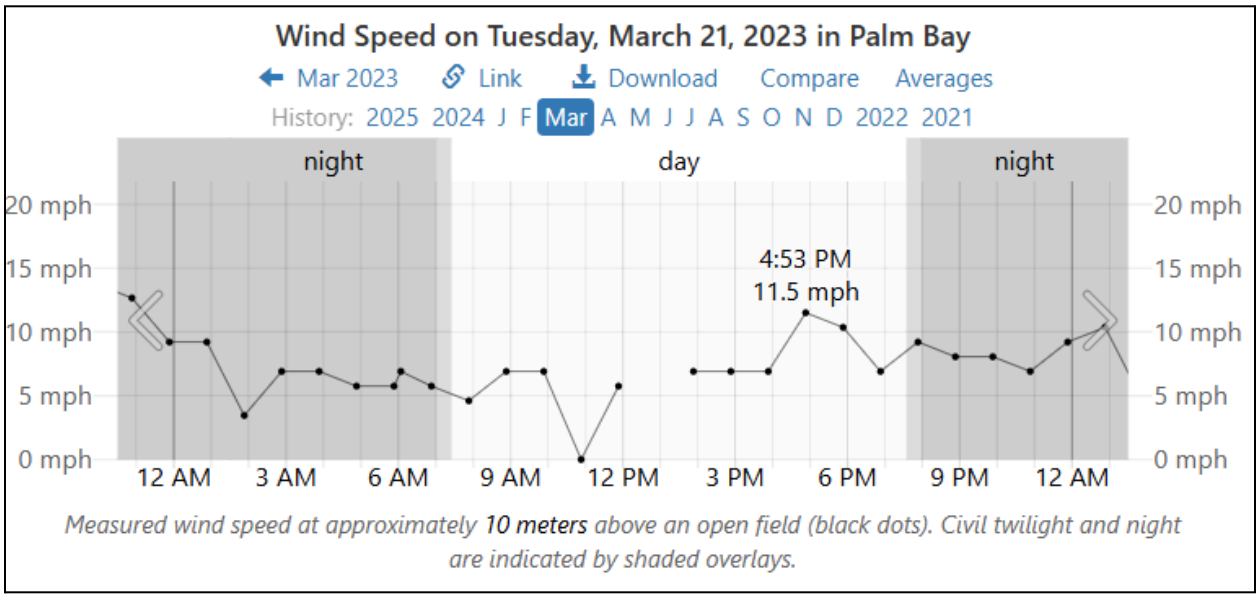


Fig 45. Plots pinpointed wind speeds at various times of the day in 2023

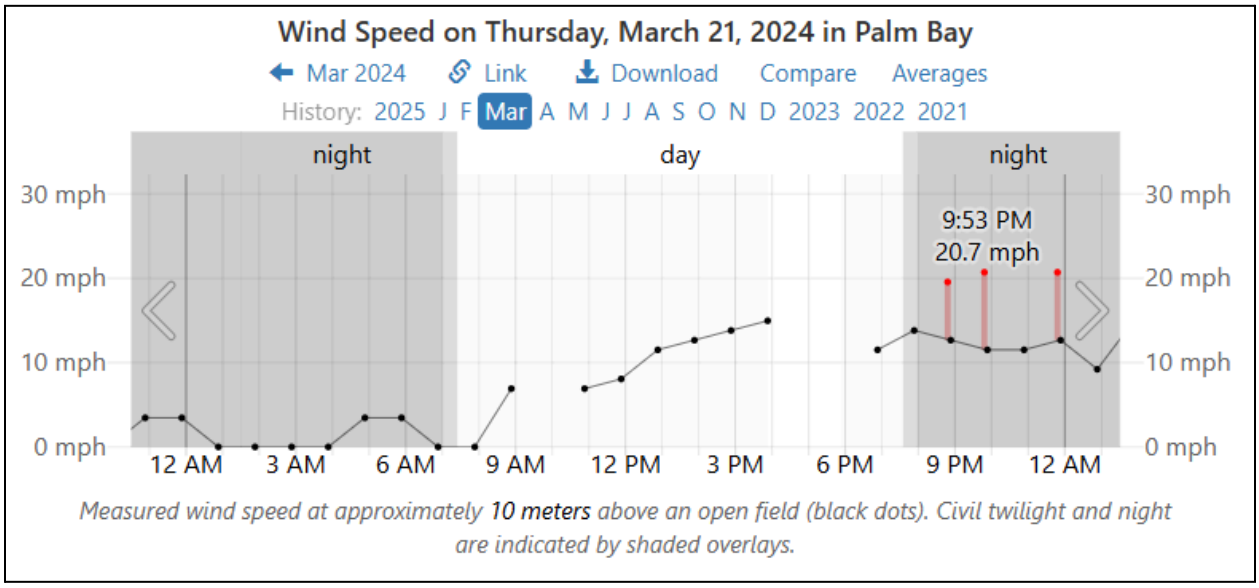


Fig 46. Plots pinpointed wind speeds at various times of the day in 2024

6.5 Open Rocket Analysis (continued)

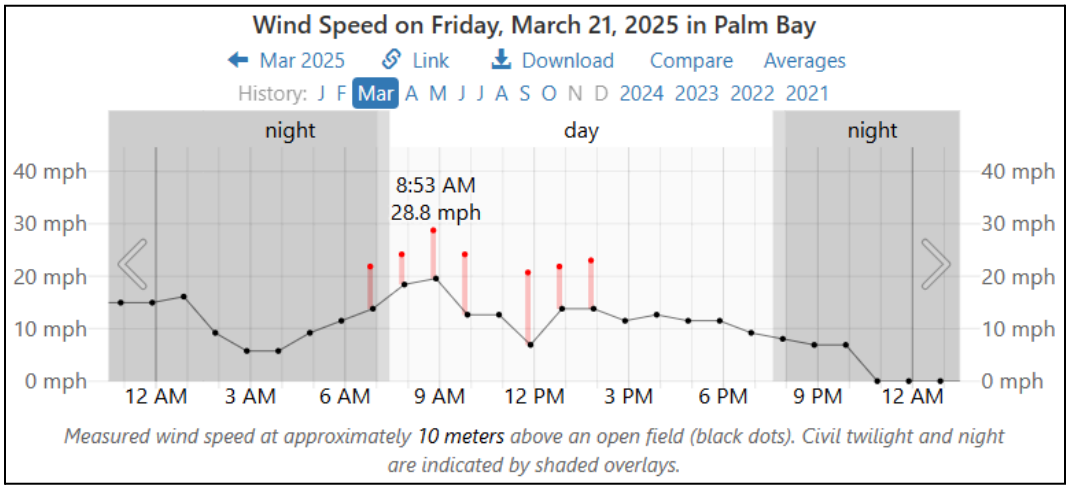


Fig 47. Plots pinpointed wind speeds at various times of the day in 2025

The targeted launch date for this team is March 21st, 2026. To anticipate likely wind conditions, an average of the past three years wind data was collected to apply to the simulation. Wind speed points were taken at 8 AM which would be around the anticipated launch time.

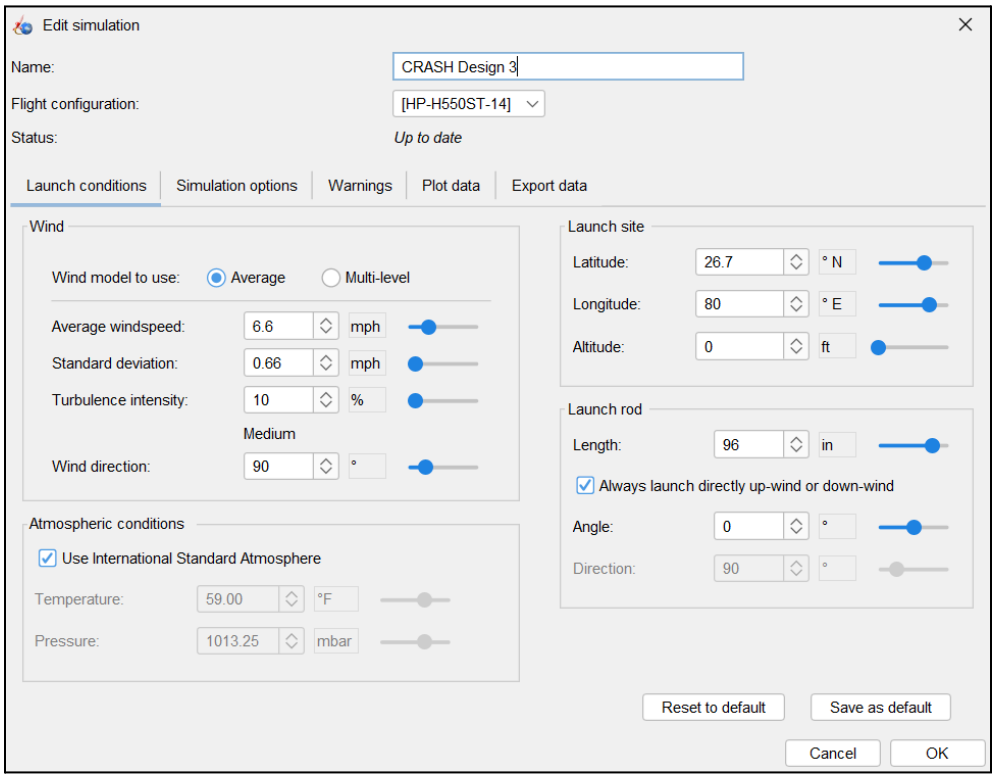


Fig 48. Combines initial launch condition findings which are applied to the simulation

6.5 Open Rocket Analysis (continued)

To analyze all conditions during flight, the following plots show each condition during launch, in flight, and once the rocket hits the ground.

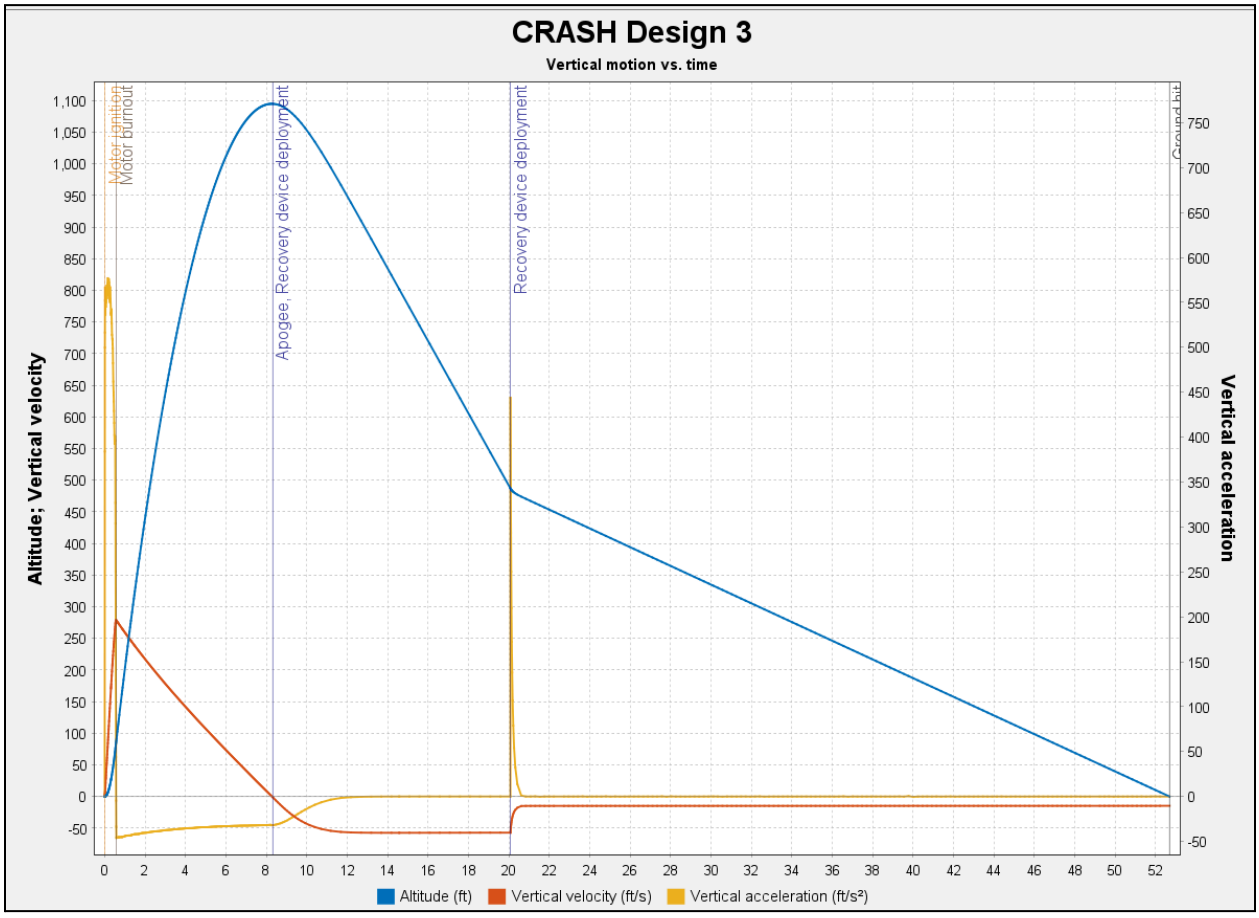


Fig 49. Demonstrates vertical motion vs. time throughout flight

Relevant Equations: [29]

$$v = v_0 - gt$$

Final velocity equation

$$y = y_0 + v_0 t - \frac{1}{2} g t^2$$

Position equation

$$v^2 = v_0^2 - 2g(y - y_0)$$

Velocity with respect to position

6.5 Open Rocket Analysis (continued)

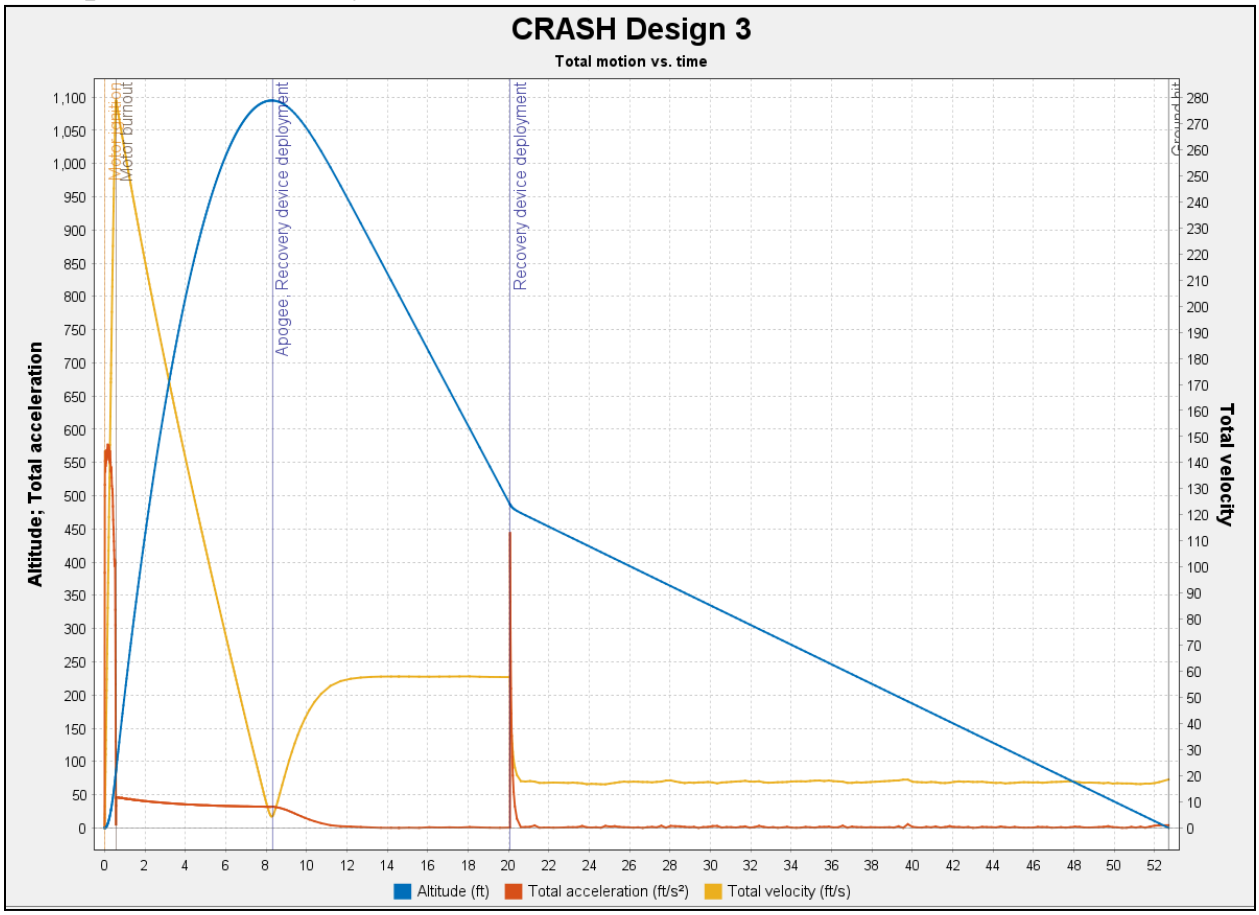


Fig 50. Demonstrates total motion vs. time throughout flight

Relevant Equations: [30]

$$v = v_0 + at$$
$$r = r_0 + v_0 t + \frac{1}{2}at^2$$
$$r = r_0 + \frac{1}{2}(v + v_0)t$$
$$v^2 = v_0^2 + 2a(r - r_0)$$
$$r = r_0 + vt - \frac{1}{2}at^2$$

Equations with respect to acceleration during flight

6.5 Open Rocket Analysis (continued)

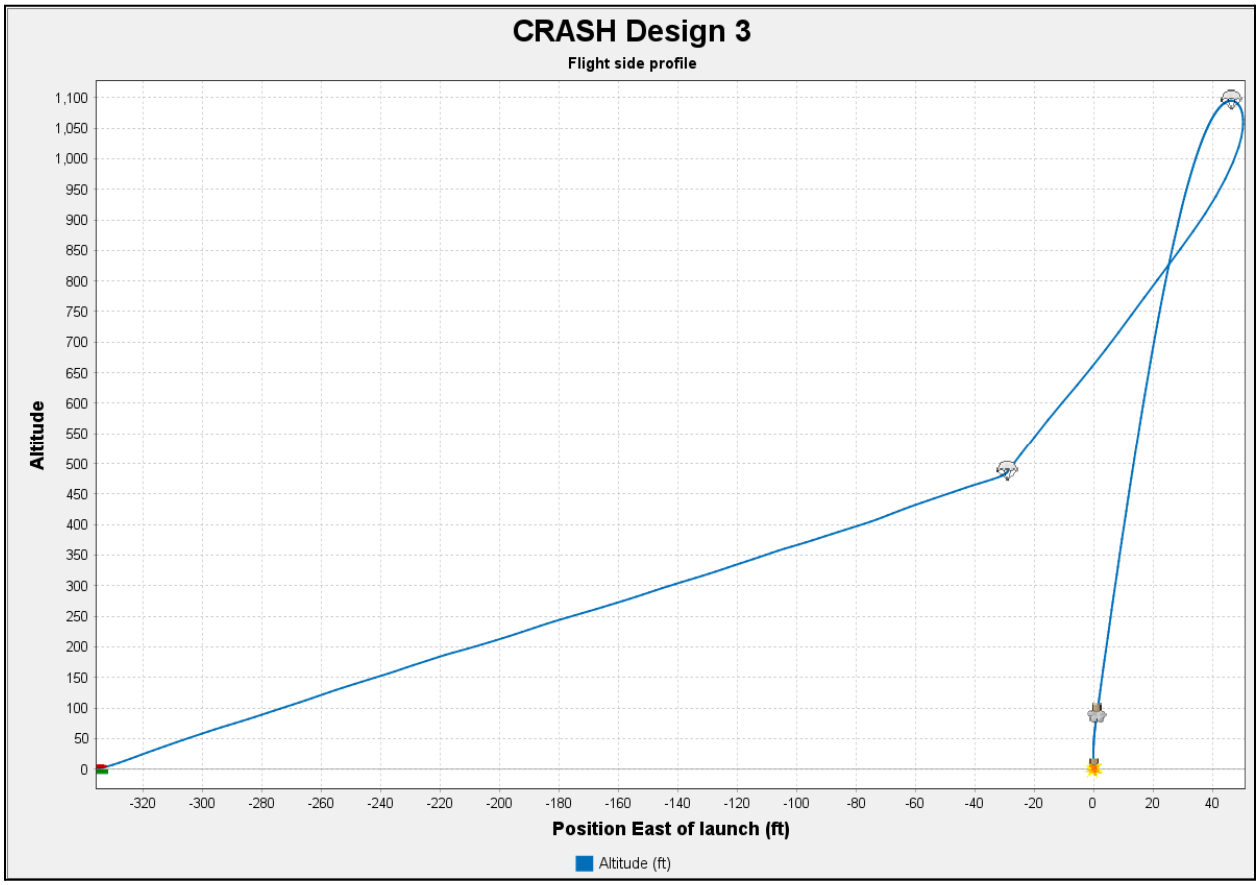


Fig 51. Demonstrates relative location and altitude throughout flight while pinpointing stages of recovery

Relevant Equations:[31]

$$\frac{V_x}{V_w} = 1 - e^{-\int_0^z n_\alpha dZ} \approx 1 - e^{-n_\alpha dZ}$$

Horizontal Wind Profile

$$z_1 = \frac{1}{2}\left(\frac{F}{m} - g\right)t^2 \qquad z_2 = \frac{F\ z_1}{m\ g}$$

Burnout Altitude (Z1) & Peak Altitude (Z2)

$$M\frac{dV}{dt} = T - D - Mg_0 \qquad with\ \ D = C_D\frac{1}{2}\rho V^2 S$$

Flight with Respect to Drag

6.5 Open Rocket Analysis (continued)

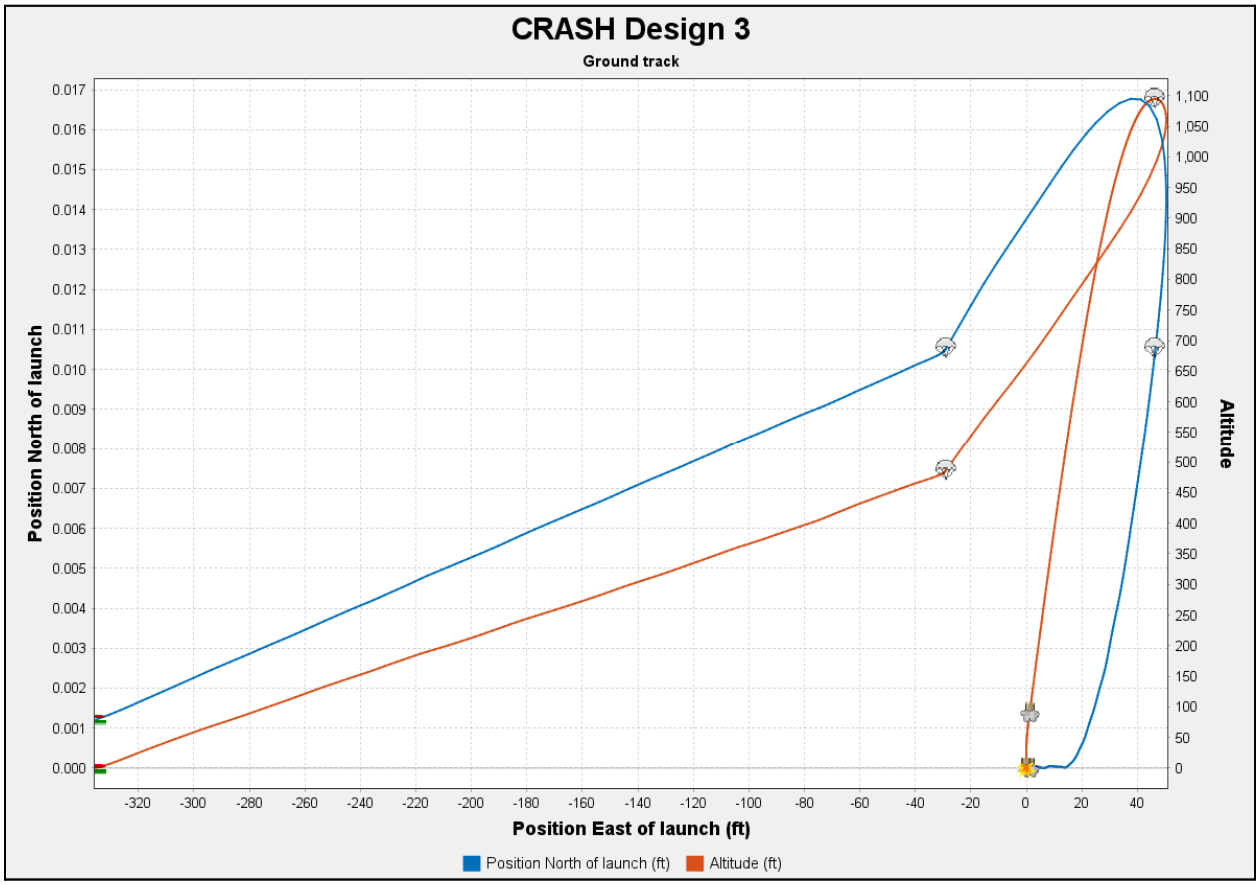


Fig 52. Demonstrates ground location with relative altitude throughout flight while pinpointing stages of recovery

Relevant Equations: [31]

$$\frac{V_x}{V_w} = 1 - e^{-\int_0^z n_\alpha dZ} \approx 1 - e^{-n_\alpha dZ}$$

Horizontal Wind Profile

$$\Delta X_i = V_w t_f$$

Wind Displacement

$$M\frac{dV}{dt} = T - D - Mg_0 \qquad with \;\; D = C_D\frac{1}{2}\rho V^2 S$$

Flight with Respect to Drag

6.5 Open Rocket Analysis (continued)

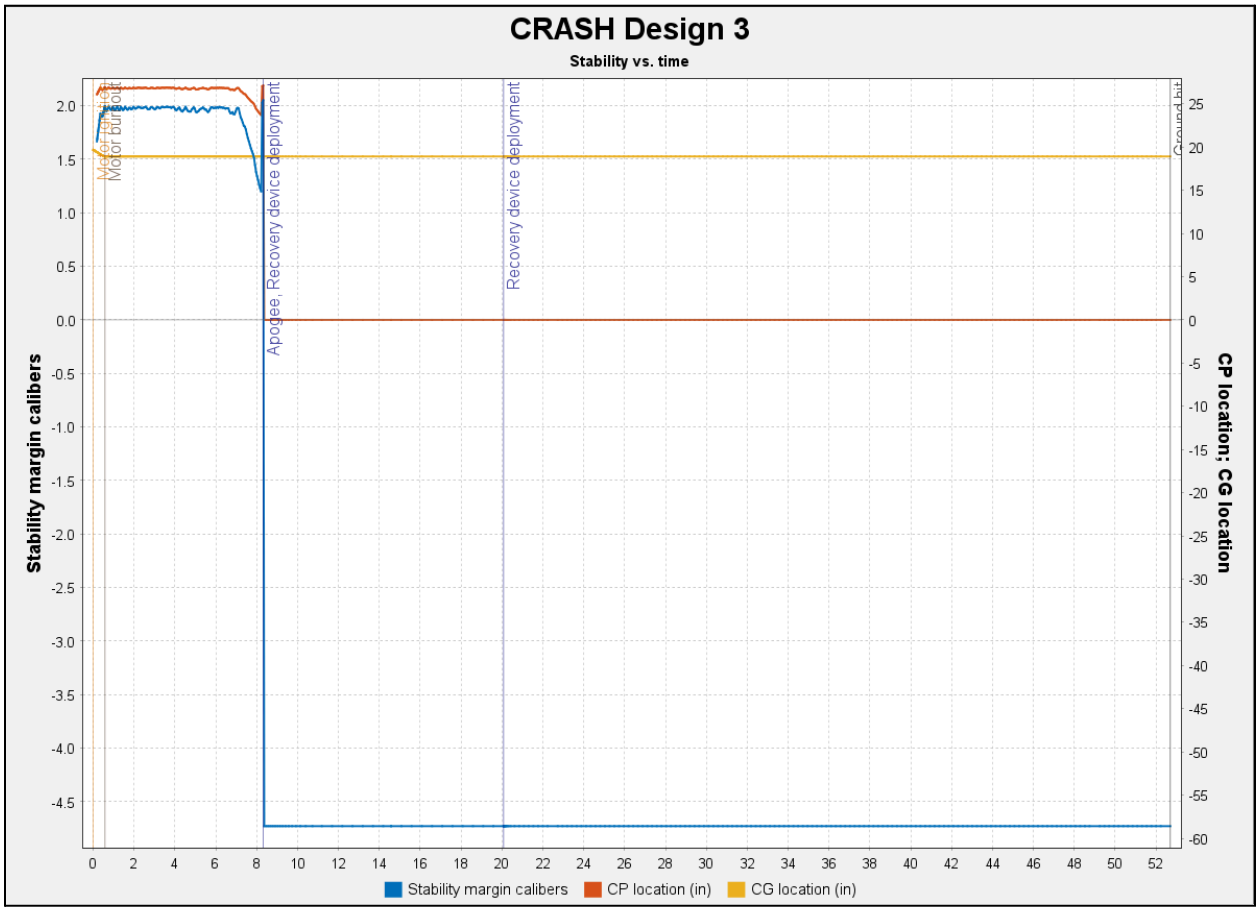


Fig 53. Plots rocket stability during timed flight including centers of pressure and gravity

Relevant Equations:

$$A \cdot cp = [ad](\text{nose}) + [ad](\text{tube}) + [ad](\text{fins})$$

Center of Pressure Calculation [32]

$$W \cdot cg = [wd](\text{nose}) + [wd](\text{recovery}) + [wd](\text{engine}) + \ldots$$

Center of Gravity Calculation [33]

(Cp - Cg) / Rocket Diameter = Stability

Stability Calculation

6.5 Open Rocket Analysis (continued)

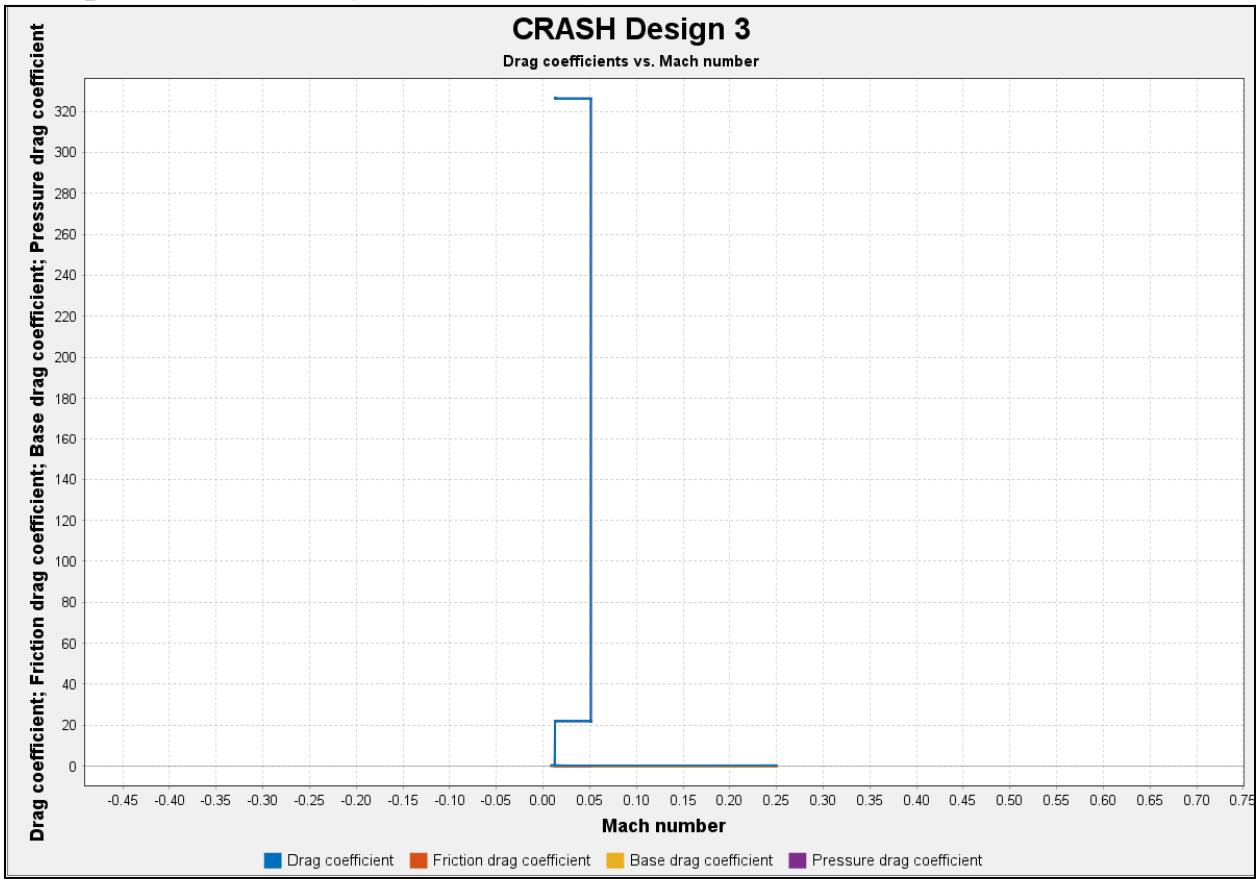


Fig 54. Plots the rocket's drag coefficient(s) relative to Mach number during flight

Relevant Equations:

$$C_{Df} = \frac{2D_f}{\rho V_\infty^2 S_{ref}} = C_f\left(\frac{S_{wet}}{S_{ref}}\right)$$

Friction Drag Coefficient [34]

$$Cd = D / (A * .5 * r * V^2)$$

Drag Coefficient [35]

$$c_p := \frac{\Delta p_{stat}}{p_{dyn,\infty}}$$

Pressure Drag Coefficient [36]

6.5 Open Rocket Analysis (continued)

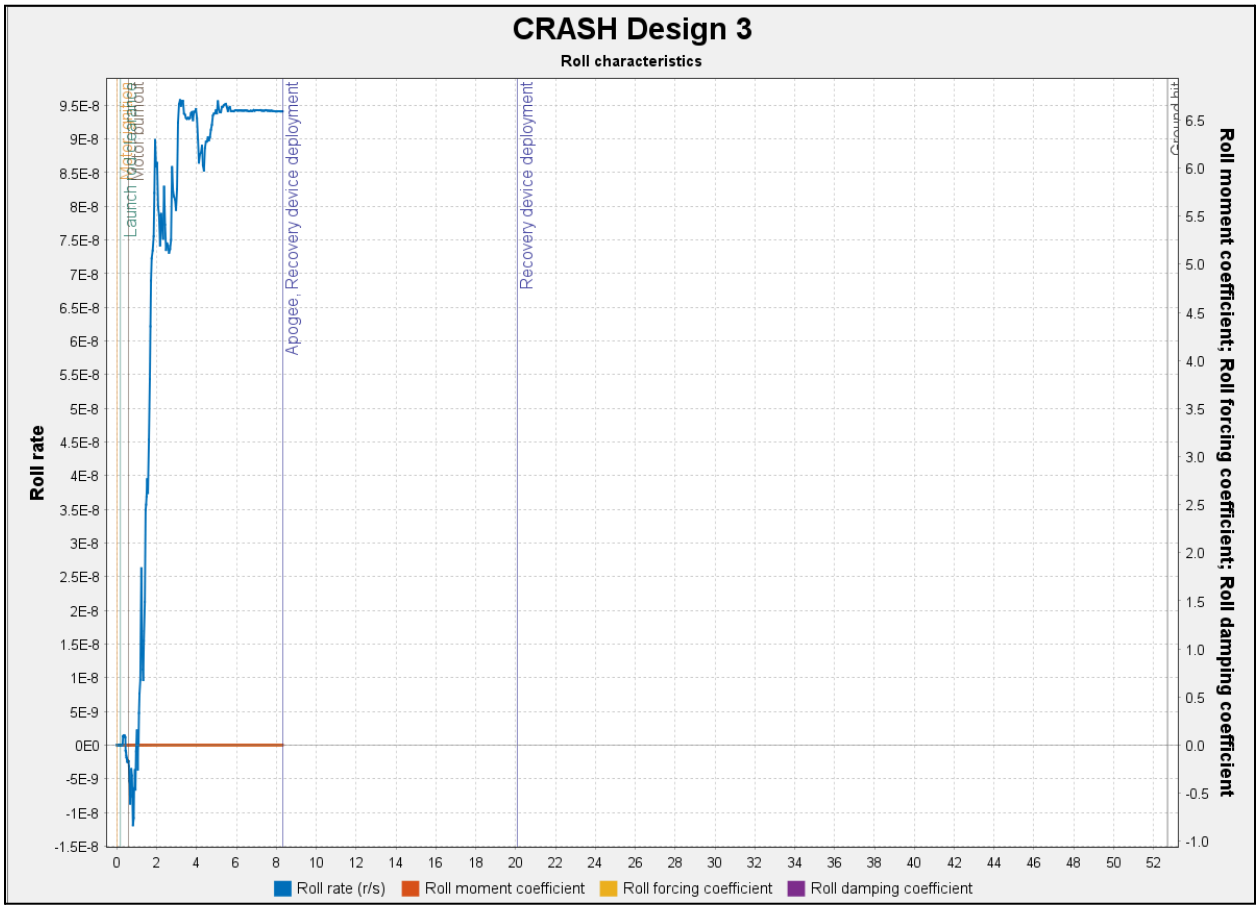


Fig 55. Plots the rocket's roll characteristics relative to timed flight

Relevant Equations:

$$M_f = N(Y_{MA} + r_t)(C_{N\alpha})_1 \delta \bar{q} A_r$$

Roll Force Moment Equation [37]

$$I_X \dot{p} + M_{X,p} p = M_{X,\delta} \delta + M_x + M_j p$$

$I_X \dot{p}$	$M_{X,p} p$	$M_{X,\delta} \delta$	M_x	$M_j p$
Inertial acceleration moment	Aerodynamic damping moment	Aerodynamic driving moment	Auxiliary driving moment (time dependent)	Jet damping moment

Rolling Moment Equation with Time Dependency [38]

6.5 Open Rocket Analysis (continued)

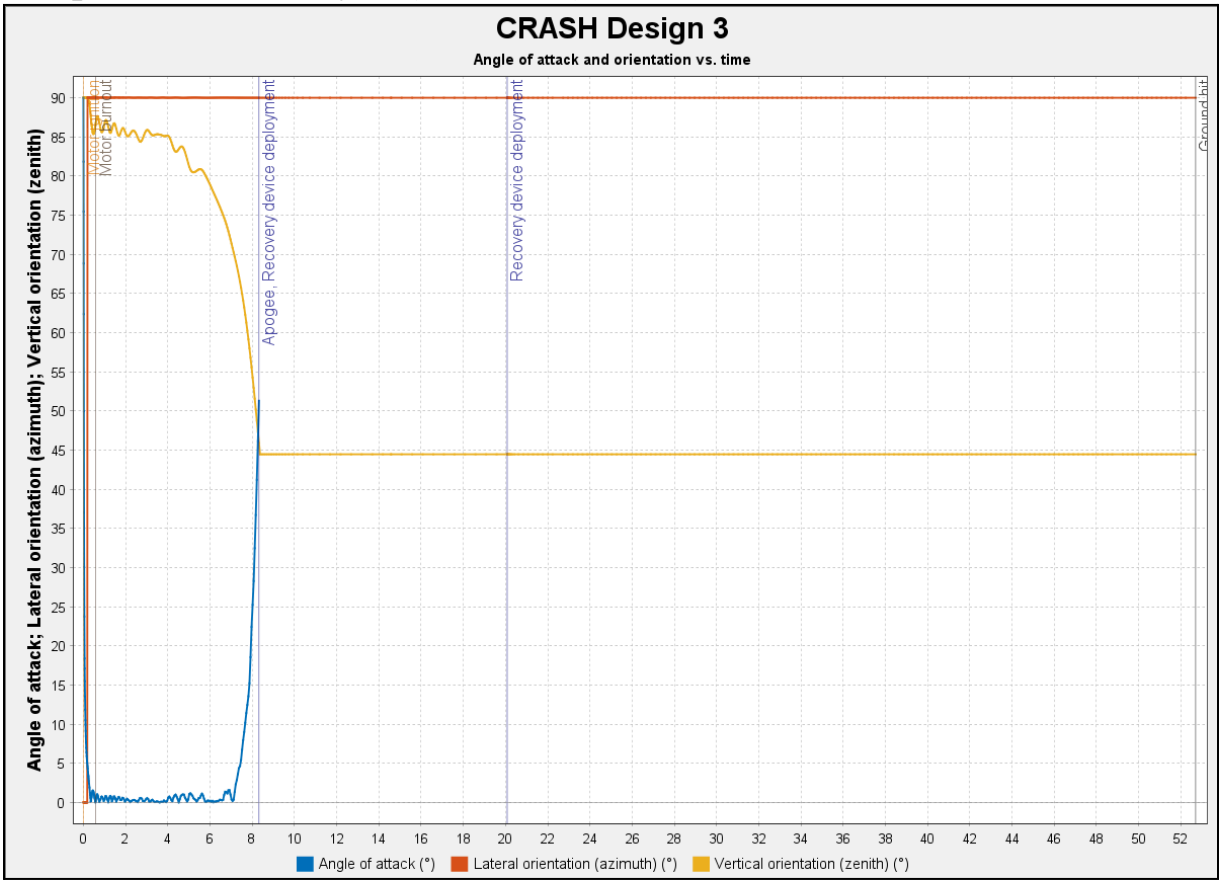


Fig 56. Plots the rocket's angle of attack relative to timed flight

Relevant Equations:

$$\tan\alpha = \frac{V_w - V_x}{V_z}$$

Angle of Attack Relative to Wind & Ground Velocity [39]

$$e_0 = \cos\left(\frac{\varphi}{2}\right)\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\psi}{2}\right) - \sin\left(\frac{\varphi}{2}\right)\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{\psi}{2}\right)$$

$$e_1 = \cos\left(\frac{\varphi}{2}\right)\cos\left(\frac{\psi}{2}\right)\sin\left(\frac{\theta}{2}\right) + \sin\left(\frac{\varphi}{2}\right)\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{\psi}{2}\right)$$

$$e_2 = \cos\left(\frac{\varphi}{2}\right)\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{\psi}{2}\right) - \sin\left(\frac{\varphi}{2}\right)\cos\left(\frac{\psi}{2}\right)\sin\left(\frac{\theta}{2}\right)$$

$$e_3 = \cos\left(\frac{\varphi}{2}\right)\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{\psi}{2}\right) + \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\psi}{2}\right)\sin\left(\frac{\varphi}{2}\right)$$

Euler Equations to Define Rocket Orientation [40]

	Warni...	Name	Configuration	Velocity off rod	Apogee	Velocity at deployment
✔		CRASH Design 3	[HP-H550ST-14]	98.7 ft/s	1095 ft	57.8 ft/s

Optimum delay	Max. velocity	Max. acceleration	Time to apogee	Flight time	Ground hit velocity
7.74 s	279 ft/s	577 ft/s²	8.27 s	52.7 s	18.6 ft/s

Fig 57. Compiles critical launch data

7.0 Flowchart, Gantt Chart, and Major Deliverables

7.1 Flowcharts

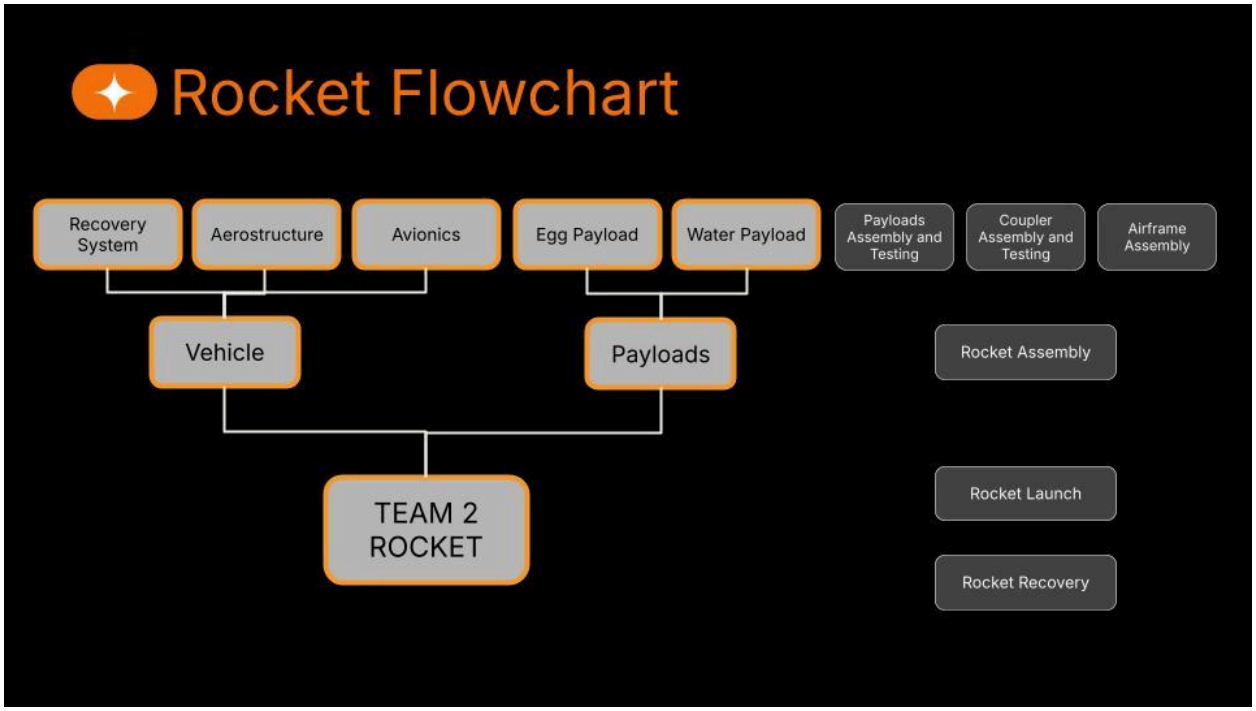


Fig 58. Rocket Flowchart

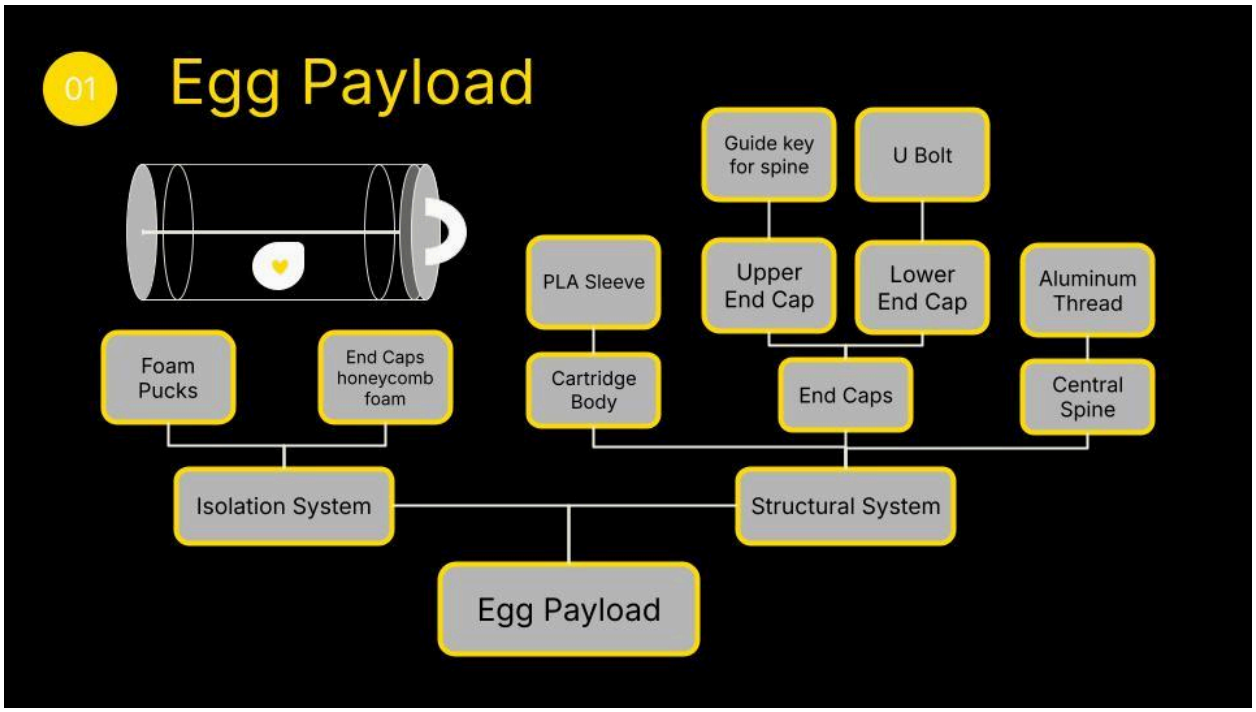


Fig 59. Egg Payload Flowchart

7.1 Flowcharts (continued)

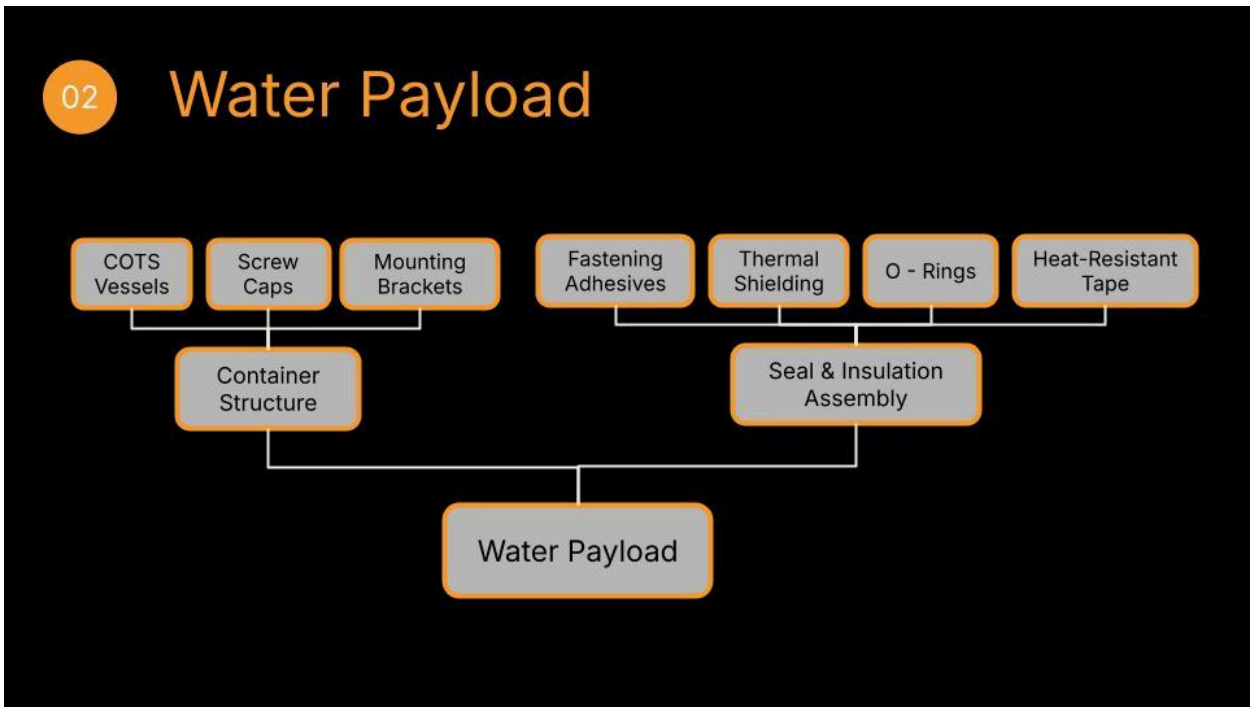


Fig 60. Water Payload Flowchart

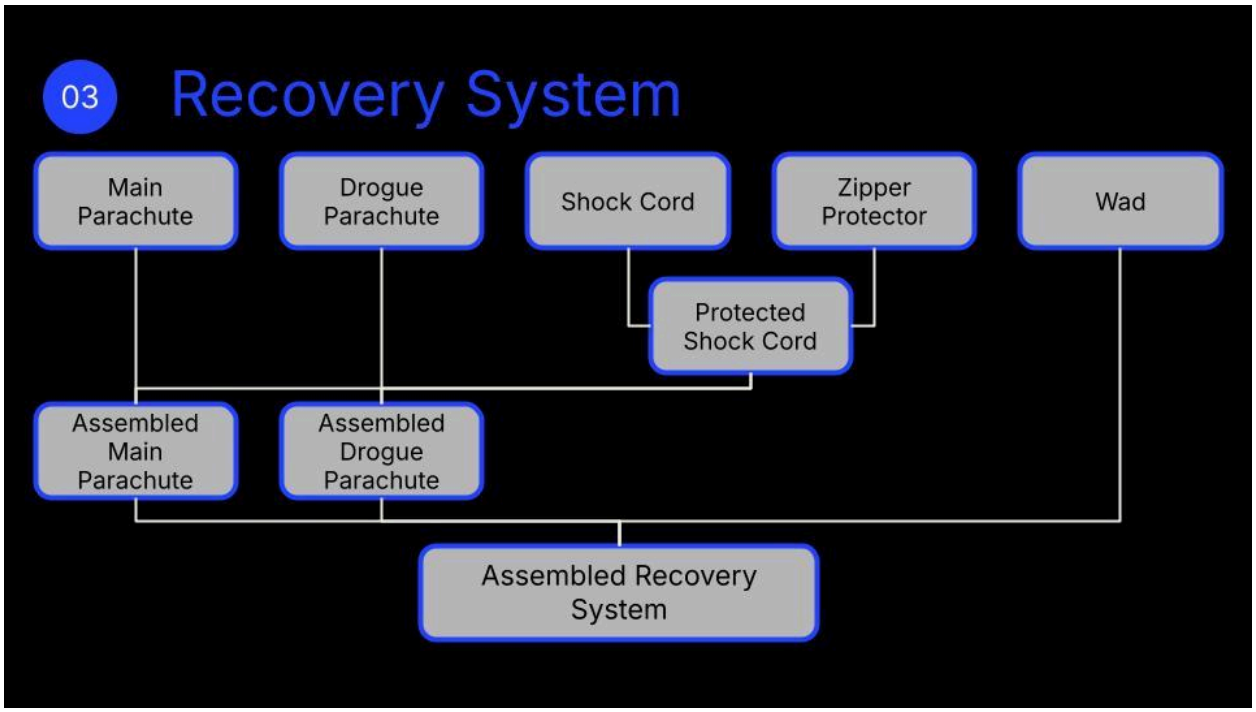


Fig 61. Recovery System Flowchart

7.1 Flowcharts (continued)

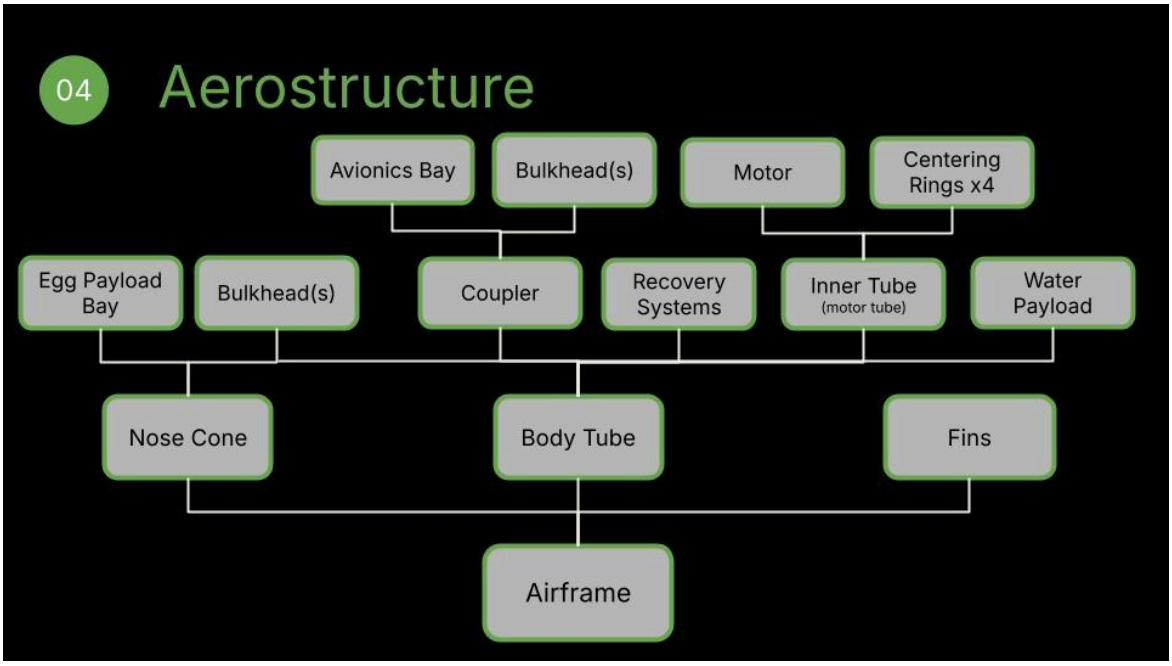


Fig 62. Aerostructure Flowchart

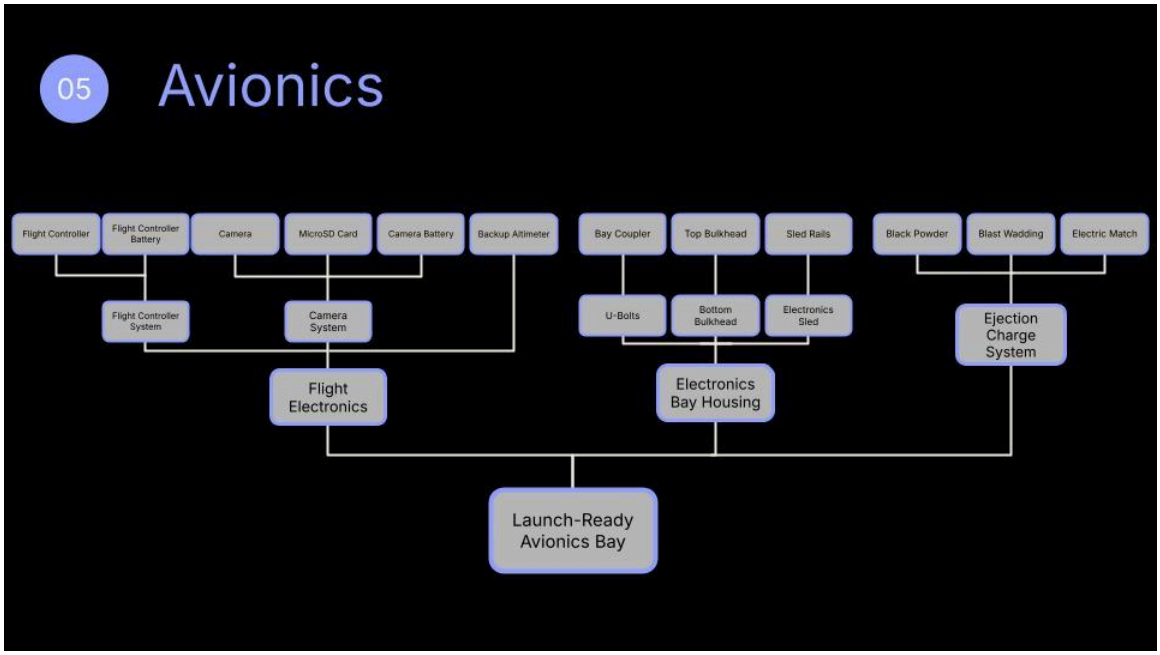


Fig 63. Avionics Flowchart

7.1 Flowcharts (continued)

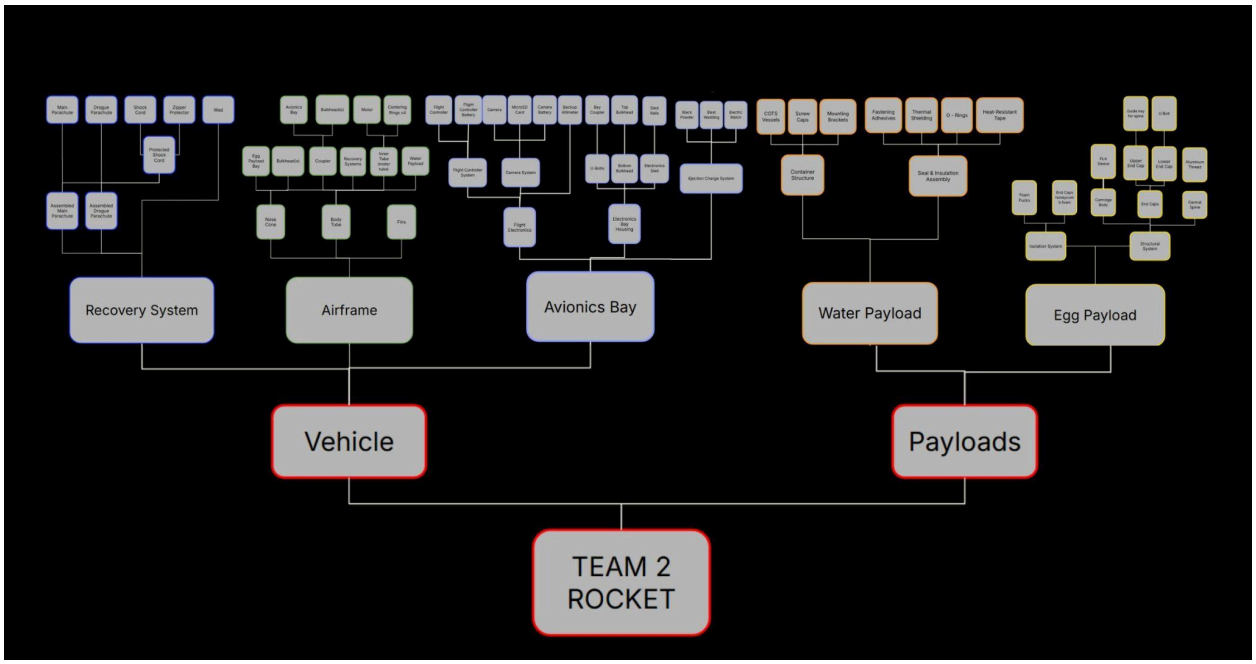


Fig 64. Integrated Rocket Flowchart

7.2 Gantt Chart

Full Gantt Chart

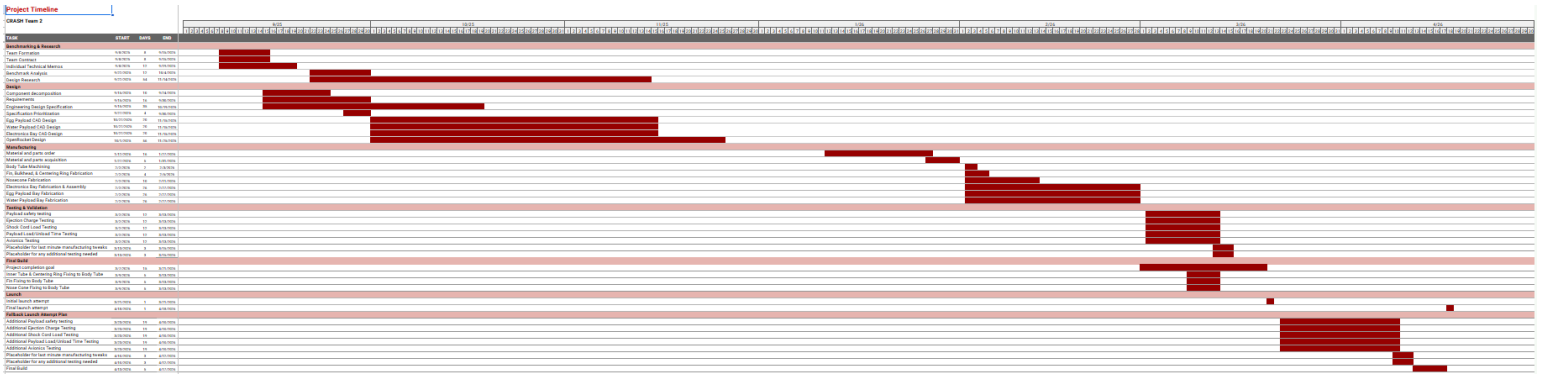


Fig 65. Full Gantt Chart

7.2 Gantt Chart (continued)

TASK	START	DAYS	END
Benchmarking & Research			
Team Formation	9/8/2025	8	9/15/2025
Team Contract	9/8/2025	8	9/15/2025
Individual Technical Memos	9/8/2025	12	9/19/2025
Benchmark Analysis	9/22/2025	12	10/4/2025
Design Research	9/22/2025	54	11/14/2025
Design			
Component decomposition	9/15/2025	10	9/24/2025
Requirements	9/15/2025	16	9/30/2025
Engineering Design Specification	9/15/2025	35	10/19/2025
Specification Prioritization	9/27/2025	4	9/30/2025
Egg Payload CAD Design	10/27/2025	20	11/15/2025
Water Payload CAD Design	10/27/2025	20	11/15/2025
Electronics Bay CAD Design	10/27/2025	20	11/15/2025
OpenRocket Design	10/1/2025	56	11/25/2025
Manufacturing			
Material and parts order	1/12/2026	16	1/27/2026
Material and parts acquisition	1/27/2026	5	1/31/2026
Body Tube Machining	2/2/2026	2	2/3/2026
Fin, Bulkhead, & Centering Ring Fabrication	2/2/2026	4	2/5/2026
Nosecone Fabrication	2/2/2026	10	2/11/2026
Electronics Bay Fabrication & Assembly	2/2/2026	26	2/27/2026

Egg Payload Bay Fabrication	2/2/2026	26	2/27/2026
Water Payload Bay Fabrication	2/2/2026	26	2/27/2026
Testing & Validation			
Payload safety testing	3/2/2026	12	3/13/2026
Ejection Charge Testing	3/2/2026	12	3/13/2026
Shock Cord Load Testing	3/2/2026	12	3/13/2026
Payload Load/Unload Time Testing	3/2/2026	12	3/13/2026
Avionics Testing	3/2/2026	12	3/13/2026
Placeholder for last minute manufacturing tweaks	3/13/2026	3	3/15/2026
Placeholder for any additional testing needed	3/13/2026	3	3/15/2026
Final Build			
Project completion goal	3/7/2026	15	3/21/2026
Inner Tube & Centering Ring Fixing to Body Tube	3/9/2026	5	3/13/2026
Fin Fixing to Body Tube	3/9/2026	5	3/13/2026
Nose Cone Fixing to Body Tube	3/9/2026	5	3/13/2026
Launch			
Initial launch attempt	3/21/2026	1	3/21/2026
Final launch attempt	4/18/2026	1	4/18/2026
Fallback Launch Attempt Plan			
Additional Payload safety testing	3/23/2026	19	4/10/2026
Additional Ejection Charge Testing	3/23/2026	19	4/10/2026
Additional Shock Cord Load Testing	3/23/2026	19	4/10/2026
Additional Payload Load/Unload Time Testing	3/23/2026	19	4/10/2026
Additional Avionics Testing	3/23/2026	19	4/10/2026
Placeholder for last minute manufacturing tweaks	4/10/2026	3	4/12/2026
Placeholder for any additional testing needed	4/10/2026	3	4/12/2026

7.2 Gantt Chart (continued)

Gantt by relevant rows

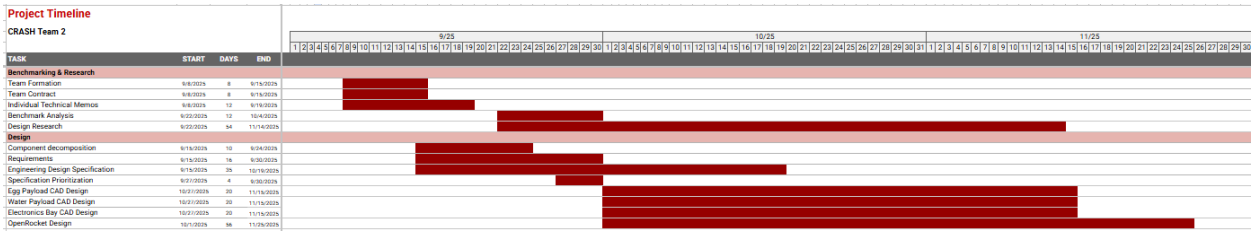


Fig 66. September to November

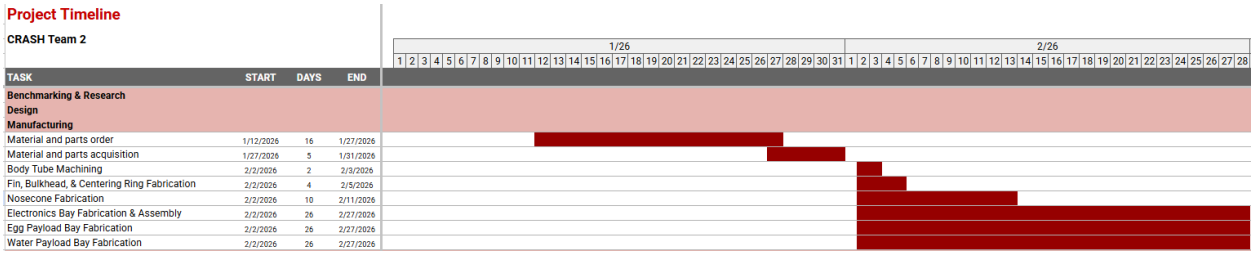


Fig 67. January to February



Fig 68. March to April

7.3 Major deliverables

Table 11. Major Deliverables

TASK	START	DAYS	END
Design			
Egg Payload CAD Design	10/27/2025	20	11/15/2025
Water Payload CAD Design	10/27/2025	20	11/15/2025
Electronics Bay CAD Design	10/27/2025	20	11/15/2025
OpenRocket Design	10/1/2025	56	11/25/2025
Manufacturing			
Electronics Bay Fabrication & Assembly	2/2/2026	26	2/27/2026
Egg Payload Bay Fabrication	2/2/2026	26	2/27/2026
Water Payload Bay Fabrication	2/2/2026	26	2/27/2026
Testing & Validation			
Payload safety testing	3/2/2026	12	3/13/2026
Ejection Charge Testing	3/2/2026	12	3/13/2026
Shock Cord Load Testing	3/2/2026	12	3/13/2026
Payload Load/Unload Time Testing	3/2/2026	12	3/13/2026
Avionics Testing	3/2/2026	12	3/13/2026
Final Build			
Vehicle Fabrication & Assembly	3/9/2026	5	3/13/2026
Launch			
Initial launch attempt	3/21/2026	1	3/21/2026
Final launch attempt	4/18/2026	1	4/18/2026

8.0 System Concept Evaluation Plans

Building a rocket of this size involves several areas where risk can affect flight performance, safety, costliness, and schedule maintenance. Major concerns of risk come from technical issues, components and their material availability, challenges during the manufacturing process and hazards/challenges faced during operation.

8.1 Risk Assessment and Mitigation Overview

One of the biggest risks teams run into is simply waiting for parts. Motors, electronics, recovery components, and aerostructure components can all take longer to arrive than expected. High-power suppliers sometimes have limited inventory or long lead times. When a shipment and supply causes delays, it can halt manufacturing/assembly, cut into testing schedules, and cause design changes at the last minute. To stay ahead of this, it is recommended to order long-lead items as early as they can and line up backup vendors just in case.

During manufacturing it is more than likely that machine users and assemblers cut a part wrong, misalign fins, create weak bonds, or contaminate composite materials which all can seriously affect the rocket's strength and flight performance. Temperature and humidity can impact adhesives and composite layups. Teams manage these risks by working in controlled build spaces, keeping clear documentation, and having team members double-check each other's work during critical steps.

When it comes to flight performance, improper mass distribution, structural weak points, or deviations from the nominal geometry can lead to instability or invalid competition altitude. Small errors in weight, center of gravity, center of pressure, fin alignment, or other components packed into the body tube can cause unexpected flight behavior. To counteract this, OpenRocket simulations must be updated regularly, and built/purchased components should be weighed and inspected to ensure they match design assumptions.

The dual deployment recovery system involves its own risks. Faulty e-matches, unreliable altimeters, battery failure, or incorrect deployment timing can lead to detrimental damage of the launch vehicle. Well-thought-out systems, thorough testing prior to launch, and proper practice of electronics organization help prevent these issues.

Coming down to launch day, weather, range availability, and safety obstacles from surrounding launches can delay launches or force launch scrubs. Weather changes, especially including wind changes, can heavily affect stability and flight performance. There must be a plan for multiple launch opportunities and adjust flexible schedules.

Safety-wise, handling motors and other flight hardware carries inherent safety hazards. Failing to follow certification rules, usage requirements, or NAR launch-site procedures can halt the project and pose danger to viewers. Proper training, documentation, and communication with launch site officials reduce these risks.

8.2 Complete Assembly Point Analysis

The competition awards points for each challenge faced. Our current plan includes potentially nine eggs, a camera, and an auditory tracker. However risks like the nine egg capsule being insufficient at protection can force us to switch to the two egg backup, the potential high cost of a camera can cause us to abandon that recording challenge, and the use of an auditory tracker may not meet the competition definition of a tracker. The add up of the current plan, assuming all goes well the team would earn 350 points. Then in the scenario where the three risks mentioned prior come to fruition, the team would still earn 180 points.

Table 12. Points Analysis

Challenge	**Points with currentPlan**	**With minimum**
9 Eggs	70	0
Camera Recording	50	0
Water Ejection	15	0
Ejection	15	15
Parachute	50	50
Dual Deployment	100	100
Tracker	50	0
Total	350	165

8.3 Test and Evaluation Strategy

Testing of the fully assembled rocket is not recommended because of the risk of being unable to replace or repair the rocket before competition day due to physically losing it or rocket failures. Instead each subsystem will be tested and evaluated separately to confirm their functionality. The egg payload will test two versions of the egg capsule by the traditional competition method of dropping from a high point. The water payload will test its releasing mechanism to confirm the ability to release the water, and evaluate through simulation if there is a risk of damaging the container. Both payloads will have practice runs for loading and unloading to ensure they are within 5 minutes. There will be a ground test for ejection charges for the main and drogue parachute, shock cord load testing, and avionics function testing.

9.0 Failure Modes and Effects Analysis (FMEA)

Potential failure modes for the vehicle and the effects of such failures were analyzed through first assessing the likelihood of each mode, then assessing the consequence of the effect for each mode, and finally assessing the detectability of each mode. The three categories for assessment: likelihood, consequence, and detectability, were assigned a rating scale to determine their relative

9.1 Egg Capsule FMEA

There are many possible failures for the egg capsule, falling into two categories: failure to protect the eggs and failure to load and unload within five minutes. Failure to protect the eggs can be caused by failure to dampen vibrations, failure to dampen impact forces, or eggs being improperly secured in the capsule. Failure to unload under five minutes can be caused by the capsule being stuck in the rocket, the capsule not opening, the loading of the eggs taking too long to stack, or the unloading of the eggs taking too long to unstack. The potential failures for the egg vessel are not hazardous and are straightforward to mitigate. Testing will show whether additional cushioning is needed for the eggs or if more securement is required. If necessary, another rod can be added, or small notches can be incorporated into the walls to better align the pucks. Issues such as the capsule getting stuck in the rocket or the capsule not opening can also be identified through testing; if needed, the capsule can simply be sanded for a better fit. The stacking and unstacking of the pucks and eggs can be practiced to consistently remain under five minutes.

Table 13. Egg Capsule Risk Analysis

Failure Modes	Effects	Likelihood (L)	Consequence (C)	Hazard Score (HS)	Detection (D)	RPN (LxCxD)	Mitigation
		Unlikely(1) -Likely(5)	Low(1)- High(5)		Easy(1)- Hard(5)	Min(1)-Max(125)	
Failure to protect the eggs							
Failure to dampen vibrations	Can cause cracks	3	5	0	4	60	Testing to check dampening is effictient
Failure to dampen impact forces	Can cause cracks	3	5	0	4	60	Testing to check dampening is efficient
Faliure to properly secure eggs	Can cause cracks	2	5	0	5	50	Test if egg pucks will rotate in the capsule or using two rods in the capsule
Failure to unload in under five minutes							
Capsule stuck in rocket	Can't load/ unload under 5 minutes	3	5	0	1	15	Test how the capsule fits in the rocket and practice loading and unloading
Capsule not opening	Can't load/ unload under 5	1	5	0	1	5	Repeated practice of opening and closing

Failure Modes	Effects	Likelihood (L)	Consequence (C)	Hazard Score (HS)	Detection (D)	RPN (LxCxD)	Mitigation
	minutes						the capsule
Egg pucks take too long to stack	Can't load under 5 minutes	1	5	0	1	5	Practice stacking of pucks
Egg pucks take too long to unstack	Can't unload under 5 minutes	1	5	0	1	5	Practice the unstacking of pucks

9.2 Water Payload Assembly

The water payload subassembly of the vehicle was analyzed using a failure modes and effects analysis to identify potential failure modes, their effects on the mission, and appropriate mitigations. The analysis covers all major subsystems of the subassembly, the HDPE vessels, the sealing system, the supporting structure, and a potential water deployment mechanism. It is important to note that any unintentional water leakage during the flight results in a disqualification from the competition, therefore all failure modes leading to water loss are treated with the highest severity. The FMEA scoring uses a 1-5 scaling for all three of the categories, likelihood, consequence, and detection, with a higher number denoting a worse performance for the respective parameter. A general risk priority number was then calculated from these parameters to gauge the relative severity for each risk. The table below summarizes the findings from the FMEA. Each risk and the justification behind the RPN will be discussed, along with the mitigation strategies.

Table 14. Water Payload Risk Analysis

Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
		Unlikely(1)-Likely(5)	Low(1)-High(5)	Easy(1)-Hard(5)	Min(1)-Max(125)	
Vessel Structural Damage	Leakage resulting in disqualification	1	5	5	25	Ensure vessel can withstand forces with margin
Vessel Bursting	Leakage resulting in disqualification	1	5	5	25	Ensure vessel can withstand forces with margin
Vessel Thermal Damage	Structural integrity of the vessel compromised	1	3	4	12	Ensure vessel is shielded from direct thermal loads
Vessel Unplanned Detachment	Loose vessel which could become a projectile	2	4	5	40	Ensure vessel is properly seated in place
Caps Improperly Sealed	Leakage resulting in disqualification	2	5	3	30	Properly inspect sealing prior to launch
Caps Loosening due to Vibration	Leakage resulting in disqualification	2	5	5	50	Ensure caps are carefully and properly installed
Caps damaged due to Over-tightening	Leakage resulting in disqualification	2	5	4	40	Ensure caps are carefully and properly installed
Supports Structural	Loose vessel which could become a	1	4	5	20	Ensure supports can withstand forces with margin

Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
Damage	projectile					
Supports Unplanned Detachment	Loose vessel which could become a projectile	1	4	5	20	Ensure supports can withstand forces with margin
Supports Improperly Manufactured	Improper fittings exacerbating other issues	3	3	1	9	Ensure the supports are manufactured within specified margins

9.2.1 FMEA Discussion

Bottles

The HDPE bottles shall serve as the containment method for the water payload. A critical failure mode for the bottles is structural leakage, such as cracks. However the likelihood of the bottle cracking or suffering abrasion under normal conditions is very low, as the bottle material is strong [41], however there is the potential for manufacturing defects to induce cracks under stress from launch [42]. Obvious structural damage should be apparent upon basic inspection, however a manufacturing defect may not be detectable at all.

Another failure mode is the vessels bursting during specific high stress periods, such as takeoff, or impact on landing. The vessels being the failure point for such a case is extremely unlikely, but not impossible. As this failure would be sudden, detection is not possible. To mitigate such an event, any circumstances that could pressurize the bottles shall be avoided.

With the vessels in close proximity to the motor, a concern is the bottles experiencing thermal distortion from the heat generated by the motor burn. The material of the bottle was selected taking this into consideration [43], resulting in the likelihood of this failure mode being low, and designing the module such that the bottles are not in direct contact with the motor assembly shall be important to mitigate this risk. As with all modes related to the vessels, detection is not possible prior to flight.

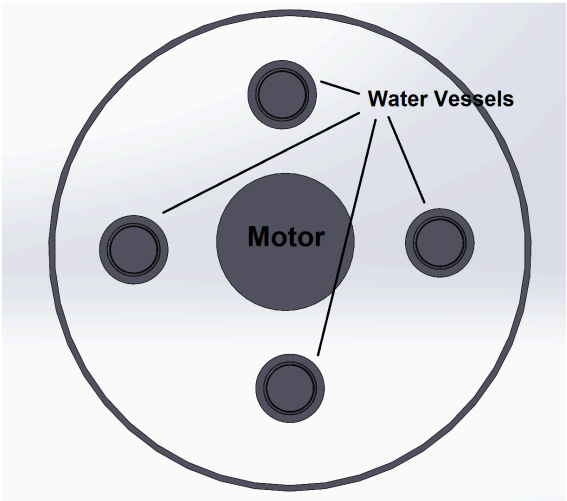


Fig 69. Cross-sectional View of Lower Module

9.2.1 FMEA Discussion (continued)

A final failure mode for the vessels is becoming dislodged from their supporting structure. A loose bottle has the potential to become a projectile if it were to escape the rocket body. The vessel must stay removable from the vehicle, so complete attachment is not an option. It shall be important to firmly secure the vessels to the frame prior to takeoff. Detection is possible, through thoroughly inspecting the mounting of each vessel before the vehicle is launched.

Sealing System:

Each vessel shall be sealed with a screw-on cap, with a secondary, redundant, sealing method. The cap and seal are critical to preventing leaks, and would be the most likely point of failure for the subassembly. The most preventable failure mode for this component is improperly installing the cap. A cap which is not fully threaded down will result in ineffective sealing, directly causing a leak [44]. The likelihood for this to occur is hopefully low, as with proper detection measures it will be caught before it is an issue.

A failure mode which could be exacerbated by the previously discussed mode, is the cap loosening due to vibrations during flight [45]. Due to the intensity of the forces experienced during flight, even a slight loosening of the sealing system could result in water leaking past the threads. Since the overall flight duration is not very long, the general likelihood of a vibration sustained for long enough to loosen the cap is low. Mitigation strategies for this mode are the same as before, with proper inspection and ensuring the sealing is properly fitted before flight.

A potential user error related failure mode could be induced by over-tightening the cap, damaging the neck of the bottle, or the cap itself. Such a crack might not be apparent until the bottle is subjected to intense flight conditions. The likelihood of this failure mode is low, as materials were selected with durability in mind. Mitigation is the same as before, with care being taken to ensure the sealing is properly fitted.

Supporting Structure:

The support structure shall consist of a series of rings which support the vessels around the motor. This structure's primary purpose is to secure the bottles, as well as the motor itself, in place during flight. A structural failure of this support system, such as a ring cracking or snapping, is a potential failure mode. If one of the rings were to break under the stress of flight, one or more vessels could come loose inside the body tube. Due to the location of the vessels, the consequence for this would not be very high, as they could not damage anything critical, however in the event they somehow escape from the vehicle during flight, the vessels could become a projectile. Therefore, robust construction of the lower module will ensure the vessels cannot escape the vehicle, if the rings were to crack.

A similar failure mode would be the mounting hardware dislodging completely during flight. Whether from vibration or other forces, if the structure completely becomes dislodged from the inner walls of the body tube, it could cause the motor to shift, which could be catastrophic for the flight trajectory of the vehicle. Mitigation of this failure mode will consist of using specific hardware to fasten the support structure to the body of the rocket.

9.2.1 FMEA Discussion (continued)

Manufacturing errors have the potential to cause a misalignment or fitting issue which leads to a failure mode. If the cutouts are off-size, they could either be too tight or too loose for the water vessels. A tight fit could put additional stresses on the entire subassembly, while a loose fitting would allow the bottles to rattle. Either case has the potential to cause critical failures. The likelihood is moderate, as small discrepancies in measurements are not uncommon. Detection would be fairly simple, as a basic inspection of the components prior to assembly would reveal any potential misalignment errors. To mitigate this risk, a thin liner could be used to accommodate slight size variations, while ensuring the vessels fit snugly into place.

9.3 Propulsion and Recovery FMEA

Table 15. Propulsion Risk Analysis

Item/Component	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
			Unlikely(1)-Likely(5)	Low(1)-High(5)	Easy(1)-Hard(5)	Min(1)-Max(125)	
Motor:	Motor explodes	Invalid competition entry	1	5	1	5	N/A
	Ejection charge fails	Drogue does not deploy	2	4	1	8	Backup ejection charge for drogue
	Motor case fails	Rocket may explode	1	4	5	20	N/A
	Motor fails to ignite	Rocket does not launch	1	5	1	5	Ensure igniter isn't damaged
	Motor falls out	Rocket loses propulsion	1	5	5	25	Ensure retaining spring and centering rings are undamaged
	Motor wall burn-through	Heat may burn through rocket tube	1	3	5	15	N/A
	Nozzle blow-out	Motor loses thrust	1	5	5	25	N/A
	Premature ejection	Target altitude not reached	1	5	5	25	N/A
Shock Chord:	Shock cord snaps	Parachute loss	2	5	1	10	Ensure shock cord isn't damaged before launch
	Zippering	Body tube damage	3	1	1	3	Ensure zipper

Item/Component	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
							protectors are placed correctly
	Shock cord gets tangled	Shock cord doesn't reach full length	2	2	3	12	Ensure shock cord is folded correctly
	Shock cord gets burned	Possible parachute loss	1	4	3	12	Ensure wad is low over motor
Parachute:	Parachute tears	Rocket can't fully slow down	3	5	4	60	Ensure parachute is intact before launch
	Parachute doesn't fully deploy	Rocket can't fully slow down	3	5	4	60	Ensure parachute lines aren't tangled before launch
	Burned parachute	Rocket can't fully slow down	2	3	5	30	Ensure wad fully covers parachute
	Parachute lines tangled	Rocket can't fully slow down	2	3	3	18	Ensure parachute lines aren't tangled before launch
	Improper packing	Parachute may not deploy	1	3	1	3	Ensure parachute is packed correctly
	Improper folding	Parachute may not deploy correct	1	3	1	3	Ensure parachute is folded correctly
	Wad plugging	Parachute may not deploy	1	5	2	10	Ensure wad can move freely in body tube
	Broken parachute lines	Possible parachute loss	2	4	1	8	Ensure parachute lines are intact before launch
	Inadequate parachute size	Rocket can't fully slow down	2	5	4	40	Test with simulators to attain correct parachute size
	Early deployment	Rocket drifts too far	4	5	5	100	Ensure altimeter is set correctly
	Late deployment	Rocket can't fully slow down	4	5	5	100	Ensure altimeter is set correctly

9.4 Aerostructure System FMEA

Table 16. Aerostructures Risk Analysis

Component	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
			Unlikely(1)-Likely(5)	Low(1)-High (5)	Easy(1)-Hard (5)	Min(1)-Max (125)	
Body Tube:	Body tube cracks	Could prevent reuse or reduce stability	2	3	2	12	Handling the rocket carefully and controlling descent velocity
	Body tube get burnt through	Cause a hole leading to decreased stability and prevent reuse	2	3	1	6	Containing motor burn or carefully positioning motor to avoid burning
	Body tube disconnects early	Invalid competition entry as altitude will not be reached	1	5	1	5	Ensure proper connection
	Body tube doesn't separate	Safety violation as rocket will have no decreased velocity	1	5	1	5	Ensure proper connection
	Body tube fails to bond with centering rings	Motor tube disconnects leading to safety hazard	1	5	3	15	Ensure proper connection & build up epoxy
	Body tube fails to properly bond with nose cone	Rocket becomes unstable and crashes	1	5	3	15	Ensure proper connection & build up epoxy
	Rail pins create crack propagation	Could prevent reuse depending on crack size	3	3	2	18	Carfully screw into rocket
	Rail pins tear through body	Tear could prevent reuse	1	3	1	3	Ensure proper connection

Component	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
							and screw is not fretting
	Body tube buckles	Instability; rocket crashes	1	5	1	5	Carefully handle rocket and avoid coming in contact with liquid
Nose Cone:	Nose cone disconnects during flight	Instability; rocket crashes	1	5	1	5	Ensure proper connection
	Nose cone buckles during flight	Instability; rocket crashes	1	5	2	10	Ensure material is stable & has a strong bond
	Nose cone cracks	Could prevent reuse or reduce stability	2	3	2	12	Handling the rocket carefully and controlling descent velocity
	Nose cone dents	Could prevent reuse or reduce stability	3	3	1	9	Handling the rocket carefully and controlling descent velocity
Fins:	Fins crack before/during ignition	Could lead to additional cracking upon landing; prevent reuse	4	2	2	16	Handling the rocket carefully and controlling descent velocity
	Fins chip before/during ignition	Could affect flight performance/stability	2	4	1	8	Handling the rocket carefully and controlling descent velocity
	Fins break off before/during ignition	Extreme hazard/safety concern	1	5	1	5	Handling the rocket carefully and controlling

Component	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
							descent velocity
	Fins break off during landing	Prevents reuse entirely; defies competition rules	2	5	1	10	Handling the rocket carefully and controlling descent velocity
	Fins crack during landing	Compromises fin integrity	3	1	2	6	Handling the rocket carefully and controlling descent velocity
	Fins chip during landing	Potentially prevents reuse depending on size	4	4	1	16	Handling the rocket carefully and controlling descent velocity
	Fins fail to bond to the body tube	Fins fall off, leading to instability	1	5	1	5	Ensure proper connection & build up epoxy
	Fin material shears	Compromises fin integrity	1	2	2	4	Delicate Manufacturing & material handling
	Fins burn during ignition	Compromises fin integrity	3	3	1	9	Containing motor burn or carefully positioning fins to avoid burning
	Fins misaligned for launch	Could lead to instability	2	5	3	30	Ensure proper alignment, connection, and epoxy build-up
	Uneven fin surfaces	Compromises fin integrity and air flow	1	2	3	6	Delicate Manufacturing & material handling
	Fins flutter during flight	Causes instability	2	5	2	20	Ensure proper alignment, connection,

Component	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
							and epoxy build-up
	Fins bend during flight	Causes instability	2	5	3	30	Ensure material is stable & has a strong bond
Bulkheads:	Bulkhead fails to bond with body tube	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
	Bulkhead fails to bond with coupler	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
	Bulkhead fails to bond with payload bay(s)	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
	Bulkhead cracks	Compromises bulkhead integrity	2	1	2	4	Handling the rocket carefully and controlling descent velocity
	Bulkhead chips	Compromises bulkhead integrity	2	4	1	8	Handling the rocket carefully and controlling descent velocity
	Bulkhead disconnects/bursts	Rocket integrity compromised causing launch failure	2	5	1	10	Ensure proper alignment, connection, and epoxy build-up
	Fretting observed between eye bolt & bulkhead	Possible disconnect of eye bolt and therefore recovery system failure	2	5	2	20	Delicate Manufacturing & material handling

Component	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
	Eye bolt tears out of bulkhead	Recovery system failure	2	5	1	10	Ensure material is stable & has a strong bond
	Bulkhead burns	Compromises bulkhead integrity	1	2	1	2	Containing motor burn/boom pool
Centering Rings:	Centering ring(s) fail to bond with body tube	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
	Centering ring(s) fail to bond with motor tube	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
	Centering ring(s) crack	Compromises centering ring integrity	2	1	2	4	Handling the rocket carefully and controlling descent velocity
	Centering ring(s) chip	Compromises centering ring integrity	2	4	1	8	Handling the rocket carefully and controlling descent velocity
	Centering ring(s) disconnect/burst	Rocket integrity compromised causing launch failure	2	5	1	10	Handling the rocket carefully and controlling descent velocity
	Fretting observed between eye bolt & centering ring	Possible disconnect of eye bolt and therefore recovery system failure	2	5	2	20	Ensure proper alignment, connection, and epoxy build-up
	Eye bolt tears out of Centering ring	Recovery system failure	2	5	1	10	Ensure material is stable & has a strong bond

Component	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
	Centering ring(s) burn	Compromises centering ring integrity	1	2	1	2	Containing motor burn/boom pool
Motor Retainer:	Female side of retainer disconnects from bottom centering ring	Lack of motor security; Possible motor loss/instability	2	5	1	10	Ensure proper connection & build up epoxy
	Male side of retainer unscrews during launch	Lack of motor security; Possible motor loss/instability	1	5	1	5	Ensure proper connection
	Retainer cracks	Integrity compromised	2	2	2	8	Handling the rocket carefully and controlling descent velocity
	Retainer chips	Integrity compromised	1	3	1	3	Handling the rocket carefully and controlling descent velocity
	Retainer burns and/or melts	Shape failure and possibly unable to hold motor	2	4	1	8	Containing motor burn
	Retainer bursts	Rocket integrity compromised causing launch failure	2	5	1	10	Handling the rocket carefully and controlling descent velocity

9.5 Avionics FMEA

This section will cover the failure mode analysis of the avionics section of the vehicle. Based on the current configuration, the avionics bay was split into the proper bay capsule itself, and the electronics that are contained within there. These are circumstances in either category where there are failures in the function of the avionics component of the vehicle based on environmental factors or functional failure. These were then assigned values based on likelihood, consequence, and detectability. From this, we can quantify the gravity of each failure condition,

9.5 Avionics FMEA (continued)

and then generate mitigation techniques through either improved build techniques or design choices.

Table 17. Avionics Risk Analysis

Item/Component	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
			Unlikely(1) - Likely(5)	Low(1) - High(5)	Easy(1) - Hard(5)	Min(1) - Max(125)	
Avionics Bay:	Avionics bay integrity failure on ground hit	High potential for electronics damage	2	4	2	16	Minimize ground-hit velocity
	Avionics bay integrity failure on recovery deployment	High potential for electronics damage	2	4	2	16	Minimize ground-hit velocity
	U-Bolt shear out	Recovery failure; potential freefall	3	5	3	45	Use thick bulkheads and separate at apogee
	E-sled improperly secured	Probable component deseating or additional vibration stress	2	3	3	18	Ensure all parts are within proper tolerance and torque
	Components improperly secured to e-sled	Probable component deseating or additional vibration stress	2	3	3	18	Ensure all parts are within proper tolerance and torque
Avionics Electronics:	Premature battery depletion	Complete loss of FC and camera functions	2	5	3	30	Size battery (mAh rating) for long pad operations
	Flight controller dropout (malfunction)	Loss of reliable flight event control and data logging	1	5	2	10	Ensure proper handling and operation procedure is followed
	Screw terminal wire de-seating	Loss of power to electrical functions mid-flight	3	5	2	30	Secure all wires with items like zipties
	Camera dropout (malfunction)	Loss of recorded video (bonus points)	2	2	2	8	Ensure large enough SSD storage and operation

Item/Component	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
	Camera fogging obstruction	Loss of video quality	1	1	4	4	Minimize temperature change and visually inspect
	Electrical component condensation failure (Humidity)	Potential loss of reliable flight event control and data logging	1	5	4	20	Minimize temperature change to prevent condensation
	Electrical component overheat failure (Temperature)	Potential loss of reliable flight event control and data logging	1	5	4	20	Ensure proper component venting and spacing
	Black powder ignition failure	Failure of primary recovery deployment	2	5	4	40	Keep powder dry and securely packed
	Audio recovery device failure	Loss of ground recovery aide	3	2	2	12	Only enable device as close to before launch as possible
	Unintentional arming switch toggle	Potential premature recovery event	1	4	1	4	Ensure arming switch is secured in either operating position

10.0 Significant Accomplishments and Future Work

Major accomplishments for the project this semester include the successful formation and development of a cohesive team and the evaluation of industry standards that parallel our project. Early efforts focused on identifying the project's needs and deriving clear requirements for the team's design to follow. Building on this foundation, the team brainstormed solutions for each subsystem and their correlating challenges, limited by cost and weight. Generating concepts for the egg payload, water payload, and vehicle configuration. A great deal of time went into understanding the physical behaviour and strength characteristics of eggs, updating the original assumptions of a very low newton limit to a much more reasonable requirement of below 250N. The team also adapted to the updated rules for the water portion of the challenge, from a safe recovery approach to an in-flight release requirement. These two changes required the team to rethink strategies and explore new designs. The team has begun preliminary engineering analysis to assess the feasibility of the current concepts and to refine the most promising designs. Potential failure modes and risks are being evaluated and documented to ensure that mitigation strategies are applied.

The team is positioned to move into more detailed design work, organized testing, and successful launch, recovery, and completion of the rocket challenge. The next phase will begin with a review of each subsystem to ensure proper integration, followed by the finalization of each subsystem design element. Once finalized, the team will compile a list of parts and submit a purchasing order. After materials arrive, each subsystem will begin fabrication and testing. The team will conduct practice for the loading and unloading of the egg capsule, and the loading of the water container to ensure the team is within the 5 minute requirement. The final step will be the full assembly of the rocket and launching on competition day.

11.0 Conclusions and Recommendations

As the team continues to gather information and refine its understanding of the project new possibilities and design directions have emerged. Throughout the semester, the team has strengthened its knowledge of motors, rocket configurations, egg strength characteristics, and innovative approaches for the water-container expulsion system. Insights into egg architecture have enabled the development of a protection strategy that uses full foam coverage to maximize safety while minimizing both foam mass and wasted volume, allowing a more efficient use of the payload cabin. The water container has been simplified, focusing on the essential mechanisms required to achieve in flight release of the water. Minimizing mass and complexity, broken down to the cap and vessel being separated by a force already in the system, the shock cord. Moving forward the most effective solutions will likely stem from informed analyses and a clear focus of the project requirements. The team is prepared to progress into more detailed development of current designs, testing, and future mission success.

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Appendices

Appendix A: ABET Design Competence Matrices and Topic Criticality Matrix

AERONAUTICAL	Critical / Main Contributor	Strong Contributor	Necessary but not a primary contributor	Necessary but only a minor contributor	Only a passing reference	Not Included in this Project	Report Section
Aerodynamics	X						6.5
Aerospace Materials	X						3.4 & 5.6
Flight Mechanics					X		6.5
Propulsion	X						3.3 & 5.3
Stability & Control		X					6.5
Structures		X					3.4, 5.1, 5.2, & 5.6

ASTRONAUTICAL	Critical / Main Contributor	Strong Contributor	Necessary but not a primary contributor	Necessary but only a minor contributor	Only a passing reference	Not Included in this Project	Report Section
Aerospace Materials	X						3.4 & 5.6
Attitude Determination & Control			X				6.5
Orbital Mechanics					X		6.5
Rocket Propulsion	X						3.3 & 5.3
Space Environment					X		6.5
Space Structures				X			3.4 & 5.6
Telecommunications	X						5.5 & 6.4

MECHANICAL	Critical / Main Contributor	Strong Contributor	Necessary but not a primary contributor	Necessary but only a minor contributor	Only a passing reference	Not Included in this Project	Report Section
Thermal-Fluid Energy Systems				X			3.3 & 5.3
Machines & Mechanical Systems	X						5.5 & 6.4
Controls & Mechatronics			X				6.5
Materials Selection	X						3.4 & 5.6
Modeling & Measurement Systems	X						5.0 & 6.0
Manufacturing	X						3.4 & 5.6

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Appendix D: Terms and Abbreviations

Abbreviation	Term
COTS	Commercial Off-the-Shelf
CRASH	Controlled Recovery and Safe Handling
HDPE	High-Density Polyethylene
TPU	Thermoplastic PolyUrethane